Yori Gidron

Behavioral Medicine

An Evidence-Based Biobehavioral Approach

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Preface

This book has been a journey into my learning, research work, and steps in behavioral medicine and health psychology. You are warmly invited to join the outcome of this journey.

Behavioral medicine is the interdisciplinary field of psychology, neuroscience, and medicine, which investigates the bi-directional relationships between psychosocial factors and health outcomes. It also investigates the underlying neurobiological mechanisms of such associations, a domain called psychoneuroimmunology (PNI). It also includes development and scientific testing of new psychological interventions to affect health outcomes. This book covers all these topics, from doctor-patient communication and patient adherence to preparation for surgery and from psychological risk factors of heart disease to psychological prognostic factors and interventions in cancer.

I included a "problem-based learning" (PBL) approach and a multi-level analysis, by beginning and ending each clinical chapter with a hypothetical case. The chapter then provides the epidemiology and biomedical background of the disease, the psychological predictors of onset or prognosis in this condition, the PNI findings possibly explaining such associations, and psychological interventions relevant to that health condition. Each clinical chapter ends by returning to the hypothetical case, applying the covered evidence to better treat that patient. The entire book takes a strict "evidence-based" approach.

Lille, France Yori Gidron

Acknowledgments

I wish to dedicate this book first to the memory of my beloved parents, Michael and Erika Gidron; may they rest in peace. We lost them also due to my and our professional ignorance. I dedicate this book also to my beloved children, Aviv, Oriane, Idann, and Aner Gidron, who I wish good health and fulfilment in life.

There are numerous people I must thank: Laure-Anne Thieren for her tolerance and help for many years, as well as my sister Ada and brother Amos and their families; my professors, Joel Norman, Richard Schuster, Ruth Kimchi, Shlomo Breznitz, Patrick McGrath, Karina Davidson, Ray Klein, Vincent Lolordo, and Michael Sullivan; my peer students, Nethanel Snow and Farhad Dastur; and many colleagues and mentors including Reuven Gal, Arik Shalev, Sara Frieedman, Moshe Farchi, Shmulik Ariad, Rivka Berger, Miriam Argaman, Harel Gilutz, Mahmoud Hueihel, Alain (Ilan) Ronson, Darius Razavi, Bianca Gordon, Johan Denollet, Ivan Nyklicek, Nina Kupper, Martijn Kwaijtaal, Flortje Mols, Manos Karteris, Brigitte Velkeniers, Jo Nijs, Martin Zizi, Reginald Deschepper, Filip Germeys, Antonio Giangreco, Alexis Cortot, Eric Lartigau, Sophie Lelorain, Delphine Grynberg, Mathias Chamaillard, Julian Thayer, Hideki Ohira, Luca Vannucci, David Spiegel (the book evaluator), and Giulia Plat for entering the references. I thank my dear past and present doctoral students, Bina Nausheen, Emilie Arden-Close, Rachel Sumner, Marijke De Couck, Danial Herzog, Bibiana Fabre, Einav Levy, Eowyn Wittenberghe, Eva Reijmen, and Laura Caton, and many master students. If I forgot someone, please forgive me. I also thank each patient and study participant who took part in my studies along this voyage, to whom we owe our knowledge. Special thanks for Redford Williams' initial important comments.

Every book is a person's own story, his or her summary and interpretation of what they know, in this case of behavioral medicine. This is my story. I hope it will serve many of you well.

Shalom Yori Gidron

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Chapter 1
Models of Stress and Methodological Considerations in Behavioral Medicine

What Is Behavioral Medicine?

Behavioral medicine is a scientific and clinical domain incorporating scientific models, empirical knowledge, research methods, and clinical skills from psychology and medicine. Behavioral medicine is the field which scientifically investigates the bi-directional relationship between psychosocial factors and illnesses and health and applies this knowledge to patient care. As such, it is a domain that is, or needs to be, taught in medical schools and in nursing schools, as part of medical education, as well as in health psychology courses and programs. Related domains are health psychology and medical psychology. While some authors may wish to distinguish between these domains, I personally think the differences are not large nor are they very important. The agenda and potential importance of these topics for science and patient care are greater than any such semantic endeavor. Topics in behavioral medicine include examining which factors predict recovery from surgery, whether anger triggers heart attacks, whether and how hopelessness can accelerate tumor progression, whether providing elderly people with control has any beneficial health impact, whether child attachment influences children's health outcomes during illnesses, what neuroimmunological factors mediate the relationships between psychosocial factors and diseases, and whether psychological interventions can improve patients' health status. I shall attempt to cover all these questions in this book, step by step, carefully but purposefully and systematically. Multiple domains from psychology converge into behavioral medicine including personality, social psychology, clinical psychology, learning and behavior, biological psychology, perception and cognitive neuroscience, and neuropsychology, to name a few. Furthermore, the topics of immunology, neuroimmunology, epidemiology, internal medicine, pathology, genetics, and psychiatry are main medical domains which converge into behavioral medicine as well. Thus, one main challenge for the student of behavioral medicine is to try to grasp this vast array of knowledge and

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information. Research methodologies from psychology, medicine, and epidemiology are heavily used in behavioral medicine, to which we shall relate in this chapter as well.

This is an introductory chapter. For readers familiar with these issues, this chapter may be mainly a rehearsal, though various findings may be new. I present a general model of stress from a psychological point of view, followed by multiple empirical examples. The basis of behavioral medicine includes understanding the processes in stress. The chapter ends with methodological issues that are essential to consider in behavioral medicine.

The Basic Concepts and Global Model of Stress and Adaptation

One of the most fundamental topics of behavioral medicine on which it heavily rests is the issue of stress and its consequences. Perhaps one of the most frequently used words pronounced by people across many countries and situations is the word "stress." This word is often also used (and misused) by patients and clinicians. This fundamental chapter and many parts of this book will aim to provide more precise definitions for concepts related to stress and adaptation and to its impact on health. Let's examine three different definitions. Baum (1990) referred to stress as an unpleasant emotional experience, accompanied by cognitive, behavioral, physiological, and biological concomitants, which are directed either at altering the stressful event or accommodating to its effects. This definition conceives stress as a multidimensional response, which has a function – to change the eliciting situation or to accommodate to it. That definition sees stress as a response, aimed to elicit coping in order to change the trigger or to adapt to it. That definition, as we shall see, includes too many of the components of stress, thus seeing it as a process. In contrast, Lazarus and Folkman (1984) conceptualized stress as a situation or event that is *appraised* as one which is demanding beyond a person's resources. This definition conceptualizes stress as an event or trigger rather than as a response. Furthermore, it links this issue to how the event is appraised or evaluated, vis-à-vis one's perceived severity of the event and one's resources for dealing with the event. Thus, here, stress is an event, which is seen as stressful if it demands more than one's resources can meet. Finally, Chrousos (2009) proposes this definition: "Stress is defined as a state in which homeostasis is actually threatened or perceived to be so." Here, Chrousos too includes both the actual and subjective perception of threat, in agreement with Folkman and Lazarus' emphasis on the subjective appraisal. However, Chrousos defines stress as a state of departure from homeostasis, usually connoting a physiological disequilibrium, thus conceiving stress as a response. Just as when you go to the furniture shop and ask for a chair while not pointing at a table, we must be clear as scientists and clinicians when discussing the issue of stress with patients or colleagues: Does stress refer to the triggering event or to the response?

To clarify this soup, I propose to distinguish between the stressor (the triggering event) and the stress response, which can be elicited seconds, minutes, hours, or even weeks after the event. The response can of course manifest as a psychophysiological state of disequilibrium, in line with Chrousos' definition. The additional crucial contribution of Lazarus and Folkman (1984) to the field is the addition of the concept of appraisal. Primary appraisal refers to a person's cognitions about a situation including whether it is positive or negative, relevant or irrelevant to him or her, and whether it is threatening or challenging. Secondary appraisal includes cognitions such as whether the situation is controllable and predictable and whether one has the resources to meet and overcome the situation. Indeed, an older but classical study clearly demonstrated how our appraisal causally affects our stress response. In that study, students were about to view a film depicting Central American youth undergoing a rites ceremony. The students were randomized to receive no preparation (control), preparation inducing negative appraisal, preparation inducing neutral (factual) appraisal, or preparation inducing positive appraisal. Students receiving the negative appraisal induction (e.g., the film is unpleasant and the youth experience fear and pain) indeed reported and objectively had higher sympathetic stress responses (as measured by galvanic skin response; Speisman et al. 1964). Thus, just as beauty is in the eyes of the beholder, stress is in the brain of the perceiver as the stress response is caused by our appraisal, among other causes. As we shall see below, our appraisal is of course also mediated by alterations in brain activity in regions of crucial impact on our short-term stress response and long-term well-being as well (Goldin et al. 2008).

What takes place between the stressor, our appraisal, and finally our stress response? This calls for another element in the chain of stress and adaptation called "coping." This too is another misused concept we will clarify as well. Coping refers to one's cognitive and behavioral efforts aimed at either changing or solving a stressor (problem-focused coping; PFC) or aimed at adapting and accommodating to the stressor (emotion-focused coping; EFC; Lazarus and Folkman 1984). As we shall now see, both forms of coping, namely, PFC and EFC, are legitimate and effective coping strategies; however, each can be effective in different types of stressors. PFC includes defining the situation, proposing various solutions, choosing the optimal one, implementing it, and finally evaluating the chosen solution's effectiveness to solve the problem. Many individuals and organizations have difficulties in some or all of these PFC stages. EFC on the other hand includes relaxation, denial, avoidance, praying, reappraisal, and catastrophizing.

Hundreds of studies have examined the efficacy of both general coping categories in relation to stress responses and adaptation. Generally, PFC appears more effective in solvable and controllable situations, while EFC is mainly effective in unsolvable and uncontrollable situations. Thus, people need to match the type of coping to the controllability of the situation, a model called "the goodness of fit hypothesis" (Forsythe and Compas 1987). Empirically, this model has received quite a bit of support in many different contexts. For example, Levine et al. (1987) examined the relationship between denial (a form of EFC) and short- and long-term cardiac health indices in cardiac patients arriving at hospital. He found that while denial was inversely related to recurrent cardiac events during hospitalization, denial was positively related to readmission to hospital during the 1-year follow-up after discharge from hospital. The levels of controllability for the patient are different in and out of hospital. Denial, a clear EFC, was helpful in hospital when the *medical staff had control* and when patients' denial could possibly reduce unhealthy excessive sympathetic responses, which may result from excessive worrying and even panic in some cases. However, denial was ineffective in the long run, after discharge from hospital, when *patients* had control and thus should *take control* over their lives, to change any unhealthy lifestyle behaviors. Another real-life study was done in the context of missiles launched at Israel from Iraq during the First Gulf War. Weisenberg et al. (1993) assessed Israeli adolescents' coping strategies while they were in the "sealed room." Citizens entered the sealed room after they heard the siren, wore their gas masks, and verified the safety of the room, as taught by security forces. Adolescents were inquired about what they did then, after taking the safety steps they could do. They found that adolescents using EFC (humor, distraction, social support) had fewer psychiatric symptoms than those using PFC (e.g., verifying their mask is well placed). Again, after doing the minimal PFC steps under their control (entering the sealed room, wearing the masks, and verifying the closure of the room), pursuing further control in such a highly uncontrollable situation may lead to much worse stress responses due to failing to alter the situation and reduce one's stress response. In such contexts, EFC is preferable. Finally, another study shows strong evidence for the matching of control and coping in relation to adaptation. Zakowski et al. (2001) assessed perceived control, coping, and depression in 72 community participants. They found that when control was perceived as low, EFC predicted lower depression than PFC. In contrast, when control was appraised to be high, PFC predicted lower depression than EFC. Together, these studies support the "goodness of fit" hypothesis.

Two additional important groups of variables affect the stress and adaptation process. External resources and impediments can alter one's appraisal, coping, and stress responses. External resources and impediments include one's time, money, additional external (stressful) events, and importantly, our social support. The latter is among the most important factors protecting us from the adverse effects of stress, a phenomenon called the stress-buffering hypothesis (Cohen and Wills 1985). In one study, Gerin et al. (1992) had participants debate an issue with two other people, when both others either apposed the participant's point of view (low support) or when one of the others agreed with the participant (high support). The participants' levels of diastolic and systolic blood pressure and heart rate increased to a lesser extent in the high support condition than in the low support condition. In a systematic review of studies on patients with cancer, levels of social support positively predict better prognosis, independent of confounders such as initial cancer stage (Nausheen et al. 2009). Social support can be viewed quantitatively – number of friends and level of support – or from a functional perspective, instrumental, informational, and emotional types of support. These can of course also affect both PFC and EFC.

The second group of variables refers to internal resources and impediments. These mainly refer to personality variables or general capacities. Resources include optimism, self-efficacy, sense of coherence (SOC), and mental resilience, while impediments include pessimism, hostility, and neuroticism, to mention a few examples. Let's examine the internal resource of SOC which reflects the ability of people to understand, connect, and predict events in their lives (comprehensibility), to be able to meet their daily challenges they set for themselves (manageability) and the perception that their efforts they invest in their daily tasks are worth and purposeful (meaningfulness; Antonovsky 1987). One study examined elderly people who were about to move to an old-age home and compared them to matched controls remaining at home. Movers have lower positive affect and lower natural killer cell activity (immune capacity relevant to cancer and viruses) compared to non-movers. However, if movers had high SOC, the adverse psychological and immunological outcomes were mitigated (Lutgendorf et al. 1999). Thus, in this case, SOC had a moderating (protective) effect on the adverse effects of relocation on the immune system (see below further information about moderation).

An example of an internal impediment variable would be hostility, the tendency to behave antagonistically, think cynically, and feel anger (Barefoot 1992). Among participants undergoing a cold pressor stress test, those high on hostility evidenced higher levels of increased heart rates than those low on hostility (Rhodes et al. 2002). Thus, in this case, stress and hostility synergistically interacted to result in a worse stress response. Numerous other examples of the effects of internal resources and impediment variables on stress responses exist, and I just provided a few illustrative examples. Two important biological variables detailed in the next chapter can be conceptualized as psychoneurological internal resource variables, namely, left hemispheric lateralization, shown to weaken effects of life events on mental health (Herzog et al. 2016), and high vagal nerve activity, shown to predict better health and increase people's biological resilience after stress (Weber et al. 2010; Zhou et al. 2016). These will be explained in Chap. 2.

We shall now put all these elements of the stress and adaptation chain into one model, using Taylor's (1995) framework (see Fig. 1.1). According to this framework, an event (the stressor) undergoes primary and secondary subjective appraisals. These give meaning and intensity to the perception of the event. In fact, one can put a bi-directional arrow between the event and appraisal since the latter can also shape the event to some extent, via our cognitive evaluations of the event. Appraisal is followed by PFC or EFC, and research shows that PFC is more suitable for controllable events, while EFC is mainly suitable for uncontrollable events. These elements are all under the influence of external and internal resources and impediment variables (e.g., social support, hostile personality, left hemispheric lateralization). The stress response or outcome can eventually result in adaptation or illness, daily functioning or disability, positive or negative mood. Figure 1.1 depicts this framework in relation to a minor traffic accident as the event. Here, the event is appraised mildly negatively, and the person uses both PFC and EFC to manage the event. Having high social support as well as an optimistic perspective and high SOC, this person has little risk for developing post-traumatic stress disorder (PTSD). We will

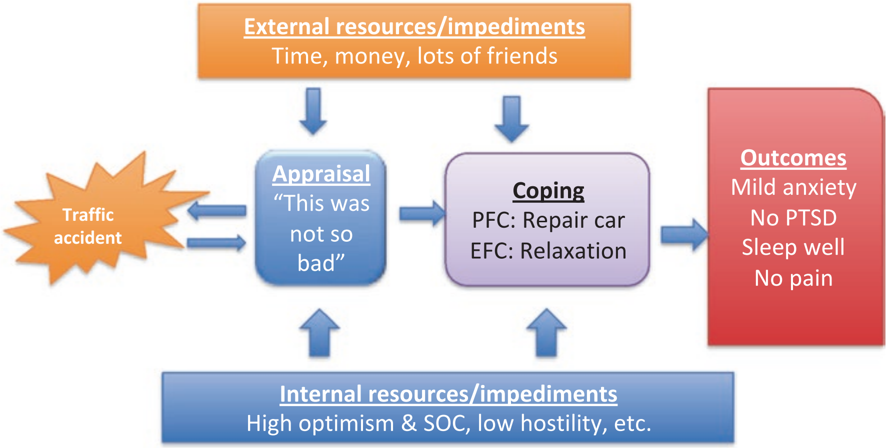

Fig. 1.1 Adapting Taylor's (1995) framework of stress and adaptation to a hypothetical traffic accident event and its possible consequences

use Taylor's (1995) stress and adaptation framework throughout the book and map onto it the scientific evidence concerning psychological risk factors and prognostic factors of the diseases we shall study in the following chapters.

Up to this point, I provided evidence for the role of elements within Taylor's (1995) stress and adaptation framework. However, one study examined together all aspects of Taylor's framework, using a semi-experimental design (Gidron and Nyklicek 2009). We administered to students written scenarios which verbally manipulated the variables in Taylor's model, except for personality. The scenarios depicted an event with different levels of objectively judged severity (e.g., you spilt coffee on your pants, reflecting a minor event, or were informed that a close person is very ill, reflecting a major event), one's appraisal (you considered the event benign or severe), coping (denial versus PFC), and one's level of social support (being lonely lately or with many friends). A brief hostility scale represented the internal impediment variable. Participants had to rate their level of distress in each of 16 scenarios, reflecting their stress response. Results revealed that event severity, appraisal, and social support significantly and independently affected distress levels. Furthermore, two interactions were found: Event and appraisal synergistically interacted in relation to distress, and appraisal interacted with coping in relation to distress. PFC led to less distress than did EFC, but only in events appraised as benign, not severe. Future studies need to examine this global framework of stress and adaptation either using real experimentally induced events and objective measures of stress responses or by conducting a comprehensive prospective field study (e.g., work setting) using diary and objective physiological measures corresponding to Taylor's framework. This brings us to the next topic.

Job Stress Models

We shall now focus a bit on one specific life domain concerning stress, namely, job stress. More and more emphasis is put today on job stress for the following reasons. First, job stress has a potentially grave effect on health outcomes such as burnout and cardiac mortality (e.g., Elovainio et al. 2006). Second, stress affects our decision-making by possibly causing us to see few choices and to choose among them the desired option in a more chaotic, rather than systematic, manner (Keinan 1987). Third, job stress has a grave economic cost (Ganster et al. 2001). For these reasons, one important topic in behavioral medicine, of relevance to occupational medicine, is job stress and its management. I will briefly review here the three leading job stress models and provide examples for their evidence.

The first and most known model is the demand-control model (DC; Karasek Jr 1979). According to this model, jobs can be classified according to their level of job demands (e.g., working hours, deadlines, night shifts) and to the degree of control workers have over their job (e.g., choice over taking breaks, how to perform the work). The DC model then posits that jobs with high demand and low control, a situation called job strain, have the worse health consequences. For example, a recent study done in 500 Iranian petrochemical workers found that high job strain was strongly associated with the cardiovascular risk factors of elevated heart rate and smoking (Habibi et al. 2015). In a meta-analysis of 13 prospective European studies with 197,473 people, who initially had no heart disease, job strain was significantly predictive of cardiac events (nonfatal myocardial infarction or coronary death), independent of sex, and age (Kivimäki et al. 2012). These associations were even stronger when excluding workers who eventually had early-onset CHD in the 3–5 years after assessing DC, thus reducing the risk of "reversed causality" in which early undiagnosed cardiac disease may have led to job strain. Furthermore, job strain is also related to job absenteeism (Schechter et al. 1997). In more recent years, social support from workers' bosses or co-workers was added to this model because of its relevance to moderating the negative effects of high demand and low control. Indeed, some research proposes that supervisor support can buffer (moderate) the adverse effects of job demands on burnout levels, in workers with low control. However, co-worker support may at times exacerbate the negative effects of job strain (Willemse et al. 2012).

The second and more recent job stress model is the effort-reward imbalance (ERI) model (Siegrist 1996). According to the ERI model, work is part of a social reciprocity context where workers have expectations of rewards (e.g., recognition, salary) for their efforts (e.g., working hours, meeting deadlines, task production). However, at times, workers could face a long-term ERI, by putting greater efforts than receiving rewards, which can be a chronic stressor. Furthermore, Siegrist claims that such an ERI situation is particularly difficult when the will for putting efforts is intrinsic, a situation he terms over-commitment. Already in 2005 van Vegchel et al. (2005) conducted a review of 45 studies. In that review, ERI was found to predict smoking and excessive drinking, cardiovascular symptoms and cardiovascular diseases, and poor well-being. A study mentioned above, used a simple scale for assessing work injustice, closely related to ERI (Elovainio et al. 2006). They found that low-moderate work injustice predicted a higher risk of cardiac death than did high work justice. In a small-scale study we conducted among Belgian doctors in Brussels, we attempted to assess ERI using the standard ERI questionnaire, as well as by a word-search test. The latter indirect test included ten words reflecting “effort” and ten words reflecting “reward,” which participants had to locate in 2.5 minutes. This was based on a “pop-out” effect, where people with certain traits or interests perceive in their environment stimuli that are congruent with their traits or states. Interestingly, while the explicit questionnaire-based measure of ERI was inversely related to presenteeism (coming to work during vacation or when ill), the indirect ERI scores were positively correlated with presenteeism (Gidron et al. 2014a). This study proposes that the manner of measuring ERI, a sensitive construct particularly during economic crises, can influence its relationship with outcomes such are absenteeism and presenteeism.

The most recent job stress model is the Job Demand-Resources (JD-R) model (Bakker and Demerouti 2007). This model is mostly an elegant reframing and integration of the DC and ERI models together. In the JD-R model, demands (e.g., working hours) are contrasted with the resources people receive, some of which are resources at work (e.g., salary, feedback) and some of which are person-oriented (e.g., recognition, support). In the JD-R model, higher demands constitute a chronic stressor, possibly leading to adverse outcomes, while higher job resources increase worker motivation and productivity. In a unique study, job resources were conceptualized as the level of transformational leadership workers have, given to them by their boss. This type of leadership is characteristic of empowering workers, letting them share some decisions and increasing their abilities, thus using workers’ own resources and increasing their job control. This is in contrast to more old-fashioned and still quite common transactional leadership, where workers earn money according to their efforts and performance. Among 262 German workers, Syrek et al. (2013) found that though time pressure (job demands) correlated positively with exhaustion and inversely with work-life balance, high transformational leadership (resources) weakened (moderated) these relationships. This study is elegant as it not only supports the JD-R model but it also considers a crucial and empowering form of leadership, transformational leadership, as the resource variable, further supporting this type of leadership’s importance for workers’ well-being.

Theories and models are essential in integrating variables in a concise and meaningful manner and in directing research and clinical work. However, all scientific and clinical endeavors must rest on sound scientific testing and empirical support. As such, behavioral medicine needs to adhere to important methodological rigor, to enable researchers and clinicians to infer with reliability and validity any conclusions concerning the role of psychosocial risk factors or interventions in health outcomes. We thus now turn to some major methodological issues upon which the science and practice of behavioral medicine rests.

Methodological Issues

Assessment in Behavioral Medicine

Assessment is a major issue in behavioral medicine, since it enables clinicians to examine patients' symptoms and health over time, in order to examine responses to treatment. Furthermore, it enables one to assess risk factors (e.g., smoking, hostility, social support) and thus identify people at risk or who are less at risk for an illness. For researchers, assessment is a major tool to investigate the associations or effects of certain variables on health outcomes. We shall follow Taylor's framework and provide examples of assessment tools for her framework's main elements. Stressors can be assessed with many types of questionnaires, from ones inquiring about major life events via the Life Events Questionnaire (Norbeck 1984) to ones assessing minor daily hassles such as the Daily Hassles Scale (Holm and Holroyd 1992). Appraisals can usually be assessed with single-item scales concerning perceived stressfulness of a certain event, using a visual analogue scale for stress (Sneed et al. 2001) and a single-item scale of perceived control over an event or over one's life or illness in general (e.g., Tovbin et al. 2003). An important empirical application of assessing one's immediate appraisal of a situation (or stress response) was done in a study in Israel, where patients arriving at the emergency room after traumatic events were asked to rate on a 1–10 scale their level of stress, twice within an hour. Those showing twice no reduction of perceived stress below 5 or an increase from below to above 5 were at increased risk for post-traumatic stress disorder later (Dekel and Kutz 2004). This exemplifies the predictive validity of simple one-item appraisal scales of one's stress appraisals. We should note that such assessment could reflect either appraisal of the situation's stressfulness or of patients' state stress response.

Coping strategies are typically assessed with numerous types of scales; among the more known ones are the ways of coping scale, the coping inventory for stressful situations, and the COPE by Carver (e.g., Carver 1997). One problem with coping scales (and other self-reported scales; see below) is that people may not be aware of their precise coping strategies since these are often semiautomatic processes. One manner of overcoming this is to use implicit tests. In such tests, the participant is unaware of the variable being measured or of the process of doing so, thus bypassing problems of presentation biases, self-awareness, and recall, inherent in self-reported measures. Indeed, one study used the implicit association test (Greenwald et al. 1998) for assessing alcohol consumption as a form of coping and its relationship with drinking frequency (Lindgren et al. 2012). In many studies, such implicit tests correlate poorly with self-reported measures of similar constructs (e.g., Nausheen et al. 2007). This is not necessarily evidence for the lack of validity of implicit tests; rather it suggests that explicit and implicit processes are different and predict different outcomes.

Outcome measures can include self-reported measures of stress responses such as the perceived stress scale (Cohen et al. 1988); burnout with the Copenhagen

Burnout Inventory (Borritz and Kristensen 1999); depression, using the Center for Epidemiological Studies – Depression scale (CES-D; Radloff 1977); or PTSD, using, for example, the post-traumatic distress scale (Foa et al. 1997). Another important outcome variable is daily functioning, which may be assessed via self-report (e.g., Duke Activity Status Index; Hlatky et al. 1989) or by observing patients' ability to walk as many stairs as possible in 1 minute (Harding et al. 1994). Biological stress responses consist of numerous measures and include cardiovascular parameters (e.g., heart rate, heart rate variability, blood pressure), neurohormones (e.g., cortisol, norepinephrine), immune parameters (e.g., number or functioning of T-cells, pro-inflammatory cytokines), DNA damage (e.g., 8-OH-dG), and brain activity measures (e.g., using EEG, MRI) in regions related to stress modulation. These outcomes need to be chosen based on the research question and relevance to the illness being investigated. Too often, simple and general measures such as cortisol are used, in relation to any disease. However, the relevance of cortisol to colds, diabetes, heart disease, or cancer is far from simple. Instead, health psychologists need to collaborate with biologists and physicians and should dare to "dive" into reading the core biological processes of a disease and test the relationships between psychological factors and disease-specific outcomes and their relevant underlying biological factors such as matrix metalloproteinase in coronary artery diseases or markers of DNA damage and expression of tumor suppressor genes in cancer. The majority of this book examines the relationships between psychological factors (including stressors) and the onset or prognosis of physical diseases, and we shall refer to the framework of Taylor and to such assessment tools throughout most chapters. One last point to consider when assessing outcomes is that they often can correlate strongly but not perfectly with each other. Thus, for example, following oral surgery, we found that post-operative pain, extent of mouth opening, and self-reported pain were all significantly correlated with each other but not perfectly (Gidron et al. 1995), suggesting that each outcome measure "tells another story." This can result from using different methodologies or from each measure reflecting other disease aspects.

Internal resources refer to multiple variables which each can be assessed using questionnaires and structured interviews or more recently by implicit tests and even physiological indexes which reflect psychophysiological resources. These include, for example, the general self-efficacy scale (Schwarzer and Jerusalem 1995), a person's typical coping strategies assessed as a general trait by the Brief COPE (Carver 1997), sense of coherence (reflecting comprehensibility, manageability, and meaningfulness; Antonovsky 1987), or trait positive affectivity assessed by the positive affect and negative affect schedule (PANAS; Watson et al. 1988). Physiological resource variables could include vagal nerve activity, indexed by heart rate variability (HRV; Kuo et al. 2005) and hemispheric lateralization (HL). HRV should be measured during a 5-minutes resting period, and HRV predicts multiple physical and mental outcomes (see throughout the book). Importantly, HRV reflects true psychophysiological resilience because people with high HRV recover from stress faster, on multiple body systems (Weber et al. 2010). HL reflects the degree of activating certain brain regions or functions associated with one hemisphere versus the other. We can measure HL by questionnaires, neuropsychological tests including the line bisection test, and electroencephalogram. Testimony of HL being an internal resource variable was found in a study showing that in people with left HL, severe life events do not correlate with mental problems, while such associations do exist in people with right HL (e.g., Herzog et al. 2017).

Internal impediment variables include scales of hostility (e.g., Gidron et al. 2001), negative attribution style (e.g., Peterson et al. 1982), or trait-negative affect assessed by the PANAS (Watson et al. 1988), to name but a few.

External resources include time or money which can be assessed using indexes of socioeconomic status and include scales for assessing social support such as the Duke social support scale (George et al. 1989). Finally, external impediments can include scales of harassment or bullying at work (Escartín et al. 2010) or perceived economic hardship, for example.

Here is the place to mention a few unique approaches for assessing various psychosocial variables in behavioral medicine and psychology in general, to overcome self-reported biases. Various variables can be assessed by implicit or indirect tests, especially for constructs which are socially sensitive to admit about, such as loneliness or hostility. The causal attribution style, which reflects the extent to which people attribute negative events to themselves (internality), to a stable cause (stability) and to a cause which has global consequences (globality), is part of the revised model of learned helplessness and depression (Abramson et al. 1978). One unique manner for assessing such negative attributional style is the Content Analysis of Verbatim Explanations (CAVE; Schulman et al. 1989). Here, the investigators use written content from which they derive quantitative levels of internality, stability, and globality. One study tested whether negative attributions assessed by the CAVE method predicted health, over 35 years (Peterson et al. 1988). In another study, we used the word-search technique, to assess people's optimism-pessimism indirectly. Based on a "pop-out" effect (where person-relevant content "appears" more easily in one's perceptual field), people had to find five optimistic and five pessimistic words in 1 minute, in a word matrix. The participants were patients with lung cancer. Those who survived during a 1-year follow-up had found significantly more optimistic than pessimistic words at study entry than those who died, independent of multiple confounders (Gidron et al. 2014b). Finally, recent years have seen the use of measures based on cognitive neuroscience, adopted to clinical and health psychology. One such measure is the implicit association test (IAT; Greenwald et al. 1998). We adopted the IAT to assess loneliness, where patients had to match words like "alone" or "table" to the construct attribute of lonely or furniture and words like "me" and "others" to the concepts self or others, as fast as possible. Cancer patients with faster associations between lonely and self had significantly higher levels of the pro-angiogenic factor vascular endothelial growth factor, while the self-report scale of loneliness was unrelated to this biological factor (Nausheen et al. 2010). Thus, indirect measures for assessing multiple constructs in behavioral medicine exist. These indirect measures are crucial because they bypass problems of presentation biases, recall biases, and lack of self-awareness, inherent in self-report measures. In addition, such measures may be at times better since they often predict objective outcomes when self-report measures of the same construct do not (e.g., Nausheen et al. 2010).

The Issue of Control

This is a concept of two different meanings in behavioral medicine, each which is central to the field's research rigor. First, this concept refers to the methodological consideration of and usually statistical control over third variables which may explain any relationships between a predictor X and an outcome Y. For example, Watson et al. (2005) found that hopelessness significantly predicted survival in patients with breast cancer, after statistically controlling for confounders such as age, cancer stage, and treatments. Here the statistical control removed the effects of these confounding variables from survival and enabled to test the relationship between hopelessness alone and cancer survival that was independent of the effects of such confounders. This is crucial for the field of behavioral medicine as it enables us to test and to infer the unique contribution of psychosocial risk factors to disease onset or prognosis. This argument is crucial when presenting to physicians and biologists results from behavioral medicine. Methodological control over third variables can be done by excluding members of one level of a variable (e.g., excluding men) for various reasons or by matching patients in a case-control study on third variables. For example, Gidron et al. (2010) found that hemispheric lateralization predicted the onset of colds, in a prospective matched case-control design, where cases (who later developed colds) and controls were matched on age, sex, and IQ. Here, hemispheric lateralization predicted colds after methodologically, rather than statistically, controlling for age, sex, and IQ.

The second type of control concerns the type of control group in experimental studies or in randomized controlled trials (RCT). Here control refers to scientists' attempts to "control" over effects of time (natural changes in the phenomenon being followed) by assessing the same outcome in both experimental and control groups over time, while in controls this is done without any intervention, usually. In addition, one can control over patient expectations (placebo effect), support and attention, or control for known effects of an intervention that is already given. Controlling for attention and expectations is done to test whether one's intervention has any additional effects beyond non-specific but important issues such as patient expectations, providing support and empathy. The choice of an adequate control group is not simple, and it also reflects the precise research questions, such as what is the additional value of a new intervention compared to existing ones. For example, Brom et al. (1993) found that a "coping and working through" intervention had no better effect than assessment alone, on PTSD symptoms, in Dutch traffic accident survivors. Both groups showed reductions in PTSD symptoms over time, demonstrating how important it is to compare the effects of an intervention to the spontaneous changes in a condition over time, using a control group. The issue of choosing what will a control group receive (usual care, another treatment, attention, or assessment only) is complex and involves ethical, methodological, and logistical considerations. Related to this issue is that often in behavioral medicine, the design cannot be double-blind but at best single-blind. Patients know whether they received one of two or more interventions; thus scientists cannot make them blind to group status. In contrast, it is possible to have the researcher assessing patients at follow-up blind to group status, thus achieving at least a single-blind RCT. Such blindness is crucial for reducing biases.

The Problem of Negative Affectivity

Negative affectivity (NA) is a personality trait reflecting the pervasive tendency to attend to, experience, and report negative sensations or emotions (Watson et al. 1988). NA is closely related to, and is often used as a synonym of, the personality dimension of neuroticism. In a critical review article, Watson and Pennebaker (1989) reviewed multiple studies and showed that NA is a "nuisance variable," explaining many correlations between self-reported negative psychological factors, such as catastrophizing, and subjective outcomes such as pain or between anxiety and quality of life. Furthermore, they found that after statistically controlling for NA, several of such correlations were no longer statistically significant. In contrast to reporting greater subjective symptoms, people high on NA do not necessarily have a higher risk of death (Costa Jr and McCrae 1987), possibly since they may seek and undergo more medical tests due to being preoccupied by their health (Hajek et al. 2017). This could enable early disease detection and treatment due to being monitored medically so often. A recent study done in Japan found that levels of neuroticism were positively correlated with telomere length, a noncoding end part of the DNA molecule which protects the DNA and, when shorter, predicts earlier mortality (Sadahiro et al. 2015). This again suggests that people high on NA may in fact have lower risk of death. However, often self-reported measures are the main assessment tools used in behavioral medicine, and these are often contaminated by NA. One methodological way to overcome these issues is to statistically control for NA when examining the relationships between psychological factors and self-reported outcomes. This was done in a study on the neuropsychological predictors of self-reported colds (Gidron et al. 2010). Another manner of overcoming the problem of NA is to replace one or both self-reported measures (predictor or outcome) with an objective one. This renders the measurements less "contaminated" by NA. Thus, for example, one can examine the relationship between catastrophic coping and health-care utilization as verified by objective medical records.

Nevertheless, is it possible that NA does in fact predict objective health outcomes, but that this depends on or is moderated by a third variable? Our team recently found a finding suggestive of such a possibility. NA has been related to unhealthy outcomes such as cardiac death (Shipley et al. 2007), but not all studies show this, such as concerning the common cold (Cohen et al. 2003). NA has also been inversely related to performing healthy behaviors such as physical activity (Wilson et al. 2015). In the next chapter, we describe the concept and studies related to hemispheric lateralization (HL), the tendency to activate parts of one hemisphere versus homologue parts in the other one. Briefly explained here, we found that NA was inversely related to healthy behaviors, but only in people with right-HL, not with left-HL (Herzog et al., unpublished data). It is possible that NA may actually predict poor health behavior and objective health outcomes, but that this may be moderated by people's HL. Future studies need to examine this issue more in depth.

Self-Reported Versus Objective Measures of Health

Though I have touched this issue in the present chapter (e.g., concerning NA), its importance renders it having its own subsection. First, before asking and attempting to answer which type of outcome, objective versus subjective, is preferable, it is important to understand that outcomes are complex, particularly health outcomes. Second, to answer this issue, we must ask another question – what is health? Health, as the World Health Organization states, includes physical, psychological, and social functioning, rather than merely lack of injury (WHO 1947, see Grad 2002). Empirically, though various aspects of health often correlate, they are not redundant in relation to each other. In a study on predictors of adolescents' physical and functional recovery from oral surgery, the three outcomes of pain, mouth opening, and daily functioning were strongly but not perfectly correlated with each other (Gidron et al. 1995). Thus, each aspect of health tells another part of the recovery story. It is important to assess health in several dimensions and to identify the predictors of each aspect of health. However, at times, it is also important to identify the predictors of objective, rather than subjective indices of recovery. For example, positive affectivity predicted both subjective and objective measures of colds, while negative affectivity predicted subjective indices of colds only (Cohen et al. 2003). Again, psychological factors "contaminated" by NA will tend to predict better subjective rather than objective indices of health. From a professional and scientific point of view, when trying to communicate and convince our colleagues from the biomedical sciences about the importance of behavioral medicine, it is often important to show that psychosocial factors predict onset or prognosis of objectively measured diseases, independent of known risk factors. This is especially correct when the diagnosis of the disease in question rests upon objective measures as in coronary heart disease or cancer. Nevertheless, identifying the predictors of patients' subjective suffering or quality of life is also a very important scientific and clinical endeavor. Thus, it is recommended to use both objective and subjective health outcomes, but not to substitute one for the other.

Study Designs

This issue is crucial for the inferential validity of studies, the ability to justifiably conclude certain inferences from research findings, as a function of a study's design. Many past or preliminary studies examining associations between psychological variables and an illness used a case-control design. For example, a study may have found that cancer patients were more depressive than healthy controls. However, such a design does not enable to infer at all the nature of the direction of association between X and Y. Does X, depression, cause Y, cancer, or does Y cause X. Finding that people with cancer have higher levels of depression can be totally understood as both depression being a risk factor of cancer but equally plausible as reflecting the possibility that the cancer may cause depression. Moving to prospective studies (cohort or longitudinal study) is an important improvement, because it cancels to some extent the bidirectionality of associations. This is correct at least concerning the factors assessed at time 1, which cannot be caused by those occurring at time 2. Thus, if depression at study entry predicts prognosis in a cancer a year later, one cannot infer with any validity at all that the prognosis later elicited the depression earlier. In contrast, statistically controlling for cancer severity and treatments at entry helps to partly eliminate the effects of these factors on the prognosis, leaving what is unexplained in prognosis to be possibly explained by or predicted by initial depression levels. However, to really infer a causal relationship between psychosocial (or any) factors and health, we need to perform an experimental randomized controlled trial (RCT), the highest level of science in health research and in behavioral medicine. Here, we randomize people to one of several conditions, one which includes trying to change (improve) the hypothesized psychological risk factor (e.g., reduce hostility or depression). Then, we examine the health changes in a future follow-up. In an RCT, the independent variable is group, while the dependent variable is the health outcome. The randomization usually takes care of baseline group differences that may occur after randomization. An even more strict design is a matched RCT. Here, the researcher matches pairs of patients before randomization, on one or more variables which are important for affecting the outcomes, and then randomizes one pair member to one of the conditions; the other one then automatically belongs to the other condition. This increases the chance that groups will not be different on baseline variables after randomization, and this increases the statistical power. This also enables to possibly detect statistically significant differences in the outcomes, even if the samples are small (e.g., Gidron and Davidson 1996; Gidron et al. 1999). In addition to answering the question whether associations between variables in clinical health psychology are causal, the experimental design in such settings provides evidence for the benefits of new therapies, of crucial importance for patients. In addition to clinical benefits, the experimental design also enables researchers to try to answer fundamental questions concerning biological mechanisms in behavioral medicine. These could include the effects of various types of appraisal on physiological stress responses or the effects of various emotional regulation skills on brain activity. Having mentioned the RCT, it is important to note that at times, perhaps due to ethical or logistic constrains, it is impossible to perform an RCT. In such cases, an AABB design could be warranted, where baseline values are measured twice before and then twice after an intervention. This design only controls for the effects of time on the measured outcome. Such a design was used in a study examining the effects of a mastery program on parental use of seat belts for their children in vehicles (Gidron and Hochberg 2003).

The design of a study, together with adequate sampling, reliable and valid assessment tools, and statistical consideration of relevant confounders, all contribute to the inferential validity of a study: Does a psychological variable predict or cause a health outcome?

Generality Versus Specificity of Effects of Psychological Factors on Health

I shall end this chapter with an open and relatively unanswered question. Do certain psychological factors affect the risk of specific diseases or of several diseases in general? For example, is life stress a risk factor of several diseases or only of colds? Stress and stress responses were found to predict colds (e.g., Cohen et al. 1991) but also risk of cardiac death (Elovainio et al. 2006). How do such general relationships occur? If this is a general phenomenon, do such relationships depend on co-exposure to other disease-specific risk factors such as environmental factors (e.g., pathogens), which increase the risk of a cold for example, or co-exposure to genetic disposition, which then increases the risk for the specific gene-related disease? This proposes that researchers in behavioral medicine may need to consider interaction effects between psychological factors and genetic or environmental factors when trying to predict illnesses. This issue also echoes the "nature-nurture" issue that focuses on such interactions. An example of a "nature-nurture" result is a study conducted by my team showing that hostility, a personality facet, correlated positively with severity of coronary artery disease (CAD), but only in people with a first degree relative with CAD, while such a relationship was absent in people without such relatives (Gidron et al. 2002). As we shall see throughout this book, several psychological factors (e.g., stressors, hostility, depression) predict multiple diseases. Future studies should identify the other co-occurring conditions or factors that interact with a specific psychological factor in relation to the risk of a specific or several diseases.

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Chapter 2
The Biology of Stress

Introduction

In Chap. 1, we learned the basic concepts and models of stress, and important methodological issues, all which form part of the foundations of behavioral medicine. We shall now turn to the next crucial foundation block – the biology of stress. Here, I will provide first the general background and information about the biological effectors of the stress response. This will be followed by reviewing central and psychophysiological moderators of stress. I will then introduce the important topic of neuroimmunology – the bidirectional relationships between the immune and nervous systems. This forms the basis for "mind-body" associations, the heart of behavioral medicine. These will then be followed by empirical examples for the effects of stress on immunity and on DNA integrity. Read the chapter bit by bit.

The Biology and Effectors of the Stress Response

This is a complex issue involving multiple signals and neural and physiological systems, central and peripheral. Readers wishing to have a deeper understating of this issue are advised to read the important reviews by Cerqueira et al. (2008), Chrousos (2009), Herman (2012), Kumar et al. (2013), and McEwen (2017). First, the acute stress response (ASR) utilizes two main neuroendocrine systems, namely, the hypothalamic-pituitary-adrenal (HPA) axis and the sympatho-adreno-medullary (SAM) system. The hypothalamus responds first by secreting corticotropine-releasing hormone (CRH) with signals emanating from the paraventricular nucleus in the hypothalamus. This then stimulates pro-opiomelanocortins (POMC) in the pituitary. The POMC include adrenocorticotrophic hormone (ACTH), β-endorphin, and α-melanocyte-stimulating hormone (α-MSH). Stress also stimulates in parallel the locus ceruleus (LC) of the SAM system to secrete

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norepinephrine (NE), which activates multiple peripheral sympathetically mediated responses. The SAM system and HPA axis can also stimulate each other in a mutual positive feedback manner (Chrousos 2009). Indeed, there is evidence for the HPA axis to stimulate the SAM axis since ACTH causes increases in NE (Valenta et al. 1986). Furthermore, this is bidirectional, where stimulation of the LC increases ACTH secretion (Ward et al. 1976) and HPA axis activity (Chrousos 2009). Following this response, cortisol is secreted from the adrenal cortex and NE from the adrenal medulla (Tsigos and Chrousos 2002). Additional central changes include the activation of the amygdala by NE, the limbic brain region which reacts to fear and threat. This can also activate the hippocampus, to store threatening memories (Kumar et al. 2013). Hippocampal activation can in fact downregulate the HPA axis (Herman 2012). Different subregions of the amygdala appear to respond to different types of stressors. While the basolateral amygdala responds to more homeostatic and reflexive stressors like blood loss, the medial amygdala may respond more to "anticipatory" stressors such as restraint stress in animals (Herman 2012). The amygdala excites the HPA axis (Herman 2012). As we shall see in the chapter on post-traumatic stress disorder (PTSD), this process is key to the pathology of that disorder and possibly to the acute stress response in general. The SAM system consists of the LC, a brain stem region whose turnover of NE increases during stress (Korf et al. 1973), as well as peripheral sympathetic pathways innervating most visceral organs and key arteries as well. These innervations are key to brain-body and "mind-body" associations, the foundation of behavioral medicine.

The peripheral biological stress response includes increased heart rate (HR); faster respiration; increased oxygen to the brain or injured cites; increased nutrients to the brain, heart, and skeletal muscles; sweating; pupil dilation; complex alterations in immunity (see below); and activation of crucial regulatory loops (see below; Chrousos 2009). As mentioned above, though the HPA and SAM systems positively reinforce each other, the HPA axis has a "built-in" negative feedback loop, aimed to limit the duration and intensity of the acute stress response (ASR). This negative feedback loop includes inhibitory signals from the adrenal cortex which arrive to the hypothalamus and pituitary. However, several mental health conditions including PTSD are characterized by abnormalities in this negative feedback loop (e.g., Yehuda 2005).

Psychophysiological changes mediated by central modulation during the ASR include increases in arousal, vigilance and alertness, cognition, attention and aggression, inhibition of various vegetative functions such as feeding and reproduction, and activating modulatory negative feedback loops (Chrousos 2009), which we shall now discuss. McEwen (2017) proposes that when the body responds and adapts to environmental stress, via the SAM, HPA, immune, and hormonal changes, it achieves allostasis or adaptation. However, when these responses are overused and these systems become dysregulated, allostatic load occurs.

Central Regulation of the Stress Response

A main element in the stress response and in its regulation is the prefrontal cortex (PFC; Cerqueira et al. 2008). An important basal ganglia-thalamic-PFC network of connections exists (Uylings et al. 2003). This network could be of crucial relevance to stress since the basal ganglia are involved in movement, possibly contributing to the fight and flight response, while the thalamus receives multiple sensory inputs, possibly from threatening or calming signals. Signals from the hypothalamus (which regulates neuroendocrine stress responses) to the ventromedial PFC (vmPFC) are also important. Indeed, the vmPFC is crucial in regulating responses to stressors, including uncontrollable stressors (Amat et al. 2008). In fact, animals exposed to uncontrollable stress did not develop the expected "learned helplessness" if their vmPFC was stimulated first (Amat et al. 2008). Finally, ipsilateral connections exist between the PFC and other regions including the hippocampus (Cerqueira et al. 2008), which could serve to evaluate incoming information in relation to past experiences and coping skills encoded in memory. Appraisal and integration of these signals and regulating the activity of such regions by the PFC can be crucial during stress and homeostasis. The medial PFC appears to regulate behavioral and physiological stress responses, but in a stressor-dependent manner. Furthermore, the pattern of the ASR depends on subregions within the PFC which have been activated. While stimulation of the ventral PFC enhances sympathetic stress responses, the more dorsal parts of the PFC inhibit them via activation of the calming parasympathetic stress responses (Cerqueira et al. 2008; Radley et al. 2006). It is indeed thought that the ventral PFC has a role in the initiation and in the sympathetic segment of the stress response, while the dorsal PFC plays a role in the later, parasympathetic and calming segment of the stress response (Cerqueira et al. 2008). McEwen (2017) summarizes studies showing that chronic stress causes debranching and shrinkage of dendrites in the medial prefrontal cortex, testimony of neural consequences of chronic stress.

Additional brain regions modulating the sympathetic stress responses and more generally the autonomic nervous system (sympathetic and parasympathetic arms) include the insular cortex, various regions of the cingulate, and multiple frontal regions including the ventromedial PFC (vmPFC; Critchley 2005; Thayer et al. 2012).

Two additional points are crucial to understand the brain processing of stress. First, the more superficial cortical regions, which are affected by stress, are mainly innervated by limbic structures such as the amygdala and hippocampus. The damaging effects of stress are not in one region independent of others; rather they emerge as changes in the *connectivity* between regions (Cerqueira et al. 2008). We shall return to this issue in the chapter on emergency mental health and PTSD. The second point is that neuronal loss due to stress appears to occur mainly in brain regions rich in glucocorticoid receptors (GR), while in contrast, neuroprotection occurs under stress in regions rich in mineralocorticoid receptors (MRs; Sousa et al. 2008). In a study using transgenic mice with overexpression of MRs in the forebrain, such mice were found to have reduced anxiety-like behavior and reduced cortisone (cortisol in animals) in response to stress (Rozeboom et al. 2007). Future research should examine whether the relative distribution of GR versus MR in humans predisposes people to mental health problems versus resilience after stress.

This topic becomes more interesting when we combine the psychological and neurological levels in relation to stress. For example, researchers can examine whether different forms of appraisal (catastrophizing versus a neutral appraisal) or vagal breathing versus usual breathing may differentially activate such regions and thus result in different stress responses. One study which addressed a similar issue was conducted by Goldin et al. (2008). They assigned students to either view a film of plastic surgery without instructions (control) or asked to suppress their feelings and facial expressions while viewing the film or asked to adopt a professional reappraisal (taking a doctor's perspective). While the reappraisal group began with elevated amygdala activity, this declined with time. In contrast, the suppressing students showed the opposite pattern of amygdala activity (Goldin et al. 2008). That study shows elegantly how one's cognitive and behavioral appraisal toward a situation is translated to central modifications in brain regions which modulate the ASR.

Some frontal brain regions are crucial moderators of various stress responses, and again, their activity correlates with other important psychological factors as well. Taylor et al. (2008) investigated the relationship between personal psychosocial resources and brain activity during stress. They conducted a pioneering study examining the neural correlates of psychosocial resources as moderators of biological stress responses. They first administered eight measures of "positive psychological" constructs (e.g., optimism, self-esteem, mastery, autonomy) and found them to load onto one factor they termed "psychosocial resources." They then exposed participants to a stressor and found that levels of psychosocial resources were mildly inversely significantly correlated with cortisol reactivity to the stressor ($r = -0.22$, $p = 0.014$). Importantly, they measured participants' brain activity differences between when performing an affect-labeling versus a gender-labeling task, using functional magnetic resonance imaging (fMRI). Levels of psychosocial resources were positively correlated with activity differences in the right ventrolateral prefrontal cortex and inversely correlated with activity differences in the left amygdala. In contrast, levels of cortisol reactivity to stress were positively correlated with differences in the left amygdala activity (Taylor et al. 2008). These results are based on a comprehensive assessment of psychosocial resources, experimentally induced HPA changes, and neuroimaging. They demonstrate that psychosocial resources have an important protective role in HPA stress responses, possibly occurring via modulation by higher frontal regions that modulate limbic regions (amygdala) governing threat perception. Such findings also have vast implications for helping people in the context of exposure to life threat, a topic we shall revisit in the last chapter of this book.

Importantly, the two hemispheres seem to regulate the stress response differentially. The acute stress response is mainly mediated by the right hemisphere, while the left hemisphere may moderate it possibly via interhemispheric inhibition (Cerqueira et al. 2008; Sullivan 2004). Under a state of calmness, the left hemisphere exerts inhibitory control over the right hemisphere. However after chronic stress, there is reduced activity in the left cortex paralleled by increased volume and dendritic growth in the right hemisphere, including the PFC and the bed nucleus of the stria terminalis (Cerqueira et al. 2008). Importantly, there are reduced inputs from the hippocampus to the PFC. These morphological changes are under the influence of cortisol because they can be induced by external administration of this hormone (Cerqueira et al. 2008). One MRI study showed that chronic administration of cortisol led to more volume loss in the left cingulate cortex than in the right one (Cerqueira et al. 2005). Looking at more natural settings, students were found to shift as a group on average from left hemispheric activity, during a relatively calm period in the semester, to greater right hemispheric activity, during a more stressful exam period. Of greatest interest for behavioral medicine, the more they shifted to the right, the more they were also ill (Lewis et al. 2007). All these changes contribute to an increase in sympathetic tone and HPA axis dysregulation, manifested by psychophysiological changes (e.g., increased heart rate, reduced heart rate variability (HRV), increased depression; Cerqueira et al. 2008; Lewis et al. 2007).

Another aspect of the central modulation of stress is related to hemispheric lateralization (HL) as a trait. HL refers to one's general tendency to use functions of or activate certain regions of one hemisphere, versus homologue regions in the other hemisphere (e.g., Davidson 2004). As mentioned above, research shows that mainly the right hemisphere mediates the stress response and that following chronic stress, there is reduced volume of the left prefrontal cortex (PFC) and an increase in the right bed nucleus of the stria terminalis (BNST; Cerqueira et al. 2008). Importantly, two lines of evidence propose that left HL may constitute a moderating and resilience factor. First, infants separated from their mothers, and then reunited with them (the "stranger situation"), show faster behavioral recovery if they have left HL (Cited in Davidson 2004). Second, in a series of three studies, we recently found that while in people classified as right HL, past life events were significantly and positively correlated with mental distress, these correlations were mostly nonsignificant in people with left HL. Furthermore, a brief experimental stressor led to increases in self-reported distress only in participants identified by a neuropsychological test as right HL but not as left HL-oriented people (Herzog et al. 2014). More recently, we found that in Israelis exposed to missile threat, such threat correlated with post-traumatic stress disorder symptoms only in those with right HL, not left HL (Herzog et al. 2017). These findings propose that the left hemisphere and presumably specific regions in it may moderate and even protect us from the effects of stressors on our mental health. We shall return to this issue in the chapter on emergency mental health.

A final issue to cover here is the lateralization of the sympathetic versus parasympathetic response. Clearly, sympathetic stress responses are modulated more by the right hemisphere (Cerqueira et al. 2008). However, conflicting evidence exists about the lateralization of the parasympathetic response, which we shall soon review below. Some authors suggest, based on intracarotid artery injection studies, that the right hemisphere governs and increases the parasympathetic activity (Thayer 2009). In contrast, a more recent study examined changes in basal vagal tone, in patients with a behavioral variant of frontotemporal dementia. Using a lateralization index of neuronal deficits, the researchers found a clear correlation with greater left versus right atrophy and reduced vagal tone (Guo et al. 2016). That study demonstrated clear left hemispheric control of parasympathetic activity.

The Vagal Nerve: The Body Pagoda

In Japan, a country plagued by earthquakes, often the traditional Pagoda, remained intact. Upon investigation, it became clear that this happened because the building's structure rotated around its central beam, enabling it to react to the shock and resume to its original structure after the shock ceased. Our body has such a key homeostatic gatekeeper, which is a major branch of the parasympathetic nervous system, the vagus nerve. This 10th cranial nerve originates from the brain stem and innervates multiple visceral organs including the heart, liver, and lungs. Eighty percent of its fibers are afferent (ascending), while 20% are efferent (descending; Ogbonnaya and Kaliaperumal 2013). Its afferent signals terminate in the nucleus tractus solitarius. At present, the main method for measuring its activity noninvasively is via heart rate variability (HRV), the change in inter-beat intervals between normal heartbeats. HRV is strongly correlated with actual neuronal activity of the vagus nerve ($r = 0.88$; Kuo et al. 2005). Being a key biomarker of the parasympathetic nervous system, HRV levels are inversely related to sympathetic activity (Vlcek et al. 2008). Importantly, this nerve shows clear homeostatic roles which reflect psychophysiological resilience. In a pioneering study, Weber et al. (2010) exposed people to the Trier social stress test (which includes a public speech and a mental arithmetic, both under social pressure). They measured reactivity and recovery in three body systems – cardiac (DBP), hormonal (cortisol), and immune (the inflammatory cytokine tumor necrosis factor alpha, TNF-alpha). In all three systems, those with high baseline HRV returned *faster* to pre-stress baseline levels than those with low HRV. These results clearly demonstrate the homeostatic and moderating role of this nerve in stress. Throughout the book, we will reveal how this nerve's activity predicts multiple diseases and how it may slow the onset or progression of many diseases, because it inhibits crucial factors which contribute to diseases. The vagus nerve is also a major bridge in the "mind-body" connection, and HRV is inversely related to multiple physical diseases and adverse psychological states (e.g., Zhou et al. 2016). Recently, we developed a model proposing that the vagus may be pivotal for preventing and treating chronic diseases because its activity is related in the brain to regions which modulate behavioral risk factors of diseases, because HRV predicts lower risk of many chronic diseases, and because the vagal nerve inhibits oxidative stress, inflammation, and sympathetic hyperactivity (Gidron et al. 2018). Hence, it is crucial for behavioral medicine. This brings us to the next issue.

Basic Neuroimmunology: The Hardware of Behavioral Medicine

The immune and nervous systems communicate and bidirectionally influence each other, as we shall now learn. This is perhaps the core mechanism and main arena by which psychological factors can influence physical health and vice versa, and thus this topic forms the neurobiological foundation of behavioral medicine. Both immune and nervous systems can "speak" and understand each other system's language via responding to the other language's chemical signals or ligands, via respective receptors (Leiter et al. 2016; Ek et al. 1998). Let's begin with the "top-down" paths or brain-to-immune communication. First, the main lymph organs, where the interactions between pathogens and the immune system take place, are innervated by the sympathetic nervous system (e.g., Madden et al. 1995). This innervation can influence the functioning and trafficking of the immune cells. Second, many immune cells express receptors for neurohormones such as for cortisol or norepinephrine. One study found for example that suppressor T-cells (regulatory T-cells) express a high amount of beta-adrenergic receptors compared to T-helper cells (e.g., Khan et al. 1986). This could mean that the sympathetic nervous system may particularly affect the regulation of T-cells, rendered by T-suppressors. Functionally, experimental studies demonstrate causal effects of neurohormones on immunity. In one study, T-cells were challenged with tetanus and then incubated with or without epinephrine and norepinephrine. Especially epinephrine led T-cells to shift their cytokine profile from a "Th-1" pattern (producing now lower IFN-gamma) to a "Th-2"-like profile (producing now more IL-4, IL-5, and IL-10; Agarwal and Marshall Jr 2000). T-helper cells are a key player in activating both B-cells and cytotoxic T-cells. Furthermore, norepinephrine was found to suppress the activity of natural killer cells (Takamoto et al. 1991), of importance in fighting viral infections and tumor cells.

However, a higher level of brain-to-immune modulation exists, exerted by hemispheric lateralization (HL). Numerous studies found that left HL is related to or causally (in animals) induces increases in several immune parameters, while right HL is related to or induces immune suppression (see review by Sumner et al. 2011). Davidson et al. (1999) found a positive correlation between left HL and natural killer cell activity in students. Furthermore, this correlation was particularly strong in regions of the frontal, parietal, and temporal lobes. Meador et al. (2004) conducted a semi-experimental study in epileptic patients undergoing protective brain lesions for their disease. They found that those with left side surgery later had immune suppression, while those with right side surgery later had immune potentiation. Together, these studies propose indeed that parts of the left hemisphere enhance cellular immunity, while parts of the right hemisphere suppress cellular immunity (Sumner et al. 2011).

Immune-to-brain communication occurs in a different manner. Three main ways exist for peripheral immunity to influence the brain. First, peripheral immune signals can penetrate the brain in regions devoid of the blood-brain barrier (BBB;

Dantzer et al. 2000). However, generally speaking, the brain is an "immune-privileged" organ, namely, it is protected from direct immune penetration. Second, a signaling by prostaglandins (mediators of inflammation) on both sides of and into the BBB appears to be another route of such immune-to-brain communication (e.g., Davidson et al. 2001). Importantly, particularly the presence of low-grade inflammation is signaled to the brain via a third route: the vagus nerve. This occurs via receptors for interleukin-1 (IL-1), a main inflammatory signal, on paraganglia of the vagus nerve (Ek et al. 1998). This then triggers two important negative feedback loups, which operate to regulate peripheral inflammation and immunity, which otherwise could be toxic. First, the HPA axis is triggered, resulting with cortisol producing a systemic anti-inflammatory response (Tracey 2009). Second, the descending vagus appears to have an anti-inflammatory effect, though this route is still under investigation (Tracey 2009). The precise manner of this effect is unclear. It occurs in the spleen, utilizing both vagal (parasympathetic) and subsequently sympathetic fibers finally arriving at the spleen. Norepinephrine signals, from the final sympathetic branch, ligand on their receptors, expressed on a subset of T-cells residing in the spleen. These T-cells then produce the vagal neurotransmitter acetylcholine (Ach), as if serving as a Trojan horse for the vagus in the spleen. Ach then binds to splenic or systemic macrophages in the circulation, by liganding to the nicotinic alpha-7 subtype of Ach receptors, and this suppresses the production of pro-inflammatory cytokines (Rosas-Ballina et al. 2011). However, recent studies qualify this final path and the interested reader is recommended to read a more recent review on this topic (Martelli et al. 2014).

We will return to this fundamental nerve-immune modulation since it is the basis for the possible neuromodulation of various chronic diseases (De Couck et al. 2012; Gidron et al. 2005) and for possibly understanding how psychological factors may affect onset or prognosis of illnesses via the vagus nerve. Finally, in a study conducted by my colleague Hideki Ohira and his colleagues, we exposed people to a brief stressor and measured immunological and hormonal changes together with simultaneous brain activity. Only in those with high HRV (high vagal activity) were there correlations observed between brain activity (e.g., cerebellum, cingulate cortex) and peripheral immune and hormonal changes. In contrast, such brain-periphery associations were lacking in people with lower HRV (Ohira et al. 2013). This informs us that the vagus may play a role in immune-brain synchronization during stressful periods. Such synchronization may be important for health.

One clear example of how peripheral immunity affects mental functioning comes from a pioneering study by Reichenberg et al. (2001), who administered a small portion of a bacteria (lipopolysaccharide, LPS), to healthy people, in a double-blind within-subject experimental design. They found that levels of LPS-induced cytokines correlated positively with mild negative mood and memory impairments. This forms the basis for understanding why we may feel sad or function cognitively less well during a flu (Bucks et al. 2008) and how various physical diseases which "proliferate" by inflammation also increase depressive symptoms. We shall revisit this issue in the following chapters.

Stress and Immunity: More Complex Than Meets the Eye

This topic is crucial for several reasons. First, it provides an important basis for the domain of psychoneuroimmunology (PNI) because it can reveal the mechanisms underlying the relationships between psychosocial factors and many diseases, either affected by immunity such as colds and HIV or affected by inflammation – most chronic diseases. Indeed, in the past two decades, the role of excessive immune responses in the form of inflammation and in multiple diseases including cancer, dementia, and heart disease (e.g., Ross 1999; Mantovani et al. 2008) has been discovered. Thus, relationships between psychosocial factors and parameters of inflammation can also possibly explain in part the roles of psychological factors in diseases strongly affected by inflammation. This topic is also of educational importance in my eyes, because too often psychologists, doctors, and laypeople oversimplify it by simply stating that stress reduces your immune system. However, as we shall see now, the relationship between psychosocial factors, stress, and immunity are far more complex, yet fascinating.

Already in 1993, Herbert and Cohen (1993) conducted an early review on this topic and concluded that psychological stress is associated with most immunological parameters and particularly with viral activation. Furthermore, they also concluded that “objective” measures of stress (e.g., reporting exposure/not to a hurricane) were more strongly related to immune parameters than were subjective (appraisal based) measures of stress. In a later comprehensive meta-analysis of over 180 published studies, Zorrilla et al. (2001) examined the associations between stress, depression, and immunity. They concluded that naturally occurring stressors (e.g., hurricane, tornado) were associated with increased Epstein-Barr virus titers, reduced lymphocyte proliferation to mitogens, and a reduction of cells which express the receptor for IL-2, a cytokine which plays a role in the functions of T and NK cells (Becknell and Caligiuri 2005; Boyman and Sprent 2012). Furthermore, while strong acute stress was associated with increased NK cell activity, chronic stress was related to suppressed NK cell activity. Finally, depression also had a complex pattern of associations. Depression was associated with increased neutrophil counts and decreased lymphocyte counts, increases in interleukin-6 (IL-6), and reduced NK cell activity and lymphocyte responses to mitogens. These results point at the very complex nature of the relationships between psychosocial factors and immunity. This relationship depends on the nature of stress (acute/chronic, objective/ subjective), type of psychological factor, and type of immune parameter being measured. The review shows that “stress” is related to suppression of some immune parameters but to overexpression of others.

Obviously, we cannot review all published studies on psychological factors and immunity. Thus, for the sake of brevity, comprehensiveness, and readers’ intellectual interest, I chose some studies which used various methodologies, variables, and settings to present the complexity and variety of findings on stress and immunity.

An entire line of research has examined the effects of maternal pregnancy stress on neonate development in general and on neonate immunity specifically.

Tuchscherer et al. (2002) compared the immune development of neonate piglets whose mothers were chronically stressed during pregnancy (by restraint stress) or not (controls). Lymphocyte proliferation to certain pathogens (e.g., concanavalin A, lipopolysaccharide) was significantly lower in the neonates whose mothers were chronically stressed, than in control neonates, 1–7 days after birth. Furthermore, the weight of the T-cell-producing organ, namely, the thymus, was significantly smaller in the indirectly stressed than in control neonates. These results demonstrate structural and functional changes in neonates' immunity following maternal stress. The implications of these results for human neonates' immunity and development following maternal pregnancy stress could be vast and may have implications for public health as well.

Now let's shift from piglets to medical students. Maes et al. (1998) compared the cytokine profiles of Belgian medical students during a calm period (midsemester), a stressful period (exams), and then a calm period again (after vacation). They also considered their level of anxiety and stress-proneness, reflecting a between-subjects semi-experimental design. While low-anxious/low-stress students exhibited higher anti-inflammatory cytokines (e.g., interleukin-4) during the stressful period, high-anxious/high-stress students exhibited high levels of pro-inflammatory cytokines during the stressful period (e.g., interleukin-6; Maes et al. 1998). This important study reveals how an external event together with people's trait stress responses interact to influence their pattern of cytokine responses. These results need to be examined among patients undergoing various stressors, in diseases where cytokine responses play a role in the etiology and prognosis of their disease, such as rheumatoid arthritis or cancer.

In an older study, Sieber et al. (1992) examined the effects of type of stress (controllable versus uncontrollable noise) and personality aspects (optimism, desire for control) on levels and activity of natural killer cells (NKCA). This paradigm is used often in humans to induce and examine the effects of temporary learned helplessness. Uncontrollable noise stress led to significantly lower NKCA than controllable stress, especially among participants high on optimism or desire to be in control. Segerstrom (2005) reviewed this line of research and concluded that in controllable and short-term situations, optimism is positively correlated with cellular immunity, while in uncontrollable, chronic, or complex situations, optimism is *inversely* related to cellular immunity. Segerstrom (2005) explained this pattern by proposing that in controllable situations, optimists engage in and try to solve a problem, thus removing the stressor. This possibly explains the positive association between optimism and immunity in such situations. In contrast, in uncontrollable situations, pessimists give up, which helps to reduce the stress response, while optimists continue in a futile manner to try to solve an unsolvable problem, possibly increasing their stress response and explaining the inverse association between optimism and immunity in such circumstances.

The studies reviewed so far were conducted in healthy people. However, do psychological factors correlate with immunity in people with an immune-related disease? This would have more direct health implications. In human immunodeficiency virus (HIV), the virus damages primarily CD4 T-helper cells, which can impair both cytotoxic T-cells and B-cell functioning, given the crucial role of T-helper cells in immunity. Ullrich et al. (2003) examined the combined effects of concealment of being gay and of levels of social support, on gay men's levels of CD4. They found an interaction between concealment and social support – in patients with high social support, concealment was associated with lower CD4 levels, but in patients with little social support, concealment was unrelated to CD4 levels. Here too, the social context and communication coping strategy emerge to be pivotal – in those with high social support, concealing a highly significant issue about their life (being gay) may induce more conflict and chronic stress, which could increase cortisol levels, possibly resulting in lower CD4 levels. However, this study used a correlation design and cortisol was not measured. Thus, any such attempt to explain this pattern of associations must be read with caution.

Most past studies assumed and examined linear associations between various psychosocial factors and immunity. However, the basic concept of homeostasis reflects a multicomponent complex system in which multiple parameters are changing between extremes and eventually return to an optimal level. The mere existence of negative feedback loops in the body (e.g., the HPA axis) is a proof of such quest for optimal levels. Chrousos (2009) proposed the known inverted U-shape relationship between level of demands or stress and homeostasis. The famous Yerkes-Dodson law proposes such an inverted U-shape relationship between stress and performance (Yerkes and Dodson 1908). Given such a system, and the fact that psychological factors are related to both immune enhancement and suppression (Zorrilla et al. 2001), we examined the possibility of a nonlinear U-shape relationship between depression and immune levels. Indeed, we observed a U-shape relationship between prevalence of depression and total white blood cell (WBC) count in Israeli elderly people, independent of multiple confounders. In patients with relatively low WBC (<6000), prevalence of depression was 40.5%; in those with middle-level WBC (6000–10,000), prevalence of depression was 17.1%; while in those with elevated WBC (>10,000), prevalence of depression was 37%. This could have resulted from different types of depression, with atypical depression possibly associated with hypocortisolemia resulting in high WBC, versus typical depression associated with high cortisol, resulting in low WBC (German et al. 2006; Gold and Chrousos 2002). Again, given the cross-sectional design in that study and lack of measuring cortisol, such inferences can only be suggestive. Nevertheless, more studies need to consider these complex systems as nonlinear homeostatic ones and examine nonlinear relationships between psychosocial factors and immunity. Furthermore, studies may also wish to examine the boundaries of linear and nonlinear relationships in such systems and the relevance of these types of relationships to health.

I would like to provide examples from the world of job stress. A study conducted on $n = 92$ Chinese nursing students examined their levels of anxiety, anxiety about sleep quality, general maladjustment, quality of life, and social support, in addition to various immunological parameters. Of course, students' anxiety levels were higher when at the clinic than when at school. Importantly, levels of anxiety and anxiety about poor sleep quality were inversely correlated with their levels of CD4

T-helper cells and with CD3. CD3 is a marker of T-cells and a T-cell receptor co-activator essential for T-cell activation (Lei et al. 2015). Though a correlational design, this study extends past ones to the Chinese culture to students' study-related stress and shows that these may affect an important part of the T-cell behavior.

Bellingrath et al. (2010) examined the immunological responses of $n = 55$ teachers to the Trier social stress test (giving a public speech + mental arithmetic). The immunological outcomes included lymphocyte counts of natural killer (NK) cells and CD4 T-helper cells, as well as several pro-inflammatory (TNF, IFN-g, IL-6) and anti-inflammatory (IL-4, IL-10) cytokines. This study is important in that it measured a wide array of immunological parameters including cellular immunity and markers of inflammation and its regulation. High effort-reward imbalance (ERI) and high overcommitment were related to lower levels of NK cells, and high overcommitment was associated with lower post-stress levels of CD4 cells. Finally, high ERI was associated with higher ratios of TNF-a/IL-10 and of IL-6/IL-10. These results are crucial because they reveal complex associations between a naturalistic chronic stressor, namely, teachers' stress, and their reduced lymphocyte levels on one hand together with an increased pro-inflammatory profile on the other hand. This also shows the complexity of patterns of associations between stress and immunity. These results pave the way for possibly understanding many findings presented later in this book on stress and infectious and chronic diseases.

One of the most comprehensive studies done in PNI was done by Melissa Rosenkranz et al. (2005) on $n = 6$ patients with asthma. They exposed patients to physiological challenges (asthma-inducing antigen versus a bronchoconstrictor) and to a psychological challenge (conducting the emotional Stroop Task with neutral, emotionally negative, and asthma-related words). In parallel, brain activity was measured together with inflammatory changes (eosinophils and TNF-a) and asthma-related body changes (reduction in breathing capacity, FEV-1). The asthma antigen led to greater inflammation (TNF and eosinophils) than the bronchoconstrictor. Importantly, levels of the change in activity in the anterior cingulate cortex correlated positively with increased TNF-a and increased eosinophils when contrasting exposure to the asthma-related words compared to negative words, under exposure to the asthmatic antigen compared to the bronchoconstrictor (Rosenkranz et al. 2005). This is an outstanding study since it demonstrates in one study the unique and intricate associations between brain, immune, physiological, and psychological dimensions in the context of an illness. It reveals how a crucial brain center between the limbic system and cortex, namely, the anterior cingulate, is crucial in mediating immunological and physiological responses to emotional stimulation related specifically to participants' illness.

Finally, I will provide evidence from an experimental intervention study done on patients with advanced breast cancer. These were randomized to expressive and experiential group psychotherapy (EEGP) or to a waiting list control condition. Before and after the 3-month intervention, patients underwent a social stressor, in which their immunological responses were measured. Patients in the EEGP evidenced significantly lower increases in stress-induced NK cells and NK cell activity, than controls (Van Der Pompe et al. 2001). These results show that an intervention, aimed to help patients with expressing their concerns and emotions over cancer and to provide them support, helped to mitigate stress-induced immune changes, in a parameter of importance to cancer.

To summarize, psychosocial factors are associated reliably with immunity, and this possibly involves the neuroimmune pathways mentioned above. These associations were observed in healthy and ill people and have profound implications for understanding how psychological factors predict and may even affect our health.

Stress and DNA Damage

In most discussions on health, scientists and clinicians often ask whether a cause is genetic or environmental or genetic or psychological. However, such a dichotomy assumes no relationship between both sides and assumes that psychological factors and triggers have no effect on genetic integrity. This assumption is incorrect. Numerous studies have shown in experimental studies in animals that stress can elicit DNA damage and that psychological factors are significantly correlated with various DNA damage indices in humans (see review by Gidron et al. 2006). Three general types of DNA damage exist, small base chemical alterations, single-helix lesions, and double-strand breaks, with increasing severity of potential consequences (Hoeijmakers 2001). Luckily however, nature has equipped us with four general types of DNA damage repair mechanisms including base excision repair, nucleotide excision repair, recombinational repair, and mismatch repair. Without such repair mechanisms, our risk of cancer and other maladies would have dramatically increased because many cells undergo DNA damage as part of their normal cellular life. I will review now a few examples of studies on stress and DNA damage.

In a pioneering study, Adachi et al. (1993) exposed rats (viewers) to other rats who received electric shocks (receivers), in a separate cage. Compared to control non-exposed rats, the shock-viewing rats had higher levels of 8-OH-dG, a marker of general oxidative stress-induced DNA damage, 6–18 hours later. This important study showed that a “pure” psychological stressor can induce DNA damage, and it has important implications for the literature on health consequences of caregivers. The latter are often exposed to the suffering of relatives, over which they have often limited control (e.g., in the case of caring for a demented relative). Caregivers were found to have a higher risk of mortality than matched controls (Schulz and Beach 1999), and greater DNA damage possibly from their exposure to their relative’s suffering may be one pathway leading to this grave outcome in the caregivers.

In an even more original study, Irie et al. (2000) examined whether DNA damage can be induced by classical Pavlovian conditioning. They exposed rats to a carcinogen, nitrilotriacetate, systematically paired with an inert product, namely, saccharin, or to various control conditions. Levels of 8-OH-dG were significantly higher in rats after exposure to the conditioned stimulus saccharine alone (following its pairing with the nitrilotriacetate), compared to the control conditions (Irie et al. 2000).

What may be the possible implications of such findings is difficult to know. However, it is possible that we may be exposed in life to a DNA-damaging product like smoke or radiation, which could be systematically paired in time and place with inert stimuli. Can exposure to the inert stimulus alone, once systematically paired to the carcinogens, induce DNA damage, and what are the long-term health consequences of such a "real-life paradigm"? Researches in psychobiology and behavioral medicine may wish to answer such questions.

Is there evidence for the effects of psychological stress on DNA integrity in humans? Irie et al. (2001) measured levels of 8-OH-dG in 276 Japanese men and 86 Japanese women. Effects of confounders including age, smoking, alcohol consumption, and body mass index were considered. In women, hostility, fatigue, wishful thinking coping, and anxiety were positively correlated with levels of 8-OH-dG, while in men, number of working hours, self-blame coping, and recent loss of a relative were positively correlated with levels of 8-OH-dG. Though based on a cross-sectional correlational design, these results reveal important correlations between psychological factors and DNA damage in the work context and point at sex-specific correlates.

In a study done by our team at University of Southampton (Sippit, unpublished Masters thesis), we measured the marker 8-OH-dG in British teachers, in the morning and at 1700, on a working day and in the weekend. In addition, we assessed personality using a brief scale of the Five-Factor Personality Inventory (Gosling et al. 2003). We found a triple time × day × agreeableness interaction: levels of agreeableness were associated with less DNA damage only on weekend days in the evening. This reveals how context and personality may interact in relation to DNA integrity.

Finally, in a pioneering study, Epel et al. (2004) examined the effects of chronic stress (mothers caring for a chronically sick child) on telomere length. The telomeres are situated at the edges of the DNA megamolecules and protect them. Their length decreases with each cell replication, thus also reflecting aging, but telomere length is also associated with cancer. Both duration of caregiving and perceived stress were inversely correlated with telomere length (Epel et al. 2004). This suggests that chronic stress is associated with a marker of aging.

A topic of relevance to stress and DNA damage is the field of behavioral genetics and its role in behavioral medicine. In such studies (mentioned in Chap. 4 on coronary heart disease), researchers examine single nucleotide polymorphisms in genetic loci of genes known to be important in either psychosocial factors (e.g., stress responses), diseases, or both. This helps to identify people at highest risk for health problems given both environmental stressors and genetic polymorphisms. The work of Williams and colleagues is introduced in Chap. 4 concerning this topic. However, more work is needed in other domains of behavioral medicine such as in psycho-oncology, to unravel such issues. For example, while much evidence exists showing that psychological factors predict cancer prognosis, less evidence exists showing that psychological factors predict cancer onset (Garssen 2004). It is possible that here too, psychological factors may synergistically interact with DNA damage or polymorphisms in predicting risk for cancer. This awaits testing.

Together, these studies suggest that psychological stress and psychological factors can affect health and disease also possibly via affecting DNA integrity. To the best of my knowledge, no or nearly no study has examined the health consequences of such findings. Future studies should examine whether the combination of chronic stress, especially uncontrollable stress, and excessive DNA damage over time predicts increased risk of cancer or other diseases. Such findings also challenge the nature-nurture dichotomy and call health researchers to consider the interactions between psychological and genetic factors in relation to health and disease.

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Chapter 3
Doctor-Patient Communication and Increasing Patient Adherence

Case Study

Mr. Smith enters into Dr. Feng's clinic in London.
Mr. Smith: "Hello Dr. Feng."
Dr. Feng: "Good morning Mr. Smith (while gazing at his computer)."
Mr. Smith: "emmm,"
Dr. Feng: "Well, you obviously came to see me for a reason, no?"
Mr. Smith: "Yyesss, sorry.. I have breathing difficulties d d d doctor."
Dr. Feng: (He looks for 2 more minutes at the computer screen.) "OK, let me ask you some questions Mr. Smith, so we do not waste more time, ah!"
Mr. Smith: "Yes doctor, sure; I am all yours.."
Dr. Feng: "Well, how old are you?"
Mr. Smith: "59 sir!"
Dr. Feng: "Have you had any lab tests or an X-ray recently?"
Mr. Smith: "Yes, here are the results doctor (takes out a paper)."
Dr. Feng: "Aha, your Troponin is very high, so is your LDL cholesterol and your BNP is in the sky!"
Mr. Smith: "Sorry, I do not understand these terms, is that a bad thing?"
Dr. Feng: "Never mind, you have.., ah I see your ECG, indeed, yes.., you have CHF! How long have you had CHF Mr. Smith, tell me the truth?"
Mr. Smith: "I… have CH…F for a year, Dr. Corry told me."
Dr. Feng: "Ah, another failing patient of Dr. Corry, of course!"
Mr. Smith: "What do you mean… am I going to die?"
Dr. Feng: "No, no, not so fast… I will help you; but I think you are not taking your medication! This is why your breathing is unwell!"
Mr. Smith: "Well, Dr. Corry prescribed to me 6 pills, and it is complicated to remember when to take each of them!"

People don't care how much you know, but they know how much you care. (T. Roosevelt).

Y. Gidron, *Behavioral Medicine*, https://doi.org/10.1007/978-3-030-18893-1_3

Dr. Feng: "These are excuses! If you want to live, let me help you and do as I tell you – take all these six pills, every day! I will see you in 2 months, and hopefully your situation will be better!"
Mr. Smith: "Aaah ok, ok doctor. I will try my best, see you in 2 months…"

No matter how advanced technology may be in helping doctors diagnose a disease, and no matter how effective vaccines or medications may be, a core fundamental element of medical care is doctor-patient communication. This is the means for evaluating a patient's situation and for planning and proposing an appropriate treatment. Furthermore, voluntarily taking medication depends on patients' adherence, a behavior finally under the patient's control. This chapter will first provide a framework for understanding and evaluating factors in doctor-patient communication and will review the impact of communication on various patient outcomes. The second part will focus on assessing, predicting, and increasing patient adherence to treatment. Let's first look at this patient's scenario.

Doctor-Patient Communication: Basic Theory

Ong et al. (1995) nicely framed the topic of doctor-patient communication as an interaction whose aims are to create a positive interpersonal relationship, to exchange information (in both directions), and to eventually help make treatment-related decisions. These three functions are crucial for and affect the medical process of diagnosis, care, and patient outcomes, as we shall see empirically below. In many societies, the topic of doctor-patient communication is still neglected in medical training, or is thought to be a side issue, of little relevance to patient health, and as something you either "have" or not. However, the empirical evidence provided below is sound evidence for the contrary, and this is a skill which can be taught (Razavi et al. 2003). Hence, doctor-patient communication must be taught in medical education and for all health professionals, as a basic and essential element of medical training, side by side with learning physiology, anatomy, and molecular biology. Doctor-patient communication is the vehicle through which diagnoses are made and treatments are conveyed. Furthermore, the influences of doctor-patient communication on multiple outcomes, including patient satisfaction, adherence, and most importantly patient health, strongly propose learning this and practicing adequate communication.

Doctor-patient communication can be categorized into types, according to the contents or functions, and these include instrumental content (e.g., "you need to walk 15 min a day"), emotional/affective ("How do you feel?"), verbal versus nonverbal, privacy or intimate content (e.g., "Do you have pain while urinating?"), level of control over the interaction, and medical content (e.g., "your tumor marker has gone down," or "how often do you have that pain?") versus everyday language (e.g., "How is the football team of your son doing?"). These categories map onto the three functions of doctor-patient communication mentioned above. Physician personality is related to the contents of doctor-patient communication. A recent study conducted in 228 general practitioners found that level of physicians' neuroticism was inversely correlated with communicating uncertainty to patients, while extraversion, agreeableness, and openness to experience and conscientiousness were all positively correlated with communicating uncertainty to patients (Schneider et al. 2014). We shall now examine the relationship between elements of doctor-patient communication and patient outcomes.

Effects of Doctor-Patient Communication on Patient Outcomes

I will review here studies which examined the association or experimentally tested the effects of doctor-patient communication on the amount of patients' recall of doctors' information or advice, patients' satisfaction, adherence to treatment, and health outcomes. When looking at the relationship between doctor-patient communication and patients' recall of information, let's first examine what patients recall from their consultation. In one study on 38 patients about to consent for undergoing *surgery*, the mean recall of 12 units of information was 3/12 (25%; Godwin 2000). Despite this, 63% were satisfied with the consent process, which indicates that the meaning of "patient satisfaction" may have little to do with the adequacy or quality of medical care. Studies must not only assess patient satisfaction. One study found that patients, whose doctors asked them to restate their recommendations, indeed remembered better the information than patients who were not asked to restate the recommendations (Bravo et al. 2010). This is a simple task for doctors to perform, and it not only can increase patients' recall but also provide the doctor the opportunity to evaluate the patient's level of comprehension of his or her clinical recommendations, and this can give the physician a moment to remember additional points which he or she forgot to tell the patient. It is important to note that one study found that only 12–20% of times did doctors inquire about patients' recall or understanding of new medical concepts (Schillinger et al. 2003). Clearly, much more needs to be done about this crucial issue during medical education and practice.

Many studies have examined the relationship between aspects of doctor-patient communication and patients' satisfaction with their treatment. Williams et al. (1998) reviewed many studies on this topic and found that numerous physician behaviors (e.g., being informative, discussing cause of problem, understanding patient's concerns) and physician's nonverbal behaviors (e.g., good listening skills, partnership building, approving patient's acts, social talk) were related to patients' satisfaction. These crucial issues cross countries, languages, and cultures. In a study on 150 Korean patients, doctors' levels of emotionality and empathy were positively and significantly correlated with patients' level of satisfaction from treatment (Kim and Park 2008). One important and often neglected topic is doctors' cultural competence. This is becoming more and more important due to migration of people between countries. One study developed a scale for patients to assess their doctor's cultural competence and found that such competence levels correlated positively with patients' levels of satisfaction ($r = 0.32$, $p < 0.001$) and trust ($r = 0.53$, $p < 0.001$; Thom and Tirado 2006).

Adherence is a broad and basic topic in medicine, which we shall refer to in the second part of this chapter. Numerous studies have examined the relationship between doctor-patient communication and patients' adherence levels. In one study on hypertensive patients, doctors' quality of communication (rated as collaborative) positively correlated with patients' adherence to antihypertensive medication, beyond the effects of patients' background factors, depression, and doctors' type of academic degree (Schoenthaler et al. 2009). A related factor is the effect of doctors' communication skills on patients' involvement in clinical decisions, partly reflecting adherence to the clinical process. This is of increasing importance as the doctor-patient interaction is gradually shifting from a paternalistic one to a participatory one. Van den Brink-Muinen et al. (2006) found that doctors who were more emotional and provided more information had younger patients who were more involved in clinical decisions. Such involvement, as we shall see below, could also influence their health outcomes.

Indeed, many studies examined the effects or relationships between doctor-patient communication elements and patients' health outcomes. Beck et al. (2002) reviewed all studies till then on doctor-patient communication and patients' health outcomes. They found that doctors' levels of empathy, reassurance, summarizing and clarifying, sharing information, and asking patient-centered questions were related to positive health outcomes. They also found various nonverbal communication aspects to correlate with positive patient health outcomes including doctors' head nodding, leaning toward patients, and having less mutual gaze (Beck et al. 2002). In one older study, Kaplan et al. (1989) observed doctor-patient pairs in whom the patient had either hypertension, diabetes, ulcer or breast cancer, and then followed them for 6 months. The amount of words in the interaction reflecting patient control was inversely correlated with blood pressure in the hypertensives and with levels of HbA1C in the diabetics. In contrast, these correlations were in the opposite direction when examining the amount of words reflecting doctors' control over the interaction and the two health outcomes. Finally, a newer study examined the impact of an interactive educational strategy between doctors and patients, on their health outcome. This interaction style included asking less educated patients their understanding of new concepts and adapting the information to the patient's level of understanding. Doctors who used such an interactive educational strategy had nearly nine times the chance that their patients' levels of HbA1C would be normal later (Schillinger et al. 2003). These studies reveal that simple use of patient-centered content by the doctor actually predicts strong beneficial health changes in patients.

In cancer, levels of hopelessness are a significant prognostic factor, independent of known risk factors (e.g., Watson et al. 2005), and in animals, helplessness causally accelerates tumor growth (Sklar and Anisman 1979). One study found that cancer patients rated the following doctor communication behaviors as most hope-providing: knowing available up-to-date information about a patient's cancer, offering most up-to-date information, and saying that pain is controlled (Hagerty et al. 2005). Physician behaviors such as being nervous, disclosing the diagnosis first to the family, and using euphemisms were seen as not promoting hope in patients. This is important information, given the prognostic role of hopelessness and the difficulty to reduce it. Together, these studies demonstrate the crucial importance of doctor-patient communication for patients' understanding their condition and treatment, their satisfaction with their care, their adherence, and ultimately their health.

Importantly, several studies show that medical students and physicians can be trained in doctor-patient communication skills. For example, a large randomized controlled trial assigned British oncologists to written feedback alone, a course alone, feedback + course, or control, and 2407 patients subsequently took part in evaluating these courses. Results found improvements on several key communication skills after the course (e.g., asking more focused and opened questions, expressing empathy, and reacting appropriately to patients' concerns; Fallowfield et al. 2002).

Throughout this chapter, doctors' empathy was mentioned. But is this important aspect of communication always beneficial? We recently reanalyzed data, in a post hoc prospective study, and examined this issue, in relation to lung cancer patients' survival. Patients assessed their physician's empathy; however, this did not predict survival. Astonishingly however, empathy interacted synergistically with the content of their consultation (receiving bad vs neutral news), independent of important confounders: in bad news consultations only, higher physician empathy significantly predicted higher risk of death than lower empathy. In contrast, physicians' empathy did not predict survival when patients received neutral news about their cancer (Lelorain et al. 2018). Furthermore, in a sub-analysis of these data, patients who rated their doctor as empathic while receiving bad news were significantly more hopeless and had significantly higher inflammation levels 2 months later, than those rating their doctor as less empathic when receiving bad news (unpublished data). As the book of Ecclesiastes tells us, "There is a time for everything, and a season for every activity under the heavens." Too much empathy in bad contexts may be adverse.

As we have seen, one main patient issue affected by doctor's communication style is patients' adherence to their recommended treatment. Because this is affected also by many other factors, the second half of this chapter is devoted to this important topic.

Assessment of and Factors Influencing Patients' Adherence to Medical Treatments

Just as doctor-patient communication is essential for the diagnosis and treatment of patients, the best medication will still always rely on patients' adherence to taking it (except when provided to patients who are unconscious as in intensive care units).

Adherence can be defined as "The degree to which patients follow the recommendations of their health professionals" (Zolnierek and DiMatteo 2009). This is my preferred definition since it reflects the fact that not only medical doctors recommend to patients activities or medication (also dietitians, psychologists, physiotherapists, nurses, etc.) and since it does not "prescribe" a precise number for correct adherence. The latter issue is crucial, since 80% adherence may be the minimum for one mediation while not for another or not relevant to a certain diet, for example.

The prevalence of nonadherence is extremely difficult to estimate because there is no consensus as to what constitutes full adherence or compliance to medical treatment. In addition, different studies use different methods for assessing adherence. One study done in hypertensive patients (Tomaszewski et al. 2014), using a sophisticated objective measure of high-performance liquid chromatography for analyzing urine, found a prevalence of 25% for total and partial nonadherence. In contrast, in a study on 374 diabetic patients in Bangladesh, 88% were nonadherent to a recommended diet, 25% were nonadherent to recommended exercise, and 70% were nonadherent to foot care (Mumu et al. 2014). These findings reveal that nonadherence is a global problem and that its prevalence varies tremendously due to multiple factors among which are also the recommended behavior to adhere to and its method of assessment.

Assessment of Adherence

As mentioned above, one reason for the great discrepancies in estimating nonadherence is the manner of assessing it. Adherence to medication, for example, can be measured by the following four methods. First is by objectively measuring the urine or blood levels of metabolites of the medication (e.g., Tomaszewski et al. 2014). While this is indeed an objective measure, its limitations include short half-time life of certain medications or their metabolites, which may be less than the time from last drug consumption till the actual test. Furthermore, metabolite levels can also fluctuate due to patient and drug pharmacokinetics. The second method is the pill count method, where the physician either examines the number of pills removed from a box, or doing so with an electronic device. While this again is an objective measure, its main limitation is that one cannot know whether the pills were in fact consumed by the patient or merely only removed from the box. Similar to this measure is to use a tachometer for measuring adherence to physical activity, when prescribing this type of behavior for lifestyle modification. However, this only measures movement, and may not reflect the full picture of types of recommended exercises. The third method is to examine the disease endpoint: objectively or subjectively. Thus, for example, one can assess pain levels in a migraine patient or measure blood pressure in a hypertensive patient. While this is of huge clinical importance, its main limitation is that changes in the diseases may or may not reflect adherence and could also result from other treatments or "spontaneous remission." While such a method directly asks a patient's perspective, it suffers from the two main limitations of self-reported measures, namely, presentation biases and memory biases.

One short, simple, and valid self-reported scale was developed by Morisky et al. (1986). Its strength is that it only includes four items; it relies on the fact that people may prefer to reply "yes," but here such a reply means less adherence; its questions reflect common barriers for adherence (e.g., feeling worse after taking medication); and finally, it has a long-term predictive validity in relation to disease status 2 years later (Morisky et al. 1986). It is recommended to combine at least two of these methods for assessing adherence fully.

Predictors of Adherence to Treatment

Generally speaking, we can divide these predictive factors into those related to the treating physician, to the patient, to the illness itself, or to the treatment. Several studies have examined variables related to doctor-patient communication and adherence, which are mentioned above. In the study on 150 Korean patients, doctors' levels of empathy and emotionality were also positively and significantly correlated with patients' levels of adherence (Kim and Park 2008). Similarly, in the study mentioned above on hypertensive patients, doctors' quality of communication correlated with patients' adherence, independent of patients' demographic factors and depression and independent of doctors' type of degree (Schoenthaler et al. 2009).

The topic receiving most research attention in relation to predicting adherence is about patient factors. These can be mapped onto various social cognitive models including the health belief model (HBM; Rosenstock 1966), the theory of planned behavior (TPB; Ajzen 1991), and the trans-theoretical model (TM; Prochaska and DiClemente 1983), among others. In the health belief model, behavior (adherence in this case) is thought to be influenced by patients' perceived susceptibility for having a condition, perceived severity of a condition, one's perceived barriers for performing a preventative or recommended behavior, and perceived benefits from performing the recommended behavior. A meta-analysis by Carpenter (2010) found consistent evidence for HBM variables, particularly for perceived benefits and barriers for performing a behavior, in relation to predicting health behaviors. Let's examine a few examples. One study was conducted in 134 patients with heart failure, a common disease among survivors of myocardial infarction. In this sample, patients' most frequent barriers included forgetting the time of taking medication, not having one's medication when going out, cost of medication, amount of pills required to take per day, and the belief that one will be ok even if missing one dose of medication. Indeed, patients' barriers for taking medication significantly predicted several indices of pill count (adherence), after controlling for other possible confounders (Wu et al. 2008). Thus, the barrier component of the HBM is a crucial predictive aspect of this model, to which we shall return below when discussing ways for increasing adherence.

The TPB has been the most empirically tested model. Briefly, the TPB contends that people will adhere to a behavior if they intend to perform it, which in turn is a function of people's attitudes toward that behavior, their subjective norms (perception of what others think about that behavior), and their perceived control and self-efficacy to perform the behavior. To exemplify the versatility of this model and the importance of behavioral medicine in the entire context of medical care, the following example will be provided. Anesthetists must perform multiple safety activities such as meet the patient prior to surgery and listen to warning signals during surgery, which represent patients' vital status. Anesthetists' adherence to such safety regulations was examined in relation to the TPB. Subjective norms were the only correlate of not missing a pre-surgical patient visit, subjective norms and behavioral control were strong correlates of pre-surgical testing of equipment, while attitudes, subjective norms, and behavioral control were all significantly inversely correlated with *silencing* the alarms during surgery (Beatty and Beatty 2004). One important critique of the TPB is that it may predict quite well people's intentions to behave, but not their actual behavior. Peter Hall and colleagues took this issue one important step further by introducing a neuroscientific aspect to this topic. They demonstrated in a few samples that intentions to adhere interact with executive functioning (EF) in predicting actual health behaviors. EF refers to a multicomponent capacity (attention switching, working memory, and inhibition) emerging mainly from the activity of frontal brain structures (Zillmer and Spiers 2001). In Halls' work, intention predicted diet and physical activity more strongly in people with high compared to low EF (Hall et al. 2008). Recently, they conducted an experimental study and found that an "intention implementation" method (see below) increased elderly women's physical activity, especially in those with initially high EF (Hall et al. 2014). These studies clearly indicate that one's EF, and thus prefrontal brain functioning, need to be considered in models of health behavior and adherence. These findings truly represent the biobehavioral approach to behavioral medicine.

The TM contends that people often go through five stages when adopting a new long-lasting lifestyle change, which include pre-contemplation, contemplation, preparation, action, and maintenance (Prochaska and DiClemente 1983). One may, according to this model, move up and back down the stages as in a spiral. The TM has been tested in many studies including one on condom use in 279 Taiwanese students. Indeed, while students in the first two stages mostly considered barriers (proposing to use a condom will make my partner angry), those in the action and maintenance stages mostly reported more benefits from condom use (being responsible and protective) (Tung et al. 2009).

Other patient variables include personality or mental state. For example, one study conducted in 167 Black American hypertensive patients found that depression correlated with less adherence to medication and that this was mediated by lower self-efficacy (Schoenthaler et al. 2009). Another study examined the relationship between hostility, depression, and the Millon Behavioral Medicine Diagnostic (MBMD; Millon and Antoni 2001) and self-reported adherence in 105 patients with heart failure (Farrell et al. 2011). The MBMD scale includes scales of medication abuse and a problematic compliance scale. This is an important issue since most heart failure patients require multiple medications due to malfunctioning of several vital systems (e.g., cardiac, pulmonary) and since this condition is more frequent today due to improved survival following a myocardial infarction. Farrell et al. (2011) found that hostility and depression, each, significantly correlated with adherence. That study reveals that a hostile personality and a depressive state are related to poorer adherence.

Treatment-related factors often include the dose or side effects. Indeed, one-daily dose of antihypertensive medication leads to higher adherence than more doses per day (Flack and Nasser 2011).

Finally, disease-related factors include aspects such as whether the disease is acute or chronic, symptomatic or asymptomatic, etc. One may initially think that having a symptomatic illness provides us with signals of the illness and possibly pain, which would increase motivation to adhere. However, a large-scale Polish study conducted on 63,221 patients with multiple types of diseases (e.g., hypertension, depression, memory loss) found that asymptomatic diseases were associated with a bit lower nonadherence (81.5%) than were symptomatic diseases (84.7%; Kardas 2011). However, while the investigator assessed adherence with the valid four-item Morisky Scale mentioned above, a severe adherence cutoff was used (all answers had to be negative to be considered adherent), which explains partly the very high rates of nonadherence. In addition, while statistically significantly different rates, the very high nonadherence rates of symptomatic and asymptomatic diseases are nearly identical and were significantly different due to the very large sample size. Nevertheless, another study also found higher nonadherence in patients with symptomatic (37.6%) than asymptomatic ischemic heart disease (22.7%; Carney et al. 1998). It is also possible that having certain sudden severe symptoms (e.g., cardiac chest pain) may distract patients from adhering to their usual preventative treatment (e.g., statins) in favor of other anti-symptomatic medication (e.g., pain killers), which may not be routinely prescribed for certain conditions. It is also possible that initial symptoms may elicit greater adherence, which may then fade with time when symptoms resolve. Future studies may need to examine the impact of symptomatic periods on patients' adherence within a given illness, within patients, rather than comparing symptomatic versus non-symptomatic conditions.

Interventions for Increasing Patient Adherence

This chapter merged the topic of doctor-patient communication with the topic of adherence because it is mainly via various forms of communication that the healthcare provider can increase patients' adherence. Let's first begin with the physicians. In a pioneering study, Inui et al. (1976) randomized doctors to a lecture on how to measure and increase adherence or to no lecture (control). Six months later, doctors who received the lecture wrote more words in their patients' files about adherence, and their hypertensive patients adhered more and most importantly also had lower blood pressure, than was the case for the patients of doctors in the control group. That study reveals how a brief educational effort in medical training in behavioral medicine can have important health impacts on patients.

Many clinicians use an "educational" approach for dealing with adherence. This could reflect the assumption that patients do not know enough about what is good for their health and that knowledge alone will increase adherence. However, studies in multiple health domains reveal that health education or awareness to the problem alone has limited effects on health behaviors or on reducing risky behaviors. For example, a review of studies found no effects for education on condom use for HIV prevention (Gallant and Maticka-Tyndale 2004). Another study on your drivers found no effects for driver education on reducing traffic accidents in this highly risky population and even some evidence to the contrary (Vernick et al. 1999). The studies reviewed above on the multiple factors which predict adherence inform us that education alone will not solve this issue and that interventions need to address the modifiable risk factors and predictors of nonadherence. Furthermore, studies show that interventions which include a psychological component work better than those which include education alone. For example, a meta-analysis of intervention trials using psychological, educational, or telecare methods for increasing adherence in diabetics, showed that only the psychological interventions yielded reductions in HbA1C (Viana et al. 2016).

Many different behavioral interventions have been done to increase adherence. In some, simple daily reminders or reinforcements are used, in line with a behaviorist approach. In a recent study using text messages available with current mobile phones, 90 patients with coronary heart disease received either text messages reminding patients about adherence + education, education control, or no special intervention. The first group then had higher objectively measured adherence to antiplatelet treatments (Park et al. 2014). This is important since it represents an easy and low-cost type of treatment for increasing adherence.

One important type of behavioral interventions is "intention implementation" (II), which is a form of cognitive-behavioral self-regulation coping: people make a realistic decision about a goal and plan how and when they shall proactively change their health-related lifestyle to regulate their behavior and eventually possibly affect their health outcomes. One study randomized 115 Brazilian cardiac patients to receive an II treatment or usual care with information about their medication (control). In the II group, patients were asked to state where, when, and how they will take their medication, to name up to three barriers, and how they will overcome them, to adhere to their cardioprotective and symptom-relieving medication. The investigators used a very strict criterion for adherence – adequate dose and care using the global adherence evaluation method. Indeed, significantly more patients in the II group were "adherent" at baseline and follow-up (18%) than in controls (8%; Lourenço et al. 2014). Another form of behavioral intervention to increase adherence is financial reinforcement. The rationale here follows the notion that a meaningful reinforcement will increase the frequency of the behavior that preceded it, in this case correct adherence. A recent meta-analysis of 15 randomized trials and 6 non-randomized trials found that a financial reinforcement protocol significantly increased patients' adherence. Furthermore, interventions which reinforced patients for a long duration, at least once a week and provided at least $50/week for correct adherence, yielded stronger adherence (Petry et al. 2012). These results are promising yet must be taken with caution since they may not apply to all countries where such financial reimbursement may not be economically feasible. Of course, health insurances can consider that in the long-term, if patients adhere to effective treatments, their clients' health should in principle be better and, hence, cost less to maintain.

Throughout this chapter, the role of patients' barriers against adherence to medication or to adopting healthy behaviors was repeatedly mentioned. Such barriers often include cognitive distortions about adopting healthy behaviors such as having too little time in relation to performing physical activity (Zunft et al. 1999). Another serious impediment is the presence of social pressures people face from their social environment for adopting unhealthy behaviors, as was found in adolescents concerning adopting unhealthy eating habits (Sawka et al. 2015). However, health education may do little to systematically change cognitive distortions and barriers or to help people resist social pressures.

Motivational interviewing (MI) includes interviewing patients about their barriers for adaptive behaviors and inquires them to propose ways to overcome these barriers. One study randomly assigned 190 African American hypertensive patients to MI or usual care (control). The MI group received monthly sessions of MI. While at baseline both groups had similar rates of adherence, based on electronic pill counts (56.2% for MI, 56.6% controls), these rates were significantly higher in the MI group after treatment (57%) than in controls (43%) (Ogedegbe et al. 2008).

Another more focused and brief form of cognitive intervention, specifically aimed at changing people's cognitive barriers and at resisting social pressures, is called "psychological inoculation" (PI). As in a medical vaccination, in PI, we expose people to challenging sentences, which reflect in an exaggerated (and humoristic) manner their own cognitive barriers or social pressures they face. These sentences are akin to the "vaccine." Then, the person is expected to systematically refute each challenging sentence (the "antibody response"), until he or she is "immune" to their effects. This process rapidly alters cognitive distortions and provides social resistance skills, leading to behavior change. Past studies showed that PI led to greater prevention of smoking in adolescents than health education did (Evans et al. 1981). Recently, we found in British and Belgian samples that PI increased self-reported physical activity, while a health education control group did not. Furthermore, in the British sample, more reductions in barriers correlated with greater posttreatment physical activity, only in the PI condition. Finally, in the Belgian sample, the degree of refutation of the challenging sentences predicted the level of physical activity behavior two months later (Dorling et al. 2018). We recently developed a fully automatized version of PI, where we provide both challenging sentences and a priori refutations, from which participants choose the best refutation. We found this automatized PI to reduce barriers for condom use and increase the tendency to use condoms, toward HIV prevention (Levy et al. under review). Future studies need to test the effects of PI on adherence of patients with complex medical regimens. If effective, PI could be added to II, and it can also utilize current technologies such as text messages or emails and be provided en masse with the automatized version.

One very important and pioneering study examined a psychological intervention based on the TPB (Wu et al. 2012). In this study, patients with heart failure, a highly fatal form of heart disease, received either a TPB intervention + feedback on their adherence from a monitoring device, a TPB intervention alone, or a usual care (control). The TPB element of the intervention aimed to increase patients positive attitudes toward the medication, to increase the importance of medication adherence seen by the patient's significant others as reported by the patient, and to increase patients' perceived control over medication and barriers. Both intervention groups had better adherence and a longer event-free *survival* than usual care (Wu et al. 2012). These outstanding results strongly show the crucial importance of adding behavioral medicine-based interventions to medical care, to increase patients' adherence to treatment, which can also positively affect their prognosis.

Returning to the Example of Doctor-Patient Communication

Based on all the abovementioned studies, a "patient-centered" style of communication is advocated. This style includes important contents and styles of communication and sees the patient as a vital source of information about his or her condition and as a partner in medical decision-making. In addition to one's socioeconomic background, health history, symptoms, and treatments, the contents need to inquire patients' barriers, attitudes, perceived social norms, and self-efficacy to perform recommended health behaviors. Concerning style, one should always begin with introducing one's self ("Hello, I am Dr. Shapira"), followed by as many open questions as possible. For example, "So Mr. Smith, what brought you here today please?" Have the patient explain their health complaint or concerns and do not judge him or her. Ask the patient what they think could be the factors which contributed to their illness. This can provide vital information and informs the clinician about the patient's level of understanding and illness perceptions, a vital predictor of health outcomes (Frostholm et al. 2007). Finally, it is crucial to set goals with the patient, since this may create more realistic goals, empower the patient, and increase his or her self-efficacy, and this predicts better health outcomes as well (e.g., better hypertension control; Naik et al. 2008). Ask the patient to summarize what was decided in the meeting if specific regimens or tasks are agreed upon. All this needs to be done with active listening, empathy, and keen interest in the patient. At Ben Gurion University, South of Israel, I had the honor to train medical students in doctor-patient communication, in a course led by my late colleague Dr. Judy Bernstein. This is called "the Beer Sheva experience," and indeed it was. Time and again, we learned so much from each interaction between patients, students, qualified physicians, and my colleagues.

Case Study Conclusion

Let's now return to the case which opened this chapter, and let's "treat" this patient better. In fact, it begins by treating the physician. The content is patient-centered, asks about the patient's perspective for his worsened condition, and introduces PI to increase adherence, all with a humane style unlike the poor example given at the beginning of this chapter.

Mr. Smith enters into Dr. Feng's clinic in London.
Mr. Smith: Hello Dr. Feng.
Dr. Feng: Good morning Mr. Smith (looking with interest at the patient).
Mr. Smith: Nice to see you doctor.
Dr. Feng: Nice to see you too Mr. Smith. How is your daughter's dancing developing?
Mr. Smith: She is marvelous, thank you for asking!
Dr. Feng: So, what brings you to me today (smiling)!?
Mr. Smith: Thank you doctor. I have breathing difficulties!
Dr. Feng: Oh, sorry to hear this. It is probably not so pleasant. You were treated by Dr. Shapira, at the hospital, right? Let me please take a look at your file (he looks for 1 minute at the computer screen). May I ask you some questions Mr. Smith?
Mr. Smith: Yes doctor, sure. I am all yours.
Dr. Feng: Why do you think you have these breathing difficulties?
Mr. Smith: I suppose it is related to my heart disease.
Dr. Feng: Well I also think so; you were diagnosed with heart failure, and often this can be manifested by breathing difficulties. Based on the lab and clinical tests in your file, it is not surprising that you may have such problems breathing.
Mr. Smith: Is it serious Doctor?
Dr. Feng: Well, one of your markers related to heart failure, BNP, is high; but with good adherence to your medication and physiotherapy, we may be able to help you indeed.
Mr. Smith: I understand doctor. Yes, I was given three different medications, but it is not easy for me to take them all the time doctor!
Dr. Feng: Mr. Smith, please tell me what are your reasons for not always taking your medications?
Mr. Smith: Well, first, I am unable to remember all the time, and secondly, I am unsure it really is worth all the efforts.
Dr. Feng; Let me try to help you please, using a new method which uses a small set of "funny statements". Please note that everything I will tell you will be totally *incorrect*, and your goal is to *reject* my sentences one by one, ok?
Mr. Smith: OK. I am ready to try this!
Dr. Feng: Mr. Smith – In the past, you were *never* able to reach small or big goals you ever set to yourself in life!

Mr. Smith: That's not right, I wanted to study business, and I did; and later I planned to start my own small factory, and I did!!
Dr. Feng: Wow, good, you rejected my sentence very well; please continue this way!
Dr. Feng: Mr. Smith, you cannot tell me even one way to remind you to take the pills correctly!
Mr. Smith: Yes I can tell you. I can put the medicine plan near my bed and make a copy on the fridge; I've been planning to do this for months, and I can also put small reminder stickers in most rooms of the house I suppose.
Dr. Feng: Very good, you rejected my sentence really well; please let's do one more!
Dr. Feng: Mr. Smith – you cannot mention to me one reason related to your family, for which you may want to take good care of yourself!
Mr. Smith: Well obviously, I should take my pills so that I can live to be with my adorable children and grandchildren (he sighs nearly tears); they are my favorite gift!
Dr. Feng: Wonderful, you really rejected well my sentences. Try to think about these sentences and perform the actions you just mentioned! I will see you in 2 months, and hopefully your situation will be better because you will take control over taking your medication better!
Mr. Smith: Ok, doctor. Yes, I will try my best, see you in 2 months.

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Chapter 4
Psychosocial Factors in Coronary Heart Disease

Case Study

Mr. Ismail is a 45 years old man and works in the car industry. He is married and has two adolescent children. He has been diagnosed with a two-vessel coronary artery disease. His mother has heart disease, and his BMI is 31. His LDL cholesterol is very high (192 mg/dl), but he does not drink alcohol, since he is Muslim. He is a devoted family person. However, his wife reports that he has many anger spells and at times bangs on the table when angry. They calm him down with a cloth and cold water. His ECG shows ST-suppression, and the angiogram revealed a 75% occlusion in the left anterior descending artery and a 70% occlusion in the left main artery. Two weeks ago, he experienced severe chest pain, and his troponin level was very high; subsequently, he was diagnosed with a myocardial infarction, and he underwent angioplasty with stents in each occluded artery. He receives statins and a beta-blocker. No further treatment was given to him.

Biomedical Risk Factors and the Etiology of Coronary Heart Disease

Coronary heart disease (CHD) is a leading cause of mortality worldwide and particularly in western countries (e.g., Veerasamy et al. 2014). As many chronic diseases, it is multifactorial: It is caused by many risk factors, which makes its prevention complex. Numerous prospective studies have been conducted, which show that elevated blood pressure, LDL cholesterol and triglycerides, little physical activity, and smoking are risk factors for developing CHD (e.g., Gordon and Kannel 1982; Wilson et al. 1998). But since these factors do not account for all CHD cases (Wilson et al. 1998), this calls for the identification of other modifiable

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CHD risk factors. The underlying basis of CHD in the majority of cases is atherosclerosis, the gradual narrowing of coronary arteries because of buildup of plaques which occlude the coronary arteries. It is important to realize that this is a gradually developing disease and is far more complex than simple deposit of cholesterol. First, the arteries are made up of four layers, namely, the endothelium which has direct contact with blood, followed by the intima, media, and adventitia. Atherosclerosis begins by injuries in the endothelium in regions where arteries are prone to be damaged such as in places where they turn. While a normal endothelium repels blood elements and enables them to flow, a dysfunctional endothelium expresses adhesion molecules which then recruit certain components in the circulation, particularly parts of the inflammatory response of the immune system – monocytes (Mestas and Ley 2008), which contribute to atherogenesis. Deposits of low-density lipoprotein (LDL) cholesterol then enter into injured sites of the endothelium and undergo oxidation. This oxidized LDL cholesterol triggers an immune response from the recruited macrophages (monocytes which entered into the artery; Habets et al. 2009; Samson et al. 2012). In fact this process is bi-directional since the oxidized LDL acts as a ligand which binds to macrophages and induces their cell death (Hardwick et al. 1996). Initially, these macrophages swallow the oxidized LDL cholesterol and become foam cells, which will subsequently undergo cell death or apoptosis. In parallel, smooth muscle cells migrate from the media to the intima and proliferate. All these processes together contribute to the developing and buildup of the atherosclerotic plaque (Ross 1999). Such repeated endothelial injuries, migration of monocytes, and attempts to repair and respond to oxidized LDL resemble a never-healing wound (Balkwill and Mantovani 2001). The response of the macrophages is a "right response in the wrong context," as a highly scholarly colleague of mine, Professor Luca Vannucci, once said. Macrophages are part of the innate immune response, and they respond in this case to oxidized LDL, LDL cholesterol which has undergone modification. The macrophages respond to changes in patterns of pathogens, in this case altered LDL cholesterol. However, the mere presence of this inflammatory response then serves as petrol to the developing atherosclerotic plaque since they produce more inflammatory markers which contribute to the plaque formation. Yet some of the immune response against oxidized LDL may be protective if induced as a vaccine. A pioneering study testing precisely this possibility found regression in atherosclerosis following an immunological "vaccine" in the form of LDL cholesterol for this disease in hypercholesterolemic rabbits (Ameli et al. 1996). It is possible that such vaccination involves recruitment of antibodies which neutralize the oxidized LDL cholesterol, thus reducing the macrophages' atherogenic response to it (Samson et al. 2012). Research on this very promising immunological approach for preventing or treating CHD should continue.

Once the atherosclerotic plaque forms, the flow of blood in the coronary arteries is restricted, which leads to ischemia – reduced oxygen to the heart. This usually results in pain, the common symptom of CHD, typically in the chest, though it can radiate to the left hand and fingers. This development then results in three types of

CHD, with increasing severity: stable angina, unstable angina, and myocardial infarction (MI). Stable angina is a condition of ischemic chest pain usually occurring during predictable triggers such as excessive efforts, demanding more blood by the heart and body than the coronaries can supply. Unstable angina is a condition where there is an increase in the severity, frequency, and duration of chest pain, and it may even occur at rest. It can result from a worsening in the degree of coronary occlusion (e.g., from 30% to 60%) or from sudden vasoconstriction of the coronaries, either causing a severe ischemia in the region perfused by that specific occluded coronary artery. Finally, an MI or heart attack is a severe ischemic attack lasting long enough to cause irreversible death of cardiac cells. This is typically diagnosed by chest pain, changes in the electrocardiogram (ECG) including a new Q-wave, ST-suppression or elevation or an inverted T-wave, and elevation of enzymes indicative of cardiac cell death (e.g., troponin).

Interestingly, 30% of patients may have chest pain without any objective signs of atherosclerosis (Lantinga et al. 1988). Typically, such patients have been found to have high levels of anxiety and neuroticism, the personality trait related to the tendency to attend to, experience, and report negative symptoms (Lantinga et al. 1988). However, severe ischemic chest pain could theoretically also result from non-atherosclerotic vasoconstriction, which was found to be related to various negative emotions such as anger (Boltwood et al. 1993), which we will detail below.

Once a person has developed a mild-moderate plaque (e.g., 30–40%), he or she may suffer from stable angina. Psychologically, this could be easier to cope with than with unstable angina or MI since the patient learns to predict the pain and avoid excessive efforts, beyond which angina may occur. In contrast, unstable angina could result with chest pain occurring even at rest, which means that this pain is considerably less predictive and controllable and thus possibly more stressful. Indeed, studies have found that mild, not severe, plaques frequently precede unstable angina or an MI (Stone et al. 2011). The question then is how can a mild plaque rapidly progress within weeks to a severe plaque of, for example, 75–90% occlusion, leading to an MI? The answers include the following three stages leading to the acute coronary syndrome (ACS), which includes unstable angina and MI. First, macrophages and smooth muscle cells residing in the plaque produce and secrete matrix metalloproteinases (MMPs), which lead to *plaque instability*, the first stage. This makes the plaque vulnerable to break or rupture, due to external forces including sheer stress, blood pressure, vasoconstriction, and other forces. Second, indeed these hemodynamic forces can cause *plaque rupture*, the second stage of the ACS. Finally, as a result of this rupture, circulating platelets migrate to the ruptured cite to "heal" it, but this is again possibly "a correct response in the wrong context." Such a response, when exaggerated, could lead to excessive occlusion, resulting in *thrombosis*, which leads to the ACS. This is the current working model of understanding the etiology of the ACS (Ross 1999). We shall return to this three-step model when trying to explain how psychological factors may contribute to the onset of the ACS. We shall first examine the evidence for the statement that psychological factors are risk factors of CHD.

Psychosocial Risk Factors of CHD

The question of whether psychological factors may trigger or contribute to the onset of CHD and specifically to the ACS has interested people from ancient history. The earliest account is thought to be by Celsus who attributed changes in pulse to fear and anger. The following is not a comprehensive review since many studies examined this topic. However, I try to bring a broad scope of studies on psychosocial predictors of CHD, of whom nearly all are prospective studies.

Among the first modern time researchers to examine this issue in a systematic manner were the two cardiologists Rosenman and Friedman. They noticed that many of their patients shared a similar behavior pattern, which included impatience, easily evoked anger and hostility, time urgency, and competitiveness. They conducted a prospective study, among the first in behavioral medicine, where initially healthy people without CHD were assessed behaviorally and then followed in relation to onset of CHD. They found that those with this behavior pattern, which they named Type-A, had a significantly higher risk of CHD than those without it, which they named Type-B. Importantly, the Type-A-CHD relationship was not explained by other known CHD risk factors (Rosenman et al. 1976). Subsequent studies were performed, which only partly supported these observations (e.g., Gallacher et al. 2003). However, three main limitations soon "chased" the Type-A story. First, in several cultures and samples, the prevalence of the Type-A behavior pattern was considerably higher than the eventual prevalence of CHD in middle-aged adults (Dembroski et al. 1989). Second, several methodologically sound studies did not find a relationship between Type-A and CHD onset (e.g., Shekelle et al. 1985; Hecker et al. 1988), though some well-designed studies did find it (Kawachi et al. 1998). Finally, subsequent studies quite consistently found that only one of the components of the Type-A behavior pattern, namely, hostility, was the "CHD-prone" aspect of Type-A (Dembroski et al. 1989; Williams et al. 1980). The attention then shifted to hostility.

Hostility is the multicomponent construct which reflects the relatively stable tendency to behave antagonistically, think cynically, and feel anger across situations (Barefoot et al. 1992). People high on hostility tend to interpret mild mishaps similar to major events. Indeed, in a study on daily provocations, levels of hostility were found to be inversely related to differences in anger scores between mild and severe provocations (Gidron et al., 1998), supporting a cognitive distortion and/or response bias in hostility. Numerous studies have observed a relationship between hostility and atherosclerosis (Barefoot et al. 1992; Williams et al. 1980) and a prospective relationship between hostility and onset of MI (Dembroski et al. 1989; Hecker et al. 1988) and all-cause mortality as well (Barefoot et al. 1989). Strangely, the etiological role of hostility in CHD has been scarcely investigated in the past two decades. A newer prospective study done on 3837 men and women found that trait and symptoms of anger and hostility as well significantly predicted 10-year incidence of atrial fibrillation (AF), irregular rhythms of the cardiac atria, in men, independent of multiple variables (e.g., diabetes, previous myocardial infarction; Eaker et al. 2004).

AF is a risk factor of myocardial infarction and CHD on its own (Ruddox et al. 2017). However, not all studies support these associations. For example, one large-scale study done on 20,550 Europeans found that hostility did not significantly predict cardiovascular disease (CVD) mortality in either men or women. A sub-analysis however did find such prediction only in people below age 60 (Surtees et al. 2005). One possibility for the mixed findings is how hostility has been assessed because this is a construct highly susceptible to effects of social desirability.

Hostility can be assessed using the Ho scale derived from the MMPI questionnaire (Cook and Medley 1954) and with the shorter version which includes hostile affect, aggressive responding, and cynicism (Barefoot et al. 1989), omitting items contaminated by neuroticism or not reflecting hostility. The Buss and Perry Aggression scale (Buss and Perry 1992) and its 8-item version (Gidron et al. 2001) can also be used to assess hostility. Scores on the 8-item scale have been related to atherosclerosis in young men (Gidron et al. 2001). However, the "gold standard" method for assessing hostility is derived from the original Type-A structured interview (SI; Rosenman 1978). In this interview, the participant is asked during 12 minutes standardized questions about his or her daily life, and in the middle of the interview, questions focus on responses to five daily provocations (e.g., how you respond to a slow driver). The interview is recorded and each answer is coded for hostility. Dembroski et al. (1989) went one step further and coded these answers for hostile content, intensity, and style. The latter reflected the manner of responding and the amount of hostility directed toward the interviewer. Only hostile style was found by that team to independently predict future CHD development (Dembroski et al. 1989). Barefoot et al. (1992) further developed this method by examining the hostile style in depth, with the Interpersonal Hostility Assessment Technique (IHAT). His team found in four samples that IHAT scores correlated at approximately $r = 0.60$ with severity of atherosclerosis (Barefoot et al. 1992). This magnitude of relationship is about double than that between cholesterol and atherosclerosis ($r = 0.23$ to $r = 0.48$; Tarchalski et al. 2003). Anger, the emotional component of hostility, and its expression, the behavioral component of hostility, were experimentally investigated in one study on vasoconstriction in CHD (Boltwood et al. 1993). In that study, patients were asked to recall and describe an event in their lives from the recent 6 months which made them angry while undergoing catheterization for determining their degree of atherosclerosis. Their level of anger was strongly correlated with degree of vasoconstriction (Boltwood et al. 1993). Finally, two meta-analyses found that hostility is an independent and significant CHD risk factor (Chida and Steptoe 2009; Miller et al. 1996).

Another major etiological factor which received a great deal of scientific attention is depression and its related precursor hopelessness. Numerous studies have shown that depression predicts onset of CHD, and several reviews have confirmed this. In a major review of 30 prospective studies which included 893,850 participants, depression significantly predicted CHD as well as specifically MI, independent of confounders. This risk was significant for follow-ups below 15 years (Gan et al. 2014). Several studies found that hopelessness, conceptually related to depression, predicted onset and prognosis in CHD, and one study compared both depression and hopelessness. When both were considered simultaneously, only hopelessness remained a significant predictor of future MI (Pössel et al. 2015). Hopelessness is indeed considered a precursor of depression in some models (Abramson et al. 1989).

Additional psychosocial risk factors for CHD include work stress factors. Two main job stress models, namely, the job demand-control (DC) model (Karasek and Theorell 1990) and the effort-reward imbalance (ERI) model (Siegrist 1996), have been tested in relation to CHD onset. As mentioned in Chap. 1, the DC model states that jobs which are characterized by high work demands and little work control (job strain) are unhealthy. Indeed, studies using the DC model show cross-sectionally a relationship between the DC model and CHD (Yoshimasu et al. 2001). Prospectively, some studies support this model, while others do not (e.g., Padyab et al. 2014; Netterstrøm et al. 2010). The ERI model states that jobs which involve high demands and little rewards (salary, recognition) lead to chronic stress responses and poor health consequences. A meta-analysis of 45 studies found ERI to be related to multiple adverse health outcomes including CHD and its symptoms (van Vegchel et al. 2005). One prospective study assessed with a brief measure job fairness, conceptually related to ERI. The researchers found that low-medium job fairness predicted cardiac mortality, independent of confounders (Elovainio et al. 2006).

We spoke until now about individual and negative psychosocial factors. However, people live in a social context, and as mentioned in Chap. 1, social support is an important modulator of the stress response. A relevant concept is social integration, a concept reflecting the frequency of contacts a person has with one's social network (family, religious circles, friends, etc.). Furthermore, most studies reviewed up to now focused largely on men. Chang et al. (2017) examined whether social integration predicted onset of nonfatal and fatal CHD in 76,362 women, free of CHD and stroke, participating in the Nurses' Health Study. The follow-up was approximately 22 years. They found that social integration significantly predicted a reduced risk of CHD. However, concerning nonfatal MI, this relationship was no longer significant and was mediated by lifestyle factors after statistically controlling for them. In contrast, social integration still significantly predicted fatal CHD events, after controlling for lifestyle factors, though the relationship was reduced in strength. This suggests that a broader concept of social support, one which exists in a person's wide social network, predicts reduced risk of fatal CHD in women, partly mediated by their behavioral risk factors (Chang et al. 2017).

Finally, another psychological factor often tested for its etiological role in CHD has been life events. In a large study conducted on 8365 Danish people, risk of CHD was significantly predicted by broken partnerships but not by job loss, independent of most demographic confounders (Kriegbaum et al. 2008). However, another large-scale Danish study (n = 8738) examined the etiological role of accumulation of major life events in CHD. This study did not find major life events to predict onset of CHD (Andersen et al. 2011). Several studies examined the incidence of MI during severe periods of social stress. For example, Meisel et al. (1991) found that during the first nights of the 1991 Gulf War, in which Iraq bombarded Israeli cities without any provocation from Israel, there were more MIs than five control periods

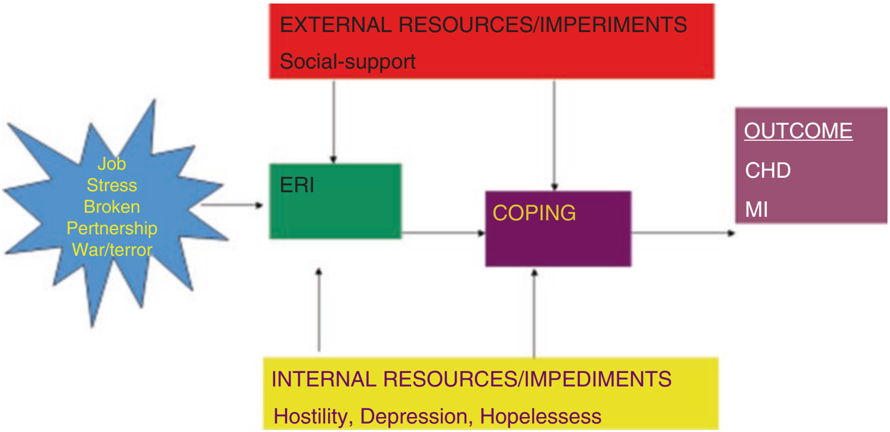

Fig. 4.1 Mapping findings on etiological psychosocial factors of CHD on Taylor's framework of stress, appraisal, coping, resources, and outcome. (Taylor 1995)

including other periods of that war. These first days were characterized by high threat and uncertainty concerning whether the missiles would include chemical weapons or not. Similarly, Feng et al. (2006) found significant increases in MIs and tachyarrhythmia, but not in unstable angina, in patients admitted in New York after, compared to prior to, the horrific 9/11 terrorist attacks.

Figure 4.1 depicts the results of the reviewed studies mapped onto Taylor's (1995) model of stress and adaptation.

Psychosocial Prognostic Factors in CHD

Psychosocial factors also predict the functional and vital prognosis in CHD. One important issue in health psychology/behavioral medicine is how patients perceive their illness and whether such perceptions influence their outcomes. This can be mapped onto the appraisal component of Taylor's (1995) framework presented in Chap. 1. One original study addressed this issue visually. The researchers asked patients to estimate their infarct size, by drawing their heart before and after their MI. Patients' estimated size of infarct in the drawing was positively and significantly correlated with their levels of troponin-1, a marker indicative of actual myocardial cell death. Furthermore, their estimated infarct size predicted their number of days for returning to work 3 months later, while troponin levels did not (Broadbent et al. 2004). Another important issue of possible influence on patients' prognosis is the way they perceive their illness, often assessed by the Illness Perception Questionnaire (Moss-Morris et al. 2002). This questionnaire assesses various aspects of perceiving an illness including its consequences, its time course, and one's control over it. In a prospective study on cardiac patients undergoing surgery, levels of negative perceptions about consequences, chronic time course, and cyclical time course were positively related to disability and inversely related to physical functioning 3 months later, independent of demographic factors and illness severity (Juergens et al. 2010).

A few studies investigated the prognostic role of different psychosocial factors together, in various samples of CHD patients. This approach enables to identify the most prognostic psychosocial factor compared to others. In a more recent study in *N* = 1268 patients with existing CHD, the variables anxiety, depression, hostility, social support, and perceived stress were examined for their prognostic role, after performing a factor analysis. In a multivariate analysis, only functional status and anxiety significantly predicted new events or recurrent hospitalization over the 9 subsequent months (Grewal et al. 2011). While these events were not strictly limited to reinfarct or cardiac death, this study suggests that for prognosis, anxiety may be a crucial psychosocial prognostic factor in CHD. However, readmissions could indeed also result from or partly reflect a person's neuroticism, overattending to and reporting of physical symptoms, as mentioned in Chap. 2. Thus, it is important to also examine more objective outcomes such as reinfarct and mortality.

Indeed, a more recent study found that among several psychological factors, only hostility significantly and independently predicted major cardiac events in patients with congestive heart failure (Rafanelli et al. 2016). The discrepancy in the results of these studies can partly be explained by the use of different psychosocial measures and consideration of different variables and by recruiting different types of patients – CHD versus congestive heart failure. Furthermore, anxiety may predict readmissions, while hostility may predict more "hard outcomes" such as major cardiac events.

Important evidence for the prognostic role of depression after MI comes from Frasure-Smith et al. (1995). In that study, *N* = 222 post-MI patients underwent assessment of depression by an interview and questionnaire (Beck Depression Inventory; BDI). Both measures and particularly the BDI strongly predicted post-MI mortality over 18 months of follow-up, independent of confounders. Finally, premature ventricular contractions synergistically interacted with depression to predict worse prognosis (Frasure-Smith et al. 1995). The latter finding is important since it shows that psychological and physiological factors can interact and also because it points at the patients at greatest risk of poor prognosis. In a following study of that research group, Lespérance et al. (2002) found that changes in depression over 1 year significantly predicted post-MI mortality. However, the main burden was due to initial levels of depression after the MI.

We spoke in detail before about hostility and anger. However, another issue often asked by clinicians is whether anger-in or anger-out is worse for health. The thought that anger-in has worse health consequences is deeply rooted in psychoanalytic thinking, yet does the scientific evidence support this assumption? In a recent study on 146 patients with heart failure, anger-out, but not anger-in, significantly predicted readmissions to hospital (Keith et al. 2017). Other studies, in other cardiac populations, support this finding.

One personality characteristic which has received much attention in relation to its prognostic role in CHD is the Type-D personality. This behavioral cluster reflects people who are high on distress (e.g., sadness, anxiety) and on social inhibition. The Type-D personality was found in several prospective studies to predict post-MI prognosis, independent of known prognostic factors (Denollet et al. 1996; Denollet and Brutsaert 1998). However, some recent studies did not replicate these early observations (e.g., Dulfer et al. 2015). A recent cross-sectional study on a large sample of $n = 4753$ Icelandic people found that people with Type-D had a significantly higher future estimated risk for a cardiac event, based on a poor lifestyle (see below; Svansdottir et al. 2013).

Up to now, I mentioned psychologically negative factors. What about positive internal resources? One such variable is sense of coherence (SOC; Antonovsky 1993), which refers to people's ability to understand and link events in their lives (comprehensibility), to deal with and meet the demands they set for themselves (manageability), and to justify to themselves their efforts (meaningfulness). In several studies, high SOC predicts better important clinical outcomes in post-MI patients. For example, SOC in general and comprehensibility specifically were found to predict better quality of life in post-MI patients. Comprehensibility also increased with time since the MI. These findings suggest that with time since one's MI, patients can increase their levels of understanding and linking events in their lives around the MI and that such increases are related to better quality of live (Bergman et al. 2012). Furthermore, Myers et al. (2011) found that SOC predicted positively levels of leisure time physical activity, independent of depression, disease severity, and other confounders. In contrast, Norekvål et al. (2010) did not find that SOC predicted prognosis in women after a MI. Thus, high SOC seems to predict good clinical and behavioral outcomes in post-MI patients, but its role concerning "harder outcomes" is unclear.

Another positive variable, which reflects more people's cognitive appraisals, is their expectations for recovery. This variable's predictive value was tested in relation to recovery from surgery (Gidron et al. 1995; Scheier et al. 1989) but also in relation to survival of CHD patients. Barefoot et al. (2011) assessed recovery expectations after angiography and followed people for 1 year in relation to functional capacity and for 15 years in relation to survival. Expectations positively predicted survival, independent of confounders, as well as predicted functional capacity. This is crucial since such a specific variable is amenable to modification via cognitive interventions.

Following Taylor's (1995) framework, the important external resource variable of social support has also been related to prognosis in CHD. In a relatively old study, Woloshin et al. (1997) assessed people's need for social support and found that need for tangible social support at home had an odds ratio relationship with post-MI mortality in the range of 3.2–6.5, independent of initial health and age. These large odds ratios are far larger than those of many typical known prognostic factors in CHD and call for taking seriously the lack of social support as a strong prognostic factor in CHD. This is also important since providing tangible support could be done either by family members (spouse, children) or by people in the

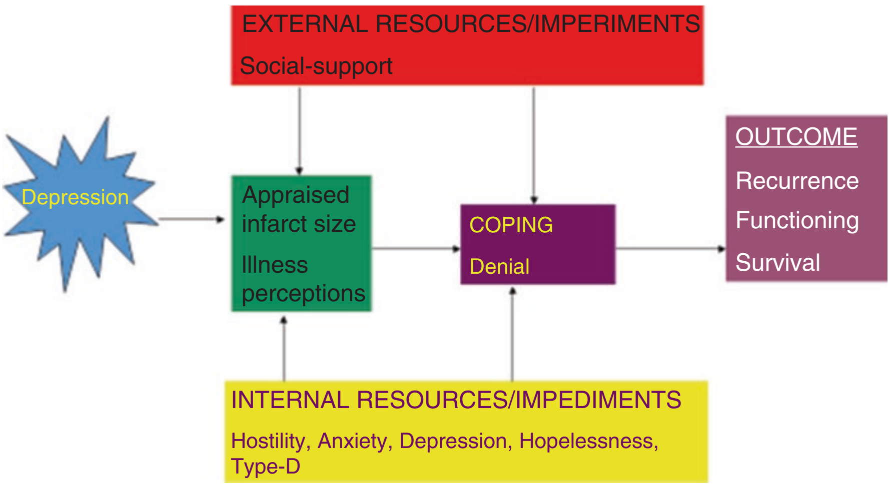

Fig. 4.2 Mapping findings on prognostic psychosocial factors of CHD on Taylor's framework of stress, appraisal, coping, resources, and outcome. (Taylor 1995)

community. For example, young university students could receive reductions in their study fees for performing weekly supervised home visits to assist in instrumental tasks such as shopping and lifting items, and such a system exists in some countries (e.g., Israel).

Figure 4.2 depicts the prognostic factors in CHD, based on the studies reviewed here and mapped onto Taylor's (1995) model of stress and adaptation. These serve to summarize the reviewed studies. The coping strategy denial was added after mentioning the study by Levine et al. (1987) in Chap. 1. In that study, use of denial in the hospital predicted a better short-term prognosis, while denial predicted more readmissions to hospital over a 1 year of follow-up. The "goodness-of-fit" hypothesis explains such differences in outcomes.

Psychoneuroimmunology of the Acute Coronary Syndrome

A major area of investigation in behavioral medicine is the study of the biological mechanisms possibly linking psychosocial factors with CHD. We will focus here on the acute coronary syndrome (ACS) which includes the MI and unstable angina. As mentioned at the beginning of this chapter, the present working model of the evolution of the ACS includes three main stages: plaque instability, plaque rupture, and a superimposed thrombosis (Schroeder and Falk 1996; Ross 1999). Let's examine the role of psychological factors in plaque instability. As mentioned above, MMPs derived from residing macrophages contribute to plaque instability of a mild (30–40%) atherosclerotic plaque. Studies have found that various psychological factors correlate with such specific biomarkers. For example, in a relatively large sample of healthy people in Sweden ($N = 402$), depression and hostility subscales were positively correlated with

MMP-9, while sense of coherence was inversely correlated with MMP-9 (Garvin et al. 2009). In another study done on Israeli ACS patients, the correlations between psychological factors and percentage of white blood cell subsets obtained at the first blood test upon arrival to the hospital were examined. Hostility and life events were positively correlated with percentage of monocytes, while perceived-control and emotional support were inversely related to monocyte percentage (Gidron et al. 2003). A more recent study by Zuccarella-Hackl et al. (2016) found that CHD patients with the Type-D personality had higher levels of macrophages producing superoxide anion. These are free radicals which contribute to the earlier stages of atherosclerotic lesions by leading to the oxidation of LDL cholesterol and contribute to endothelial dysfunction (Cathcart 2004; Libby et al. 2011). What about the effects of psychological variables on factors that lead to plaque rupture, the second stage in the ACS evolution? Among at least seven factors, two have been related to psychological factors. Boltwood et al. (1993) asked patients undergoing angiography to describe an event that made them angry during the past 6 months (the anger recall task). Levels of anger during the recall were strongly positively correlated with level of vasoconstriction in atherosclerotic regions ($r = 0.82$). Another study conducted a randomized controlled trial (RCT) testing a hostility-reduction intervention in high-hostile CHD patients. Hostility and blood pressure (BP) were significantly more reduced in the hostility intervention group compared to an attention-control group, and reduction in hostility significantly correlated with, and preceded in time, the reduction in BP (Gidron et al. 1999). These results suggest that hostility causally contributes to vasoconstriction (Boltwood et al. 1993) and to BP (Gidron et al. 1999), both which can lead to rupture of a vulnerable atherosclerotic plaque. Finally, there is evidence that hostility, anger-out, and depression are all positively correlated with platelet aggregation (Markovitz et al. 1996; Markovitz 1998; Wenneberg et al. 1997), which can ultimately lead to thrombosis and then to severe occlusion and the ACS. These findings led us to propose a psychoneuroimmunological model for the ACS (Gidron et al. 2002b). Figure 4.3 depicts most of this model visually.

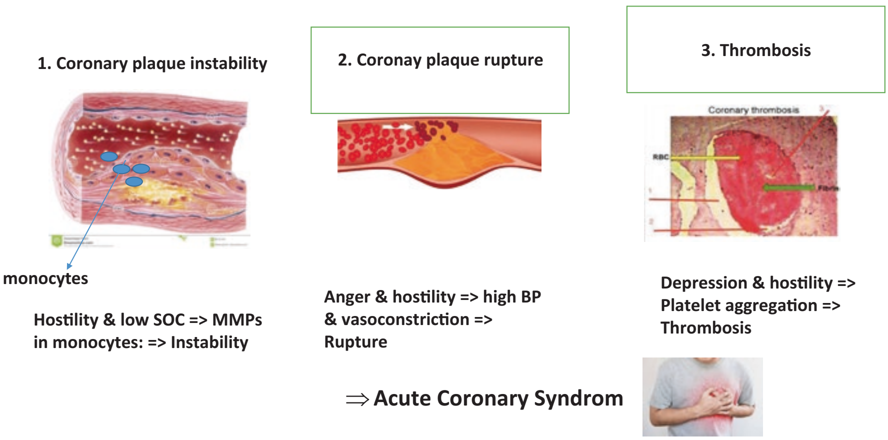

Fig. 4.3 A psychoneuroimmunological model of the acute coronary syndrome (ACS)

All these processes can be associated with one more crucial factor related both to psychosocial factors and to the processes leading to CHD, namely, the vagus nerve. Numerous studies link psychological factors such as depression to reduced vagal nerve activity, indexed by heart rate variability (Carney et al. 2001). The vagus reduces inflammation (Rosas-Ballina et al. 2011) and of course leads to vasodilitation and reduces blood pressure via the baroreflex, also since it inhibits sympathetic activity (Saku et al. 2014). Thus, reduced vagal activity may link psychosocial factors to CHD (Carney et al. 2001). Furthermore, a meta-analysis of 21 studies found that low HRV predicted a 4 times increase in post-MI mortality (Buccelletti et al. 2009).

Various psychosocial factors also correlate with biological factors which contribute to the metabolic syndrome (MetS). The MetS is a cluster of risk factors including obesity and two of the following: high blood pressure, high triglycerides, low HDL cholesterol, and high fasting plasma glucose. The MetS is on its own right a risk factor of CHD and cardiovascular diseases (Ärnlöv et al. 2010). One study examined whether hostility is associated with fasting glucose, as function of gender and ethnicity, in a sample of 565 Americans. Only in African-American women was hostility positively correlated with fasting glucose (Georgiades et al. 2009). Again, psychological factors correlate with biological factors which are etiologic to CHD, but issues such as gender and ethnic background (perhaps genetic factors) moderate such associations.

Related more specifically to such genetic issues is the line of research, led by Williams and colleagues in several studies, who associated hostility and other psychosocial factors with the central nervous system serotonin metabolite 5HIAA and genetic polymorphisms of the monoamine oxidase A promoter gene MAO-uVNTR. They found significant interactions of 5HIAA × MAO-uVNTR polymorphisms such that in men with less active MAO-uVNTR alleles, high 5HIAA was related to less hostility and a reduced psychosocial and metabolic risk. In contrast, in men with more active MAO-uVNTR alleles, high 5HIAA was related to higher hostility and a less favorable psychosocial and metabolic profile. These findings could explain the clustering of hostility with other psychosocial and metabolic risk factors in some people, which predict risk of CHD (Williams et al. 2010). Indeed, in one study, hostility interacted with family history, a simple and proxy index of genetic risk, in relation to CAD severity (Gidron et al. 2002a). In another study, Brummett et al. (2013) extended this topic to predicting actual cardiac events. They examined nucleotide variants of a serotonin receptor called 5HTR2C, the code variant being rs6318, with two main variants, namely, Ser23 C and Cys23 G. In earlier work, Brummett et al. (2012) found that people with the Ser23 C allele responded objectively and subjectively more strongly to psychological stress than those with the Cys23 G allele. In a large study on 6126 participants, those with the Ser23 C allele had significantly higher risks of all-cause mortality or MI, and this remained also after statistically controlling for confounders (e.g., number of occluded arteries; Brummett et al. 2013). Such research reveals the importance of considering both genetic and psychosocial factors in predicting risk of CHD and then in intervening with people at highest risk for the disease or poor prognosis. The field of behavioral genetics thus promises to help achieve more “tailored” or “individualized” behavioral medicine and reveals crucial behavior x gene interactions in relation to health and diseases.

Behavioral Mediators Between Psychological Factors and CHD Prognosis

Many types of behavioral mediators exist, with some being CHD risk factors such as smoking or an unhealthy diet, while others relate to medication adherence. For example, Schoenthaler et al. (2009) found that depressive symptoms were correlated with nonadherence in hypertensive patients. Another study found that sense of coherence was related to a lower risk of total nonadherence to medication in 1021 Finnish hypertensive patients, independent of important confounders (Nabi et al. 2008). One of the models explaining the relationship between hostility and health (Smith 1992) is poor health behaviors. Indeed, Siegler et al. (1992) found in a prospective study that hostility predicted smoking, caffeine consumption, lipid ratio, and body mass index. Similarly, Scherwitz et al. (1992) found in $N = 5115$ young adults positive associations between hostility and alcohol and drug consumption, smoking and caloric intake, independent of age and education level. In a more recent study, hostility was correlated with continuing to eat despite being satiated (Van den Bree et al. 2006). Finally, the Type-D personality has also been found to prospectively predict nonadherence to medication in Chinese diabetic patients (Li et al. 2016). Thus, part of the associations between psychosocial factors and the etiology or prognosis of CHD may be accounted for by their links with unhealthy behaviors. Yet this conclusion should be seen with caution since most prospective studies examining relationships between psychological factors and CHD take into account some behavioral risk factors in their multivariate analyses.

Psychological Interventions in CHD Patients

The studies reviewed above on the etiological and prognostic factors in CHD can guide clinicians and clinical researchers in developing and testing interventions which focus on these predictors, especially if modifiable. These can also be guided by the models of stress and adaptation.

Numerous types of psychological interventions have been performed and tested in CHD patients. Some aim to change their lifestyle; others aim to change psychological risk or prognostic factors such as hostility and depression. Several systematic reviews have been conducted on this topic. A relatively recent systematic review of 16 studies included randomized trials, where treatments were provided by trained staff and where the follow-up time was at least 6 months. A small effect on cardiac mortality was found, but no strong evidence was found in relation to total death reduction or risk of nonfatal infarction (Whalley et al. 2014). Another unique review attempted to investigate the effects of different components of psychological interventions or their combination, on clinical outcomes in heart disease. Interventions were grouped into usual care (control), educational (also used in some studies as control), behavioral (risk factor modification such as smoking cessation), cognitive (stress-reducing cognitive modification), relaxation, and support. Behavioral interventions were found to reduce all-cause mortality and nonfatal MI, and cognitive and/or behavioral interventions reduced depressive symptoms (Welton et al. 2009).

Often studies suffer from methodological and conceptual limitations. First, some studies do not screen for or include only patients high on a risk factor (e.g., high hostility). Second, sometimes groups differ on important prognostic factors at baseline, after randomization, which then calls for statistical control over such variables and subsequent reduced statistical power. Finally, several intervention studies included too general "therapeutic ingredients," such as support or stress management, which may not target specific psychological risk factors, as mentioned in the review of "complex" psychological interventions (Welton et al. 2009). One study tried to overcome these problems by recruiting high-hostile CHD patients and using a matched-RCT design. Pairs of patients were matched on age and hostility. They were then randomized to an 8-week hostility-reduction cognitive-behavioral treatment (CBT) or to a 1-session attention-control group. The CBT intervention included an educational component on the CHD risks of hostility, self-monitoring and skills for reducing antagonism (e.g., listening, smiling, being assertive), monitoring and skills for changing cynical cognitions (e.g., thought stopping, evidence weighing), skills for monitoring and reducing anger (e.g., relaxation and problem-solving), and, finally, a relapse-prevention component. Significant reductions in Barefoot's Ho scale, in potential for hostility observed in the structured interview and in blood pressure, were found only in the CBT group (Gidron et al. 1999). In a later study on the same patients, the CBT group was also found to have had shorter readmissions to hospital, which resulted in greater health-care savings (Davidson et al. 2007). These results were obtained despite use of a relatively small sample, possibly because of including only at-risk individuals and the matched-RCT design.

Several studies have been conducted which tested the Williams lifestyle (WLS; Williams and Williams 2011) program on both psychosocial and biological CHD risk factors. The WLS program includes ten main elements often seen in stress management programs such as problem-focused coping versus regulating one's reactions, assertiveness training, and communication skills. For example, 1 RCT conducted in Singapore randomized 58 CHD patients undergoing coronary artery bypass grafting, to the 6-week WLS program or to usual care +1 hour of stress education (control). After a 3-month follow-up, significant improvements were observed in the WLS arm compared to controls on depression, trait anger, perceived stress, satisfaction with social support, and life satisfaction, as well as improvements in resting and in reactive blood pressure and heart rate (Bishop et al. 2005). This study is important since it extends work from Anglo-Saxon countries to Asia and shows that a brief structured skills-oriented stress management program can improve both psychosocial and biological CHD risk factors in CHD patients. The long-term clinical effects of such programs need to be tested. Another RCT examined the effects of a stress-reduction intervention (e.g., relaxation methods, cognitive restructuring, risk factor education) on the risk of death in Swedish women with CHD. Controls received usual care alone. A threefold reduced risk of death was found in the stress-reduction group compared to controls, independent of confounders (Orth-Gomér et al. 2009).

A very important RCT, aimed to reduce primarily depression, was the "Enhancing Recovery in Coronary Heart Disease" (ENRICHD) trial, due to the prognostic role of depression in CHD. Between 1996 and 2001, 2481 post-MI patients, scoring high on depression or low on social support or both, were randomized to a cognitive-behavioral treatment (CBT) plus serotonin-selective reuptake inhibitor (SSRI; when indicated) or to usual care alone. After a mean follow-up of 29 months, both groups had approximately the same percentages of event-free survival (76%; Berkman et al. 2003), despite significant reductions in depression and increases in social support. In a reanalysis of this trial for sex and ethnic subgroups, only in white men were there significant reductions in cardiac mortality and reinfarcts in the intervention group compared to controls (Schneiderman et al. 2004). These results question the effectiveness and generalizability of such interventions for post-MI patients and call to tailor treatments for gender and ethnicity. In addition, the fact of providing SSRIs to patients in the experimental group only may have masked potential findings associated with the CBT alone due to effects of SSRIs.

I shall cite now two important studies using vagal nerve activation in the context of CHD. First, Arimura et al. (2017) recently tested the effects of an intravenous activation of the vagus nerve in dogs induced to have an MI by ischemia. The activation was done immediately, or 90 min, after MI, or not at all (control). Compared to controls, whose percent infarct size was 13.3%, the infarct size was significantly reduced if vagal stimulation was done immediately (2.4%) or 90 min after inducing ischemia (4.5%). These results call to urgently examine the effects of vagal nerve stimulation in people after an MI, certainly in those with high hostility, depression, or the Type-D personality. Finally, Yun et al. (2018) recently tested effects of HRV-biofeedback (HRV-B) on psychological outcomes and readmission rates of patients with coronary artery disease, using an RCT. In HRV-B, people learn to perform deep and paced breathing while receiving biofeedback on a screen concerning their level of HRV. The HRV-B significantly reduced all-cause readmissions (12%) compared to controls (25.4%) and also reduced hostility and depression. Because patients can perform HRV-B themselves, and since this intervention has no or little side effects, its use must be extended, and its effects on cardiac outcomes in CHD patients high on hostility or depression need to be tested in the future.

Finally, one important issue in psychological interventions in behavioral medicine in general and for CHD specifically is that they often have by far smaller sample sizes compared to medical RCTs. This could strongly influence psychological interventions' statistical significance, which is a function of sample size, but also a function of the effect size of the tested intervention. In one study, all existing eight psychological RCTs in post-MI patients (when the study was performed), which included death as an outcome, were compared to medication RCTs. In the latter studies, one trial of each of eight medication types, randomly selected from meta-analyses, was obtained, and together these were compared to the psychological interventions. Indeed, the medical trials had 12 times larger sample sizes on average and more often obtained statistical significance than the psychological trials. However, psychological trials seemed to have prevented three times the number of deaths/100 patients compared to medication trials (Gidron et al. 2004). This suggests that we need to consider both statistical and clinical significance (effect size) and also suggests that psychological interventions in CHD may be quite effective, but they require larger samples to have sufficient statistical power as well.

Case Study Conclusion

Mr. Ismail was diagnosed as having an acute MI with two occluded coronary vessels. His "typical" CHD risk factors are elevated LDL cholesterol, obesity (BMI >30), and family history of CHD (mother). While he received statins and a beta-blocker, no treatment was offered to him for the psychological risk factor reported by his wife – attacks of anger-out. The clinician attending him should assess his levels of hostility, and if above norms, he should be offered a brief cognitive-behavioral treatment for altering his cynical thoughts, angry feelings and antagonistic explosive behaviors, or HRV-biofeedback. Furthermore, to help him reduce weight, we could identify his barriers which impede him from eating in a more healthy manner. Then, psychological inoculation could address such barriers and unhealthy diet. Together, such treatments may cover all his risk factors. Including vagal nerve activation (possibly by HRV-biofeedback) could also serve to reduce his sympathetic responses, obesity, and inflammation, which contributed to his MI. Repeating the assessments in an AABB design would enable the clinicians to know with some validity that the changes were possibly due to the intervention.

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Chapter 5
Psychosocial Factors and the Prognosis of Cancer

Case Study

Ms. Janssens was diagnosed 2 years ago with stage 2 breast cancer. She has no signs of lymph node involvement and no metastasis. She received chemoradiotherapy before undergoing a lumpectomy. She then did some plastic surgery to restore the shape of her breast. Now she is 5 months after her surgery and is in remission. Her doctor told her she is quite optimistic about her prognosis. However, Ms. Janssesn's mother died from breast cancer, and she fears the same for herself. She spends hours in her house without working, telling herself she has no more control over her life and thinks she will die soon. She received some psychological treatment which included supportive listening from a very empathic therapist. But, there has been no change in her psychological condition. Twelve months later, she had a relapse, and she underwent curative chemotherapy. Now, she is quite helpless. The doctor tries to provide her the best new chemotherapy for her more advanced cancer.

Introduction to the Biology of Cancer

Cancer is among the most complex diseases to comprehend, but yet not impossible. While we must approach this illness with modesty, important medical improvements have occurred in the past decades. Cancer evolves following damage to multiple key genes in the DNA of cells, called the multi-hit model of cancer. Among these genes is mutation of a gene which can normally prevent cancer called p53. Hanahan and Weinberg (2011) conducted a deep analysis and literature synthesis of several hundred of scientific publications, which led them to propose several hallmarks of cancer. These include sustaining cell proliferating signals, enabling replicative immortality, avoiding cell growth suppressors (e.g., P-53), resisting cell

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death, inducing angiogenesis (creating a new network of nourishing blood vessels to the tumor), and invasion and metastasis (migration of cancer cells to other organs). Two additional emerging hallmarks are deregulating cellular energy and avoidance of immune surveillance. Other factors that contribute dramatically to tumorigenesis are the inflammatory response (Mantovani et al. 2008; Voronov et al. 2003) and the recruitment of presumably healthy cells to support tumor growth, collectively termed the tumor microenvironment (Hanahan and Weinberg 2011). The skilled reader may have noticed that the immune system plays a double-edged sword in cancer, since on the one hand, various antitumor immune cells (natural killer cells, cytotoxic T-cells) can attack cancer cells and their presence near the tumor predicts better prognosis (Galon et al. 2006). On the other hand, the inflammatory response, the recruitment of mostly innate immune cells to danger signals (mutated cells), is essential and plays a causal role in all stages of tumorigenesis (Pikarsky et al. 2004; Voronov et al. 2003). Inflammatory signals such as interleukin-1 (IL-1) and tumor necrosis factor alpha (TNF-alpha) affect multiple key intracellular processes essential for carcinogenesis including suppression of the tumor suppressor gene P-53 (Pikarsky et al. 2004). One main reason for the limited ability to produce a "vaccine" against cancer is the genetic instability inherent to cancer cells (Korkola and Gray 2010), which provides them the ability to avoid immune surveillance since the cancer "antigen" constantly changes. We shall return to this issue later in this chapter.

The basis of acquired mutations in cancer is damage to the DNA. In the short term, and if irreparable, this can lead to cell death, a protective response. However, in the long term, if not repaired and when accumulated with several mutations, this can lead to cancer (Hoeijmakers 2001). This needs to take place in key genes which regulate cell replication, cell death (e.g., P-53), or other intracellular factors that promote tumorigenesis. As we have seen in Chap. 2, psychological stress can on its own cause DNA damage as well (e.g., Adachi et al. 1993).

Cancers are then diagnosed and staged normally into four stages, using the T (tumor size), N (involvement of lymph nodes), and M (presence of metastasis) system. While stages 1–2 are considered early tumors, stages 3–4 are considered advanced cancers. It is important to know however that 90% of cancer deaths are from the metastasis of cancer rather than from the original tumor (Hanahan and Weinberg 2011). Common risk factors for cancer include excessive smoking (particularly for lung cancer), certain environmental carcinogens from pollution, little physical activity, high alcohol intake, and a high fat diet (e.g., Unwin and Alberti 2006).

Psychosocial Factors and Cancer Onset

This is perhaps among the most controversial topics in behavioral medicine. While several reviews propose that there is little to some evidence linking psychological factors to the onset of cancer (Chida et al. 2008; Garssen 2004), some prospective studies propose otherwise. Earlier studies making claims that psychological factors may be risk factors of cancer were often based on case-control designs. However, from a methodological and scientific perspective, it needs to be clear that case-control studies are seriously flawed. Finding, for example, that cancer patients have higher levels of depression than controls cannot in any way rule out the bi-directional causality where depression can "contribute" to or *result* from the cancer diagnosis. Thus, to begin to answer such a question with better methodology, one needs to use prospective studies, where psychological factors are assessed at time 1, in people without cancer, and then their health is monitored prospectively over time in relation to future cancer diagnoses. In the review of Chida et al. (2008), they examined this issue in 142 prospective studies on initially cancer-free people. The overall hazard ratio (HR) was significant but weak (HR = 1.06, 95% confidence intervals, CI, 1.02–1.11). This association was mostly driven by studies on lung cancer and when testing the role of depression. Let's look at some specific, large-scale studies. A recent study was conducted on a large sample of 13,768 French people in the GAZEL cohort. Four psychological factors were assessed including personality Type 1 (suppressed emotional expression), Type 5 (rationality and "antiemotionality"), hostility, and Type A behavior. The onset of various cancers was examined. Results revealed that the Type 1 personality predicted a reduced risk of breast cancer, and Type 5 predicted an increased risk of other cancers, while hostility predicted an increased risk of smoking-related cancers (e.g., lung, oral cavity), independent of certain confounders (Lemogne et al. 2013). Chida et al. (2008) conducted the largest meta-analysis on this topic and concluded that only for lung cancer, there is possible evidence for the relationship between psychological factors and onset of the disease. However, one could consider the major evidence concerning the role of smoking in the etiology of lung cancer (e.g., Nakamura et al. 2009) as evidence for effects of behavior on onset of this cancer, since smoking is first a behavior, translated of course directly into oxidative stress and DNA damage (Chen et al. 2015).

One study recently examined the relationship between the dimensions of the "Big Five" model of personality and cancer, in a pooled total sample of 42, 843 men and women. Controlling for age, gender, and ethnicity, none of the dimensions predicted cancer onset or cancer death, nor in relation to cancer-specific incidence (Jokela et al. 2014). This is an important study since it examined this issue using an internationally empirically based and valid measure of personality, a large sample, and since it examined this issue in several cancers in a prospective design. Thus, the conclusions from this study can be quite compelling. Another study done in 5114 German participants found that increased time urgency, originating from the "Type A behavior pattern," predicted a reduced risk of being diagnosed with or dying from cancer (Stürmer et al. 2006). To summarize, little and if at all inconsistent evidence exists for any association between psychological factors and *onset* of cancer, except for the role of smoking behavior in cancer, and particularly in lung cancer. However, it is important to recognize, as health researchers or future clinicians, that patients may have the need to attribute their cancer onset to psychological factors. For example, a recent study done on women from the French Indies found that stress, diet, and genetics were the frequently cited attributed cause of cancer (Kadhel et al. 2018). Yet the scarce evidence on the role of psychological factors in cancer onset, except for smoking, proposes that patients can be relieved from this sense of responsibility and that the public and clinicians need to be educated about this matter. It would be thus empirically unfounded, clinically a dreadful error, and highly unethical to tell cancer patients that "they brought" the cancer on themselves!

Psychosocial Factors and Cancer Prognosis

Numerous studies have examined this issue, and reviewing them all here is beyond the scope of this chapter. However, the meta-analysis by Chida et al. (2008) which included 157 studies concluded that there is evidence to show that psychological factors predict prognosis in cancer, independent of confounders. The overall HR was weak (1.03; 95% CI, 1.02–1.04). Yet, this result also remained significant when only considering the methodologically stronger studies and was particularly strong in studies which statistically controlled for multiple confounders. Life stress had the stronger prognostic role than other categories of psychosocial factors, and these associations were stronger in hepatobiliary cancer, head and neck cancer, and lymphoid and hematological cancers (Chida et al. 2008).

To nevertheless review in a meaningful manner some of the prospective studies on this topic, I shall examine various psychosocial factors according to the framework of stress, coping and adaptation of Taylor (1995), and aiming to do so in different cancers.

Breast cancer is the most prevalent type of cancer in women around the world and accounts for 23% of female cancers (Jemal et al. 2011). Recent years have seen improvements in the diagnosis and treatment of this cancer. A relatively old study done by Jensen (1987) examined the role of several psychological factors in predicting prognosis in breast cancer over 2 years. She assessed hopelessness, defensiveness, expression of negative affect, chronic stress, and day dreaming, as well as multiple biomedical factors such as cancer stage and hematocrit levels. Hopelessness had the strongest predictive significance in relation to prognosis, in a multivariate analysis. In a later important study on 578 women with early stage breast cancer, various common psychological responses to cancer were assessed. These included stoic acceptance, hopelessness, fighting spirit, denial, anxiety, depression, and emotional control. In the first follow-up of 5 years, hopelessness significantly predicted risk of death, but not after statistically controlling for initial cancer stage, while depression did remain an independent prognostic factor (Watson et al. 1999). However, at a 10-year follow-up, hopelessness was the only significant psychological prognostic factor, independent of confounders (Watson et al. 2005). Hopelessness includes perceptions of lack of control (helplessness) and negative expectancies about one's future (pessimism; Everson et al. 1996). One can construe hopelessness as part of the appraisal stage of the stress, coping and adaptation framework.

A related topic is the prognostic role of optimism. A more recent study examined this issue together with other "positive psychological constructs" and depression, in non-small cell lung cancer (NSCLC; Gidron et al. 2014). These constructs included benefit finding, mindfulness, and indirect optimism, as well as depression. Indirect optimism was assessed by a time-limited word search task in which patients had to find optimistic and pessimistic words in 1 minute in a word matrix. Such a task assumes a "pop-up" effect of identifying mood-congruent words by participants. The investigators found that only indirect optimism significantly predicted survival, independent of cancer stage, treatments, and other confounders. This study may have been the first one to test the prognostic role of an indirectly assessed measure in cancer. However, its sample (n = 78) was relatively small.

Another study on lung cancer examined the prognostic role of coping strategies. In $n = 103$ patients, the investigators assessed active coping (information seeking, problem-solving, and planning to live intensively) and depressive coping (self-pity, taking things out on others, and distancing oneself from others). Multiple confounders were considered such as stage, age, and performance status. Indeed, while depressive coping significantly predicted greater risk of death (RR = 1.91; 95% CI, 1.32–2.77), active coping predicted a reduced risk of death (RR = 0.72; 95% CI, 0.54–0.98; Faller and Bülzebruck 2002). Though the sample was not large, this study has possible important clinical implications for trying to improve prognosis by teaching patients active rather than passive coping strategies. However, as we shall see below, psycho-oncology has not been successful in translating such epidemiological findings into effective therapeutic results, to reliably improve cancer prognosis via psychological interventions.

Melanoma is a skin cancer, whose rate has dramatically increased over the past decades (Monshi et al. 2016). Its risk factors include recreational sun exposure and tanning, occupation, obesity, and smoking, and these are related to higher socioeconomic status (Jiang et al. 2014). One comprehensive study examined the prognostic role of quality of life, marital status (as a proxy measure of social support), coping by minimizing the illness, coping difficulties, anger suppression concerning the illness, and the perceived aim of the treatment (curative versus palliative), in relation to 5-year survival in melanoma. Patients who perceived their treatment to be curative survived longer than those perceiving it to be palliative. In addition, those who minimized their illness, reported greater anger over their illness, rated it as easy to cope with, and were married and had better QOL survived longer than those showing the opposite profile (Butow et al. 1999). One could interpret these results as suggesting that patients who were quite optimistic about their treatment's effects by perceiving it as curative, whose anger may have motivated them to be active about their condition, but who also put it in perspective (minimization) and could thus still maintain a better QOL, survived longer. This study is important as it reveals the complexity and richness of psychosocial factors which may be involved in cancer prognosis, as well as the simplicity of assessing one prognostic variable, namely, the perceived aim of the treatment. The results can guide interventions.

One positive resource variable, known to be related to multiple positive mental health outcomes, is self-regulation. This is the capacity to regulate one's emotions, stress responses, and behavior, to better adapt or achieve goals, greatly resembling the concept of emotional regulation. One unique study examined the prognostic role of self-regulation and of autonomic regulation, in breast and colon cancer. Self-regulation was assessed by a valid scale related to goal achievement and the ability to achieve well-being, while autonomic regulation was assessed by symptoms which reflected sympathetic dysregulation (e.g., dizziness), rest regulation, and digestive symptoms. The study found that self-regulation significantly predicted survival in a combined patient sample, independent of cancer severity, age, comorbidity, and other confounders (Kröz et al. 2011). This is an important progress in the literature as it examines the prognostic role of a resilience factor, particularly self-regulation. However, autonomic regulation was assessed by a self-reported scale, rather than by an objective measure such as heart rate variability (HRV), which directly reflects vagal autonomic regulation and psychophysiological resilience as indexed by multiple systems (e.g., Kuo et al. 2005; Weber et al. 2010).

An important aspect of appraising cancer and its fate is the perception of control. One important study did a deep analysis and assessment of control and examined its prognostic role. The results were complex. While higher perceived control 8 months after diagnosis predicted a lower risk of cancer recurrence, a higher desire to be in control predicted a higher risk of recurrence. Importantly, using an "accepting mode of control" diminished the negative effects of desire for control on recurrence risk (Astin et al. 2013). It is possible that desire for control represents a more hostile personality type (Green et al. 2005), which as mentioned in the previous chapter, has negative (cardiac) health effects (Chida and Steptoe 2009). In contrast, perceiving control over what one accepts and perceives to have control over may then lead to a more realistic active coping pattern, and such coping was found to predict positive cancer prognosis as mentioned above (Faller and Bülzebruck 2002).

Another related and important variable in behavioral medicine is self-efficacy, the perception that one can perform or achieve a certain outcome, despite the existence of barriers. One study found that physical self-efficacy significantly predicted higher survival chances, in patients with head and neck cancer (Boer et al. 1998). This is important in the context of cancer since the pain and physical disabilities related to the disease and treatment can indeed challenge one's physical abilities. Thus, such a result has important implications for psycho-oncology interventions.

What about social support? In a review of prospective studies across types of cancer, social support predicted better prognosis. This relationship was significant in methodologically better studies as well (Nausheen et al. 2009).

In Chap. 3, we learned about our recent study on empathy and cancer survival. I will repeat it and expand a bit its findings here, with the readers' (silent) permission. French patients with lung cancer assessed their physician's empathy, and we examined whether empathy predicted survival. However, empathy did not predict survival. Surprisingly, however, empathy interacted synergistically with the content of their consultation (receiving bad vs neutral news). Only when they received bad news (reflecting poor prognosis) did physicians' empathy significantly and positively predict risk of death, independent of multiple confounders (metastatic status, gender, performance status, etc.). In contrast, in patients receiving neutral news, physicians' empathy did not predict survival. When looking at what type of empathy accelerated the impact of bad new, it was the emotional empathy (listening, being compassionate) but not the empowering empathy (Lelorain et al. 2018). In further (unpublished) analyses, we found that patients receiving bad news had significantly higher levels of hopelessness and 2 months later also had significantly higher levels of inflammation (C-reactive protein and neutrophil/lymphocyte ratio) than those receiving neutral news, only if doctors were high on empathy. This tells us how complex behavioral medicine is and that the possible consequences of psychosocial factors cannot be overlooked as these interact with prognostic factors to influence psychobiological responses, which predict mortality. Prudence must be practiced when facing and treating patients, especially when breaking bad news. These results also reveal to us the importance of considering the context in which certain messages are given to patients in medicine.

Finally, quality of life (QOL) is often used as a clinical outcome in medicine in general and in oncology specifically. While health-related QOL is not strictly speaking a psychological measure, it often includes psychological components (e.g., emotional and cognitive functioning), and it is usually assessed by patients. Numerous studies examined the prognostic role of QOL in cancer, but one comprehensive study did so separately in many cancers. Quinten et al. (2014) administered to $n = 7417$ cancer patients the QLQ-C30 QOL questionnaire (Aaronson et al. 1993). Analyses were statistically adjusted for age, gender, and performance status and done separately as function of metastatic status. I will mention a few selected examples of the results. Specifically, results revealed that in brain cancer, cognitive functioning predicted survival, while in breast cancer, physical and emotional functioning, global health status and nausea, and vomiting predicted survival. In colorectal cancer, physical functioning and the symptoms of nausea, vomiting, appetite loss, and pain predicted survival. Finally, concerning three very fatal cancers, in lung cancer, physical functioning and pain predicted survival; in ovarian cancer, nausea and vomiting predicted survival; and in pancreatic cancer, global health status predicted survival. This is a crucial study because it included a large sample and examined the prognostic role of an important outcome, namely, QOL, in many different cancers, using the same QOL instrument, assessed by patients themselves (Quinten et al. 2014).

Figure 5.1 depicts the results of these studies, on Taylor's (1995) model of stress and adaptation. Here, the stressor is the diagnosis of cancer. The figure mentions the main prognostic factors reviewed here, but of course other factors found in other studies can be added.

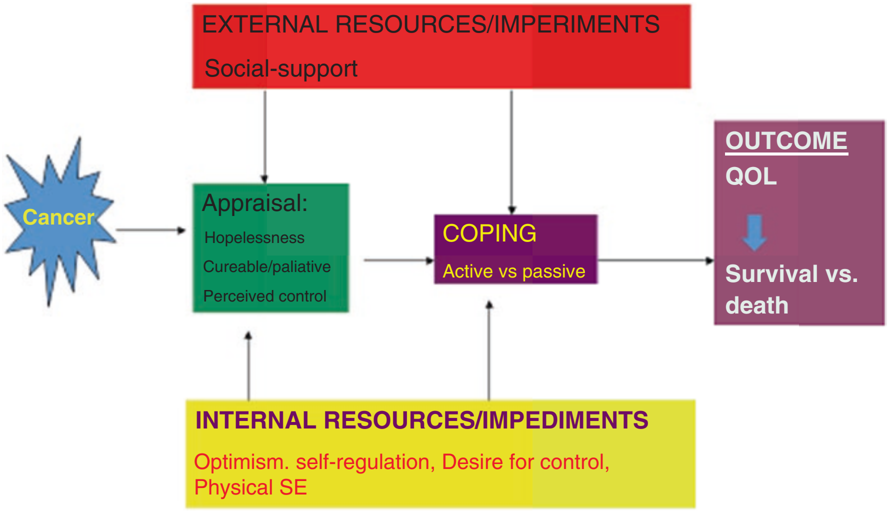

Fig. 5.1 Mapping reviewed studies of predictors of cancer prognosis on Taylor's framework of stress, appraisal, coping and adaptation (Taylor 1995)

Psychoneuroimmunology of Cancer

This topic will be addressed in three manners. I shall provide a general account based on two reviews on this topic (Antoni et al. 2006; Gidron and Ronson 2008). Then, I will provide a detailed example of one possible immunological mediator linking a psychological factor with cancer prognosis. Finally, I shall provide a newer account on the role of the vagus nerve in cancer, due to its emerging relationship with both crucial factors in cancer prognosis and with psychological factors.

Antoni et al. (2006) proposed that psychological prognostic factors such as chronic stress and depression are linked to the activity of the sympathetic adrenal medulla (SAM) system and its effector noradrenaline and to the hypothalamic pituitary adrenal (HPA) axis and its glucocorticoid effector hormones such as cortisol. Subsequently, stress may lead to DNA damage and to reduced DNA repair mechanisms, crucial in the initial steps of tumorigenesis. Stress hormones also lead to increase in vascular endothelial growth factor (VEGF), a main contributor to angiogenesis – growth of blood vessels to tumors. VEGF also negatively affects T-cells and dendritic cells, thus possibly affecting immune surveillance of cancer and anticancer immunity. Importantly, adrenaline and noradrenaline were found to increase metastasis via increasing the expression of matrix-metalloproteinase 2 and 9 (Sood et al. 2006). These factors help tumor cells break through the micro-environment to colonize in new metastatic sites. Furthermore, recent evidence also points at the role of the sympathetic nervous system in increasing expression of genes which promote inflammation on one hand and inhibition of anticancer immunity on the other. Sympathetic activity also increases the expression of genes which promote angiogenesis and metastasis (Cole et al. 2015). Concerning glucocorticoids, such stress hormones can increase the expression of survival genes in cancer cells, protecting them from effects of chemotherapy (Herr et al. 2003). Finally, multiple studies found that mental distress and chronic stress are related to or experimentally induce reduced natural killer cell cytotoxicity, and this predicts poor prognosis in several cancers (Sephton et al. 2000). Together, these mechanisms can explain the epidemiological associations between various psychosocial factors and cancer prognosis, reviewed above.

Argaman et al. (2005) looked more closely at one psychological prognostic factor, namely, helplessness-hopelessness (HH), and tried to find a possible mediator between HH and cancer prognosis, based on existing evidence. The novelty in their approach was to use three systematic hierarchical criteria for identifying a mediator: the mediator is related to HH at the brain level, the mediator is related to cancer systemically, and the mediator contributes to cancer prognosis at the in situ tumor level. They proposed that interleukin-1 (IL-1) fulfils these criteria. As mentioned above, HH has an independent prognostic role in several cancers (e.g., Jensen 1987; Watson et al. 2005). First, at the brain level, central IL-1 mediates manifestations of helplessness since blocking it prevents mice from developing learned helplessness (Maier and Watkins 1995). Furthermore, elevated brain IL-1 promotes peripheral metastases (Hodgson et al. 1999), reflecting a strong evidence for brain to tumor effects. Second, peripheral IL-1 is increased in response to uncontrollable stress (Nguyen et al. 2000), the hallmark of HH, and peripheral IL-1 predicts prognosis in cancer (Wu et al. 2016). Third, at the tumor microenvironment level, IL-1 contributes to escape from apoptosis and cellular immortality (Arlt et al. 2002) to angiogenesis (Torisu et al. 2000) and to tumor invasion and metastasis (Garofalo et al. 1995). Together, these propose that IL-1 fulfils the three criteria set for a plausible mediator between HH and poor cancer prognosis. If correct, this has clear therapeutic implications, and future studies need to examine whether reducing IL-1 can improve the prognosis of cancer patients high on HH. Recently, we observed a new related result, on the positive side of things. French patients with lung cancer were assessed for hope about their prognosis, with one item, as well as for inflammation, by C-reactive protein (CRP). Hope statistically moderated (cancelled) the prognostic role of CRP: only in patients without hope, did CRP significantly predict risk of death, but CRP did not predict death in patients with hope, independent of confounders (Gidron et al. 2018). This finding suggests that psycho-oncologists need to focus on increasing hope in high-hopeless patients, particularly in those with high CRP.

The third emerging biological mediator is the vagus nerve. The vagus nerve could also play an important role in mediating relationships between psychological factors and cancer prognosis. First, vagal nerve activity, indexed by heart rate variability (HRV; Kuo et al. 2005), is inversely related to most negative psychological prognostic factors in cancer such as hopelessness and depression (e.g., Schwarz et al. 2003). At the level of the brain, HRV is positively correlated with activity in the ventromedial prefrontal cortex (Thayer et al. 2012), a region which prevents development of helplessness in animals (Amat et al. 2008). Furthermore, based on converging evidence, Gidron et al. (2005) and De Couck et al. (2012) proposed that the vagus nerve may have protective roles in cancer because it inhibits oxidative stress (Tsutsumi et al. 2008), sympathetic activity (Saku et al. 2014), and inflammation (Tracey 2009). In cancer, these three biological factors otherwise contribute critically to cancer onset and prognosis (e.g., Entschladen et al. 2004; Mantovani et al. 2008; Valko et al. 2004). Concerning cancer progression, other groups and ours have shown repeatedly that high HRV near cancer diagnosis predicts longer survival or reduced tumor marker levels, independent of cancer stage and treatment (e.g., Chiang et al. 2013; Mouton et al. 2012; De Couck et al. 2013; Zhou et al. 2016). All these also have important clinical implications, since perhaps high-HH patients with cancer may particularly benefit from vagal nerve activation. This is currently under investigation for cancer patients in general, by our research group. Indeed, Erin et al. (2012) found in a pioneering study that an anti-inflammatory drug, CNI-1493, which mostly depends on the vagus for its action, reduced metastasis in a mouse model of breast cancer. Future studies will continue investigating the possible protective role of the vagus nerve in cancer.

Let's look at social support more closely. As mentioned above, little social support predicts poor cancer prognosis, independent of confounders (Nausheen et al. 2009). How may this happen? Studies show that oxytocin (OT) may play a role in the biological underpinnings of social support. First, OT is positively correlated with levels of social support (Grewen et al. 2005), and intranasal OT interacts synergistically with social support in reducing subjecting and objective stress responses (Heinrichs et al. 2003). Importantly, in several, though not all cancers, OT has an antiproliferative role (e.g., Thibonnier et al. 1999). Using three in vitro cancer cell line models, we recently found that administering OT reduced cell proliferation and migration, induced apoptosis, and partly reversed the tumor-promoting effects of cortisol (Mankarious et al. 2016). Taken together, OT may mediate the protective role of social support in cancer. If administering OT to cancer patients is safe, future studies need to examine the effects of OT on the prognosis of cancer patients with little social support. Figure 5.2 depicts the major elements of the psychoneuroimmunological paths in cancer prognosis.

Perhaps the most remarkable novelty in the PNI of cancer prognosis is the discovery of the association between psychosocial adversity and a cluster of genes termed the conserved transcriptional response to adversity (CTRA) gene group. The CTRA reflects on one hand increased genes signaling pro-inflammatory cytokines (e.g., IL-1b, IL-6; IL-8) and a reduced gene expression of antibody and antiviral immunity (e.g., Interferon-beta) on the other hand. CTRA has been related to various types of social adversity (e.g., Cole et al. 2012). This precise pattern of gene expression related to immunity may be especially oncogenic because the inflammatory response promotes tumors (e.g., Mantovani et al. 2008), while interferons can activate cytotoxic T-cells to fight tumors, whose in situ presence predicts good prognosis (e.g., Galon et al. 2006). Furthermore, the focus on a group of genes related to several aspects of immunity reflects better the complexity of the immune system in relation to cancer, rather than focusing on one cytokine. Antoni et al. (2016) performed a reanalysis of the RCT by Stagl et al. (2015) mentioned below, which

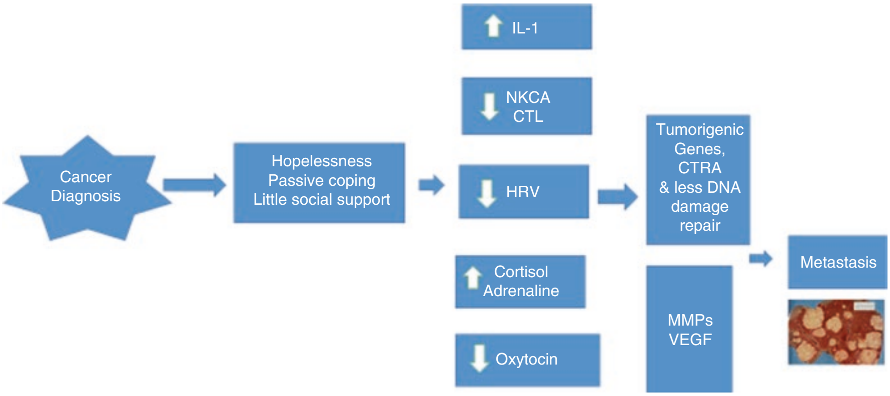

Fig. 5.2 Psychoneuroimmunological paths in cancer prognosis

examined the effects of cognitive-behavioral stress management (CBSM) on women with breast cancer, followed for 11 years. Astonishingly, while CBSM led to slight reduced CTRA, controls evidenced sharp increases in CTRA. Importantly, greater elevations in CTRA over 6–12 months strongly predicted a shorter disease-free survival in these women with breast cancer (Antoni et al. 2016). These researchers must be congratulated for revealing this crucial mechanism because it points at genetic changes which reflect crucial cancer-related immunological changes, due to improving patients' psychological situation. This has profound implications for understanding psycho-oncological processes and has vast scientific and clinical implications that physicians, psychologists and cancer biologists must learn and consider.

Adopting a Cognitive-Perceptual Paradigm for Understanding Cancer-Immune Interactions

Cellular immunity, particularly natural killer cells and cytotoxic T-cells (CTL), combats tumor cells. Furthermore, the amount of T-cells in the tumor microenvironment is positively predictive of survival from cancer (Galon et al. 2006). However, the immune system is challenged when facing tumor cells since the latter are self-originating rather than foreign cells and due to immune ignorance: at times, cancers develop in organs devoid of immune influences (e.g., testes, ovaries, the brain; Pardoll 2003; Bubeník 2005). Furthermore, the immune system may develop tolerance toward mutated cells because tumor cells often reduce their expression of MHC (thus complicating their recognition by T-cells), because they weaken their tumor-associated antigenicity (immune editing) and since cancer cells produce cytokines such as transforming growth factor beta (TGF-b) and PDL-1, which suppress anticancer immunity (Bubeník 2005; Dong et al. 2017; Pardoll 2003). These all constitute a “camouflage” and escape of cancer cells from anticancer immunity.

However, at times, by looking at a complex problem in one scientific domain via a paradigm from another domain, one can bring new insights and simplify communication between clinicians and researchers. A few years ago, I was fortunate to experience nearly total “scientific ignorance,” when attending a cancer biology conference in Crete. However, I used the glasses I had at the time, trying to make sense of the immensely complex world of cancer biology. Out of my ignorance, I developed with a prudent tumor immunologist, Prof Luca Vannucci, a model adopting the figure-ground perspective from cognitive-perceptual sciences to understand the interactions between cancer and the immune system during oncogenesis, a perspective which also has implications for immunotherapy. The figure-ground perspective tries to explain how we distinguish one stimulus from other similar ones. For example, how do we notice a frog or a lizard which are hiding from dangers by use of amazing camouflage, by resembling to similar leafs and stones in their environment? This is a situation of figure-ground, in which the degree of spatial impenetrability of elements in a visual scene inhibits grouping of boundaries, and includes a process of feedback between mentally represented boundaries (“how things look like”) and observed surfaces arriving actually at one’s retina. All these enable detection of a visual stimulus in a surrounding environment (Grossberg and Pessoa 1998). In the figure-ground arena, there are three players: the figure (the sought stimulus), the background (visually similar and camouflaging environment), and the viewer, who may have a mental representation of the figure from past exposures to it. Cohen (1992) intelligently described the immune system as a cognitive model because it can be “primed” by previous exposure to antigens, it has memory (a learned and preserved immune pattern defense), and it has a “mental representation” of itself and of the antigenic world it has been exposed to (manifested by one’s selected pathogen receptors and available immune memory). We took this paradigm of immunity as a cognitive system a step further by mapping the figure-ground model on the steps of cancer development. In carcinogenesis, the figure is the initial transformed (mutated) cell/s, and the background is initially made from healthy surrounding cells and the stroma. In this system, the viewer changes and is initially the local innate immunity (monocytes and neutrophils arriving at the scene). Eventually, cells of acquired immunity are recruited such as T-helper and CTL cells. However, due to the challenges mentioned above (e.g., reduced MHC expression, increased TGF-beta), anticancer immune cells can fail to “see” tumor cells. Tumor cells could then proliferate and attract more innate immune cells, forming an inflammatory response. Then, systemic immunity, the new viewer, could shift from an antitumor Th1 immune response to an anti-inflammatory Th2 immune response, which tragically favors cancer progression, due to this shift.

How can the figure-ground model help to change this unfortunate sequence of events? To detect the frog better, we could either pale its camouflaging background, increase the visibility of the frog, or sharpen the mental representation of the viewer concerning how that particular frog looks like. Indeed, Vannucci et al. (2008) found that rats given cancer in a germ-free environment (paling the inflammatory noise often existing in the gut and the thus reducing antigenic competition) had a stronger anticancer immune response and subsequently developed less cancer, than similar rats given the same tumor in a regular germ environment. Indeed, many studies show that taking anti-inflammatory medication (paling the inflammatory background) reduces tumor incidence (e.g., Jalving et al. 2005). Alternatively, several ways exist for increasing the visibility of the tumor. Thermotherapy, heating the tumor cite, led to larger NK cell activity and to more “chaperoning” of tumor antigens to dendritic cells, which then present these antigens to T-cells for tumor eradication (Ostapenko et al. 2005; Paulus et al. 1993). Furthermore, providing cancer-bearing animals interferon alpha (IFN-a) increases the re-expression of MHC-1 on tumor cells (Yang et al. 2004), thus increasing their antigenicity (visibility) for the anticancer immune system viewer (T-cells). Finally, some studies have found that vaccinations, which “educate” the viewer how tumor antigens “look like,” have had some promising results (e.g., Mataraza and Gotwals 2016). More importantly, some studies found that *combining* various biological treatments has a stronger effect on reducing tumor progression than each treatment alone, and these combinations are precisely in line with a figure-ground perspective. Duff et al. (2003) found that rats given both an anti-inflammatory agent + IFN-gamma had smaller tumors than those given each alone or no treatment. Together, these findings propose that steps taken in accordance with the figure-ground model, to pale the tumor background, enhance its figure antigenicity, and sharpen the viewer’s “representation” of the tumor figure via vaccines, could work synergistically in favoring anticancer immunity toward combating existing tumors (Gidron and Vannucci 2010). Future studies need to examine this model more in depth and identify its limits of veracity. If correct, it may increase the effectiveness of immunotherapy which is limited due to all the challenges mentioned above.

Psychological Interventions for Cancer Patients

Having reviewed multiple studies and reviews showing that psychosocial factors predict cancer prognosis as well as studies proposing the neuroimmunological mechanisms linking such factors with cancer prognosis, the next logical step is to examine whether psychological interventions can improve cancer prognosis. When talking about prognosis, this usually means “hard” outcomes such as survival, disease-free survival, or tumor size. However, this has become a very controversial topic due to several reasons. First, as shown below, the empirical evidence on this topic is inconsistent, and second, numerous methodological issues impede serious progress in this important topic. These shortcomings include issues such as limited sample size and not calculating the required sample size before the study in relation to prognostic outcomes. In addition, many studies include all patients consenting to take part, often without any screening for levels of a psychosocial risk factor such as hopelessness or little social support. This can lead to floor effects. We must not assume that every cancer patient is hopeless or depressed. Finally, many of the interventions are non-specific and do not target psychological factors which predict cancer prognosis (Coyne 2008). It is important to note that this debate has gone far beyond scientific issues, into more inter-researcher intrigues, which I do not support at all. Another critical review identified 627 intervention studies, which were analyzed by 10 methodological quality criteria. Importantly, only ten studies had survival or immunological measures as their outcomes, and it identified at the time only one study which received a "good" score on methodological rigor (Newell et al. 2002). This sadly points at the poor methodology in this topic of psycho-oncology, which needs to be corrected. An earlier systematic review found only eight studies meeting specific scientific criteria (most were randomized controlled trials), which tested the effects of psychological treatments on cancer prognosis (Edelman et al. 2000). Only three of the eight studies (37.5%) found that psychological treatments affected survival time. In contrast, a more recent review showed that 8 of 15 trials (53.3%) found a survival benefit from psychological treatments in cancer (Spiegel 2012). The contrasting results of these reviews are evidence of large inconsistency in the scientific literature. A recent review examined 28 randomized controlled trials done specifically in women with non-metastatic breast cancer. Evidence for benefit was only found in relation to reducing mental distress but not in increasing survival, and the quality of evidence was not satisfactory (Jassim et al. 2015). The most recent meta-analysis questioned previous ones about the inclusion of trials and on the statistical metric used and recommended to use only the hazards ratio for comparing RCTs. They included trials in which the control group received usual care only and identified 12 RCTs with 2439 patients in total. Results revealed that psychological interventions had an overall significant positive effect on survival time. This effect was mostly observed in trials with fewer married patients, in those older than 50, who mostly received cognitive-behavioral therapy at an early cancer stage (Mirosevic et al. 2019). Finally, Chong Guan et al. (2016) identified 17 clinical trials, which focused on psychological outcomes. These trials reduced short-term anxiety (under 12 months) and improved quality of life and coping. We will now provide a few specific examples in relation to "hard" outcomes – prognosis.

In a pioneering randomized controlled trial (RCT), Spiegel et al. (1989) found that supportive and expressive therapy (n = 50) was related to longer survival time in women with metastatic breast cancer, compared to those receiving usual care alone (n = 36). The intervention taught women self-hypnosis for pain reduction, expressing feelings about their illness and how to be assertive and provide social support to each other. While serving as a milestone study, it did not provide an intervention which focused on a specific prognostic factor of cancer, and it did not include only patients scoring high on any psychological prognostic factor. Nevertheless, this was a study of major influence in psycho-oncology since it was the first to propose that psychological interventions may improve cancer prognosis, using an RCT design.

Another RCT was done in patients with primary stage melanoma. Here, a psycho-educational intervention was provided to 38 patients, while 28 served as controls. The intervention included health education, learning relaxation, problem-solving, and support, resembling current stress management interventions. Improvements in immune measures related to cancer (NK cells), psychological outcomes (e.g., active coping) and survival, were seen in the experimental group compared to controls (Fawzy et al. 1993). However, the same limitations seen in the previous study were seen here as well. In addition, the improvements on the immunological and psychological factors were unrelated to the survival benefit, questioning the mechanisms and underlying assumptions of the study. This issue raises questions including whether simply teaching people stress management strategies, which improve psychological and immunological outcomes, can affect cancer prognosis, and why? It is also possible that other mechanisms, not measured in that study, such as increased vagal activity or reduced inflammation, could partly explain their intervention's effects.

One older study by Morgenstern et al. (1984) compared patients, who attended with their families a psychological program, to a control group of patients matched on age at diagnosis, stage, surgery, and sequence of malignancy. The intervention included guided imagery, meditation, and group discussions. Initially, the psychological group showed a survival benefit; however, this advantage was fully mediated by and did not remain significant after controlling for the time from cancer diagnosis to seeking psychological help, which differed between groups. Though using a matched controlled design, the main flaw of that study is its lack of randomizing patients. Nevertheless, this study was unique in that it included family members in the intervention, which often have significant distress from the cancer of their relative.

Similarly, Edelman et al. (1999) randomized patients with metastatic breast cancer to cognitive-behavioral therapy (CBT; n = 60) or to usual care only (control; n = 61). The CBT included thought monitoring and restructuring, communication skills, and relaxation. Improvements in depression and self-esteem were seen in the CBT group, but these were not maintained at follow-up, and no survival benefit was found in a 5-year follow-up. Here too, patients were not explicitly screened on any psychological prognostic factor as an inclusion criterion.

Two important RCTs were performed around 2007–2008. First, Spiegel attempted to replicate his early trial by including a larger sample ($n = 125$). Surprisingly, women with metastatic breast cancer, who received supportive and expressive therapy, were not found to survive longer than those in usual care alone. However, a post hoc analysis indicated that there was a strong survival benefit of the psychological intervention only for women with an estrogen receptor (ER)-negative type of tumor (Spiegel et al. 2007). Though partially with negative findings, this study revealed an important issue: psychological treatment (as is the case with medical ones) may be effective for subtypes of patients, depending on the type of cancer they have. As the authors explained, advances in chemotherapies for women with breast cancer together with little advancement for those with ER-negative cancer could have explained their pattern of results. Future studies should examine additional reasons (e.g., hormonal) which may account for the specificity of such observed effects.

Andersen et al. (2008) randomized 227 women with breast cancer to a psychological intervention or to assessment only. Patients were followed for a median follow-up period of 11 years. The psychological intervention aimed to reduce distress and improve positive health behaviors, quality of life, and patients' adherence to treatment. The intervention included progressive muscular relaxation, problem-solving, support seeking, assertiveness training, strategies to improve one's diet, and finding ways to help cope with treatment side effects. This intervention led to a significant increase in overall survival, recurrence-free and breast-cancer-specific survival, compared to controls. This was a large-scale RCT which is one of the only ones to perform a sample size-power analysis concerning effects on recurrence, prior to the study. It is possible that a combination of a highly powered sample size together with a psychological intervention which aims to improve psychological factors and adherence to treatment contributed to this trial's success. Future studies need to extend this crucial work to other cultures and cancers, and examine its generalizability.

Perhaps one of the most recent studies in this domain has been the study by Stagl et al. (2015). In this RCT, $n = 240$ women with non-metastatic breast cancer received either cognitive-behavioral stress management (CBSM) or a psycho-education control intervention. The CBSM included reappraisal techniques, coping strategies, assertiveness training, and relaxation skills. In a median follow-up of 11 years, all-cause mortality was significantly lower in the CBSM group than in controls, independent of biomedical confounders. This study was well-powered and included a structured intensive intervention, both which may have contributed to the positive findings in this RCT. As mentioned above, Antoni et al. (2016) demonstrated elegantly that the gene cluster CTRA accounted for the observed affects, where controls evidence an increase in CTRA, while the CBSM group showed no worsening in this marker of high pro-inflammatory cytokines and low antiviral immunity. This entire line of research by that group of researches demonstrates the methodological and scientific elements that are needed for conducting such important research in psycho-oncology.

Nevertheless, the specific studies reviewed here, together with the reviews mentioned above, provide evidence for an inconsistent picture at best. Future RCTs need to plan and then conduct more a priori moderator analyses to highlight for whom psychological treatments may be beneficial. This could include consideration of baseline psychological symptoms or subtypes of cancer (e.g., Spiegel et al. 2007). Furthermore, to the best of my knowledge, no RCT specifically targeted at the inclusion criteria and in the contents of the intervention, psychological factors of prognostic roles in cancer, such as hopelessness or little social support. These issues together with pretrial planning of sample sizes in relation to survival may help the field of psycho-oncology progress. Finally, we may also consider testing in future studies biological interventions for patients high on a psychological prognostic factor, such as IL-1 reduction or HRV biofeedback for high-hopeless cancer patients. This would place behavioral medicine in the mainstream work of medical sciences and highlight the crucial evidence-based interactions among the biological and psychological levels in cancer.

Case Study Conclusion

Ms. Janssen clearly shows signs of hopelessness, since she perceives to have no control over her life and is pessimistic about her future. However, her levels of HH need to be assessed with a standardized scale, quantitatively, to be sure and to monitor their change following psychological help. In addition, it is possible that a combination of brief CBT with psychological inoculation (PI) focusing on her lack of perceived control may help reduce this response and increase her daily functioning and quality of life. These could also have a positive effect on her levels of perceived and actual social support. Whether these can actually improve her prognosis remains at present time unclear. An example of a challenging sentence for the PI about lack of control could be:

Therapist: "Please refute fully my following incorrect sentence!" "Ms. Janssen, I am sure you have no control over which clothes you wear in the morning, because of your illness!"

Ms. Janssen: "No, no, it is only me who decides what I wear, doctor!"

Therapist: "You rejected my sentence very well, please continue that way…".

In addition, teaching her to perform HRV biofeedback could calm her down; increase her vagal nerve activity, which may improve her prognosis given this nerve's prognostic value (Zhou et al. 2016); and also activate the ventromedial prefrontal cortex (Thayer et al. 2012), which may help to reduce her helplessness (Amat et al. 2008).

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Chapter 6
Psychological Risk Factors of the Common Cold and Its Behavioral Consequences

Case Study

> Mr. Jackson is a 42-year-old man who works in an auditing firm which does auditing for 57 companies in New York. He comes to your GP clinic and complains about having a 4th cold this winter. You order multiple blood tests concerning immune measures, liver functions, and inflammation, as well as inquire about his nutrition. He explains that he eats meat three times per week and never has fruit. Then he tells you that he works 62 hours a week, including many weekend days, gets home at around 20:00, sees his small children for only half an hour, and has numerous conflicts at work and at home with his wife. These have lasted for six months. You wonder what may be the cause of his repeated colds. The lab tests only showed abnormal levels of low lymphocytes.

The Incidence, Consequences and Pathophysiology of URTI

Upper respiratory tract infections (URTI) are among the most common type of illness from which, in an accumulated manner, people suffer in total between 1 and 2 years of their life span (Rowlands 1995). This constitutes an incredible amount of illness and health and economic burden in a person's life. An URTI includes rhinorrhea (runny nose), nasal congestion, and sneezing, which could be accompanied by fever and sore throat and feeling ill (Van Kempen et al. 1999). While preschool children can have 6–12 URTI a year, adolescents and adults have on average 2–3 episodes per year (Turner 1997). One study estimated the prevalence of flu, the more severe form of URTI, to be at 20% per year in adults (Gross et al. 1995). The vaccine against flu can be quite effective; however, this depends on one's age. In adults below 65 years of age, 70–90% will be protected from flu via a vaccine; however, this

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number dramatically declines to 30–40% of protection by a vaccine in people above 65 years of age, even if there is a good match between the vaccine and the virus (Fukuda et al. 1999). Though an URTI is usually time-limited and does not constitute a severe illness, it can impair one's functioning and has a tremendous economic cost. In the USA alone, URTI are responsible for 26 million days of school absence in children and 23 million days of work absence in adults (Turner 1997). Importantly, the initial infection is often followed by a bacterial invasion which is responsible for complications from the URTI such as sinusitis, otitis media, or pneumonia, where the latter could be lethal in some people (Van Kempen et al. 1999). For example, in the influenza URTI, coinfection by *Staphylococcus aureus* may result in death in up to 42% of elderly people (Brooks et al. 1991). The pathogen responsible for the URTI varies with seasons and geographic location. However, rhinoviruses are thought to cause up to 80% of the common colds, especially in the autumn (Rossmann and Palmenberg 1988).

A main way in which the rhinovirus attacks the human body is by binding to its main binding receptor – intercellular adhesion molecule 1 (ICAM-1). In 90% of the times, this occurs in the adenoids (Greve et al. 1989). Thereafter, it is uncoated and releases its viral RNA for replication. Following this replication, it will spread to the nose and pharynx. The rhinovirus spreads from person to person via direct and indirect contact, mainly by infected respiratory secretions. Hands are the main vehicle of transmission since using hand-cleaning products reduces transmission and also because the virus can maintain its virulence for 3 days on plastic surfaces. Furthermore, contaminated hands touching the eyes or nose render an efficient route of infection (Winther et al. 1984).

The host responses include inflammation and immunological and neurological changes. In the first days post infection, an increase in neutrophil count can be observed (Levandowski et al. 1988). In contrast, the response of lymphocyte counts to colds has been found to be inconsistent (Van Kempen et al. 1999). However, the typical symptoms seen in the URTI are orchestrated by inflammatory cytokine reactions to the infection. These include a Th-1 immune profile with increased natural killer cell activity (NKCA); interferon gamma; interleukin (IL)-1, IL-6, and IL-8; and immunoglobulins (Johnston et al., 1993). The cytokines recruit and activate neutrophils at the infected site (Van Kempen et al. 1999). IL-6 plays a pivotal role in the symptoms and in the immunological battle against URTI-inducing agents, since it helps mature B-cells to produce antibodies and activates T-cells as well (Van Kempen et al. 1999). Activation of T-cells could potentially serve both B-cell and cytotoxic T-cell activation. Finally, there is a correlation between symptom severity and IL-6 levels (Zhu et al. 1996). Importantly, rhinoviruses upregulate the expression of their binding receptor ICAM-1 to cells via cytokines (Van Kempen et al. 1999). The cytokine IL-1b plays a major role in the host response to the infection. First, levels of IL-1b were higher in symptomatic infected people compared to infected but asymptomatic people and to non-infected controls (Proud et al. 1994). Furthermore, IL-1b increases recruitment of immune effector cells to the infected site via increasing ICAM-1 expression. However, by doing so, it also helps the rhinovirus spread more easily by adhering more easily to cells via this receptor

(Van Kempen et al. 1999). One more recent study on experimentally induced URTI found that initial reduced telomere length of immune cells, particularly of CD8-CD28 T-cells, significantly predicted increased risk of objective URTI and subjective cold symptoms (Cohen et al. 2013). This is important since, though the host response plays a role in the inflammatory response to URTI infections, basal integrity of cellular immune cell functioning may be crucial for prevention of having the illness despite being infected. Another important player in antiviral immunity and in fighting URTI is the natural killer (NK) cell. Mice with a hyperactivity of NK cells were found to survive more when exposed to a lung virus (Narni-Mancinelli et al. 2012). During URTI induction, NK cells were found to be activated by Type 1 interferons, which then led NK cells to produce interferon gamma (IFN-g) and granzyme-b (Hwang et al. 2012), both crucial in the lysis of target virally infected cells by NK cells.

Psychological Risk Factors of the URTI

This topic has perhaps the most compelling evidence for the relationship between psychological factors and an illness. The work of Prof. Sheldon Cohen has been pioneering and inspiring in this field and, in my humble opinion, has helped to put behavioral medicine on the world map. The first groundbreaking study was that of Cohen et al. (1991) in which nearly 400 people were experimentally infected by URTI viruses. Before infecting the participants, the investigators assessed negative affect, perceived stress and major life events, and converted these parameters into one stress index. They also considered 18 confounders including baseline immunity, lifestyle factors, and depression. A nearly perfect positive linear association was found between the stress index and actual risk of developing the URTI later, independent of these many confounders (Cohen et al. 1991). In a following study, they examined which type of stressors predicts the URTI. Using a similar experimental paradigm, it was found that chronic stressors (>30 days of duration), and particularly those involving interpersonal or job stressors, predicted higher risk of colds, rather than more acute stressors, lasting less than 1 month (Cohen et al. 1998).

What about personality and affective style? Negative emotional style (NES) reflects neuroticism and the tendency to attend to and perceive, experience, and report negative affect and body symptoms. In contrast, positive emotional style (PES) reflects the general tendency of being active, being interested, and having a positive mood. Who develops a cold, people high on NES or people low on PES? Another study by Cohen et al. (2003) examined the relationship between these affective personality styles and colds. They found that people low on PES were at greater risk of developing an objective URTI and reporting subjective cold symptoms. In contrast, people high on NES were only at risk of *reporting* cold symptoms but not for actually developing the objectively measured URTI. This is an important finding for behavioral medicine since, often, people with high NES, who are highly neurotic, over-attend health clinics but may often have little objectively defined diseases, and they may live as long as people low on NES (Costa Jr and McCrae 1987). This result could be because they are being medically tested and cared for to a greater extent.

What about other mood factors? Kim et al. (2011) assessed depressive symptoms in $N = 1350$ South Korean workers, who then underwent a self-reported assessment of experiencing the common cold during a 4-month follow-up. In both males and females, depression was predictive of a significantly higher risk of reporting having the common cold, independent of multiple confounders such as smoking, alcohol consumption, marital status, and job type. Though based on a self-reported measure of colds, this study included a large sample and a prospective design, statistically controlled for confounders and extended the research mostly done in western countries to the Far East.

Other research teams have also revealed important relationships between psychological factors and the risk of the URTI. In one complex study focusing on job stress, the role of the job-demand control model and self-efficacy in the risk of the URTI was examined. This model contends that jobs including high job demands (e.g., deadlines, many working hours) and little job control (e.g., not deciding how to do one's work) lead to burnout and risk of physical illnesses (Karasek Jr 1979). However, not all workers respond well if they receive more job control, and this could depend on their will for such control. Schaubroeck et al. (2001) found that job demand was related positively to URTI. However, while high job control weakened the job demand-URTI relationship in workers with high self-efficacy, high job control increased the job demand-URTI relationship in those with little self-efficacy. This study demonstrates that job stressors are also URTI risk factors and that providing more job control may reduce the risk of URTI from job stress only in workers who can benefit from such control, those with high self-efficacy. This study also reveals the complex associations between situational variables, internal resource variables, and illnesses, in this case the URTI.

A more recent study examined the roles of job stress in a large sample of South Korean workers ($N = 1241$). High job demands, low job control, and little social support prospectively predicted increased risk of the common cold during a 6-month follow-up (Park et al. 2010). This study too extends and shows that this model's elements are relevant to health and particularly to URTI also in Far Eastern cultures.

Often, people report becoming ill when finally taking a vacation, termed leisure sickness (LS). This refers to developing sickness during weekends and vacations. Do psychosocial factors predict LS? Within the framework of a large representative Dutch sample, Vingerhoets et al. (2002) conducted a case-control study and found that 3.6% and 3.2% of men reported LS on weekends and during vacations, respectively. In contrast, 2.7% and 3.2% of women reported LS on weekends and during vacations, respectively. Importantly, LS was associated with reporting difficulties in the transition between work and nonwork, with stress associated with leisure traveling, with workload, and with a high sense of achievement and responsibility. While not solely focusing on URTI, LS included URTI, and the latter were frequent during vacations (Vingerhoets et al. 2002). Unlike the other studies reviewed here, this particular one was not a prospective study; however, its topic is unique and often people ponder upon this issue; hence it is mentioned.

Since URTI may have fatal consequences in the elderly (Brooks et al. 1991), it is also important to examine the relationship between psychological factors and the URTI in elderly. We conducted a prospective study among elderly people residing in an old-age home near Tel-Aviv. Most residents of the old-age home used the same health clinic, enabling us to perform this study in a more controlled manner. In total, $n = 70$ elderly people participated in the study. Levels of hostility, depression, social support, and life events were assessed at the study entry, together with other confounders. The attending nurse in the clinic assessed URTI with a valid questionnaire. Hostility and lack of physical activity were independent risk factors of the URTI during follow-up (Gidron et al. 2005). Though URTI was assessed by a questionnaire, it was rated by a nurse, in a prospective study on a socioeconomically homogeneous cohort.

Another important population to investigate is children, since they have high numbers of URTI (Turner 1997) and due to their subsequent high days of school absence. Cobb and Steptoe (1998) examined in a unique study the relationships between life events, social support, and coping at baseline, with URTI assessed over 15 weeks, in 55 boys and 61 girls between 5 and 16 years old. Furthermore, daily hassles and mood were assessed repeatedly during the study. The results revealed that problem-focused coping was inversely related to duration of URTI, while avoidance coping was positively related to duration of URTI. Furthermore, life events significantly interacted with social support such that only in children with high social support were life events positively related to occurrence of URTI, while this did not happen in children with little social support. This form of interaction is not in line with the buffering hypothesis of stress and social support (Cohen and Wills 1985) where one would expect life events to predict URTI only in those with little, not high, social support. An interesting interaction also emerged between daily hassles and avoidance coping: Hassles were positively related to risk of URTI in children with high avoidance, while hassles were inversely related to risk of URTI in children with little avoidance coping (Cobb and Steptoe 1998). These results reveal the importance of considering stressors, social support, and coping, as well as their interactions, in relation to risk of URTI. The patterns of interactions also call for examining such interaction effects more often, to identify the most at-risk individuals and those at least risk. Such findings have possible important implications for public health and prevention via psychological interventions.

Finally, a more recent study conducted by Cohen and his colleagues (2015) found that perceived social support attenuated the association between interpersonal stressors (conflicts) and risk of URTI. Interestingly, this stress-buffering finding was greatly explained by reported hugs as the index of social support. Thus, first, the amount of hugs was on its own inversely related to risk of URTI, and second, daily tension predicted more risk of URTI only in people reporting fewer, rather than more, hugs. This study supports the stress-buffering hypothesis of social support (Cohen and Wills 1985) and identifies a simple, socially acceptable form of social

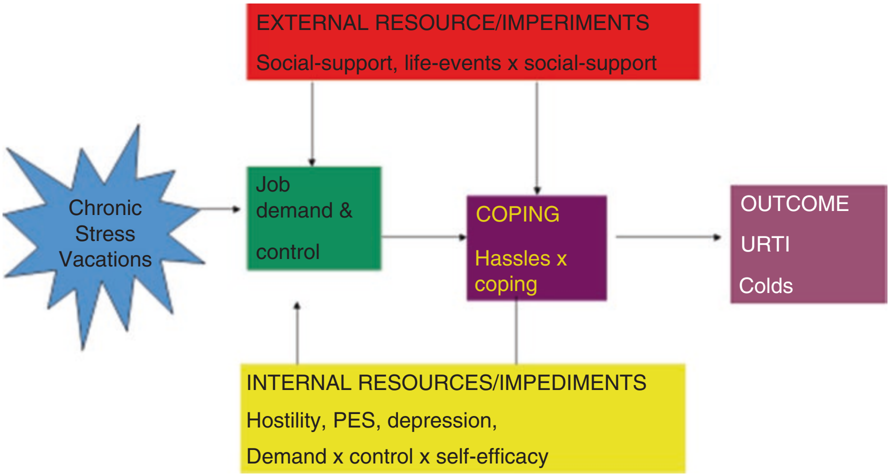

Fig. 6.1 Psychosocial risk factors of colds, mapped on Taylor's model of stress, coping, and outcomes

support, namely, hugging, to be protective in the context of an infectious illness. What a nice way to end this part of the chapter.

In a meta-analysis of 27 identified prospective studies, Pedersen et al. (2010) found a significant association between psychological stress and the risk of URTI (overall effect size correlation of 0.21). Importantly, this association was not moderated by type of stressor or how URTI was measured. Figure 6.1 summarizes the findings reviewed here in relation to the psychosocial predictors of URTI and colds, mapped onto Taylor's (1995) stress, coping, and adaptation framework.

Cognitive-Behavioral Consequences of the Common Cold

Having the URTI has high costs for society since it reduces quality of life (Bucks et al. 2008) and because it is a major cause of job absence with high economic costs (Bramley et al. 2002). Numerous studies have examined the cognitive consequences of the URTI as measured on simple and more complex cognitive tasks. These are typically measured via reaction time (RT) to perform a task according to instructions concerning various stimuli presented on a computer screen. Slower RT has been shown following URTI (e.g., Smith et al. 1993), and impaired memory was shown as well (Capuron et al. 1999). In a case-control prospective study, our team examined the effects of naturally occurring URTI on more detailed cognitive functions and emotional processing. In a sample of $n = 80$ British participants, we initially measured baseline performance on cognitive and emotional processing tasks. Later, 21 participants developed the URTI, and these were compared to controls who remained well, to whom they were matched on age, gender, and IQ. Both ill and matched controls were reassessed for performance. While controls significantly improved on speed of memory and on a face-word label matching task (reflecting prefrontal activity), ill patients showed significantly smaller improvements on those tests (Bucks et al. 2008), reflecting an impaired learning effect possibly due to the cold. In a more recent study, $n = 360$ people were measured for various RT tests, and alertness was self-reported as well. Participants who later developed colds had lower alertness and slower RTs than healthy controls. Furthermore, those who initially reported more negative life events and who later developed the URTI had the worse impairments (Smith 2013). These results show a synergism between stressors and the URTI in relation to reducing alertness and cognitive performance.

PNI Mechanisms of the URTI

Several lines of evidence propose the mechanisms by which psychological factors may be related to the risk of a ^{2}x^{2} URTI. Vedhara et al. (1999) compared people who care for a demented family member (as a model of chronic stress) to matched controls, not in such a situation. As a group, the chronically stressed (caregiving) people had significantly lower levels of immunoglobulin G (IgG) than controls. Furthermore, levels of cortisol were significantly inversely correlated with levels of IgG ($r = -0.216$). Through a cross-sectional study, this finding proposes an immune suppression mechanism possibly induced by chronic stress, via the HPA axis. To remind us, Cohen et al. (1998) found that more chronic stressors predicted the URTI. The findings of Vedhara et al. (1999) echo and may partly be explained by those of Dickerson and Kemeny (2004), who found in a meta-analysis that uncontrollable or interpersonal social evaluative stressors were most strongly related to stress hormonal changes. Caring for a chronically ill person has many uncontrollable and social pressures. Taken together, we may speculate that particularly work and interpersonal stressors, which are often only partly under our control, increase HPA axis activity and cortisol, leading to immune suppression and a higher susceptibility to colds.

Another line of research of relevance to PNI mechanisms comes from our group where we examined whether hemispheric lateralization (HL) predicts the URTI, because left HL is related to immune potentiation, while right HL is related to immune suppression (e.g., Davidson et al. 1999; Sumner et al. 2011). In reanalysis of the study mentioned above, we recruited 80 cold-free British adults and assessed their baseline HL using a neuropsychological test of visuospatial (right) versus verbal (left) recognition memory. Of the 80 participants, 21 subsequently had a URTI during a 2-month follow-up, and they were matched on age, gender, and IQ, to 21 people who did not develop colds at the same time. Indeed, those developing colds during follow-up had significantly higher right HL at baseline, independent of age, gender, and IQ, as well as independent of neuroticism and health behaviors

(Gidron et al. 2010). Since people shift to right hemispheric activation during stress (Lewis et al. 2007) and because right HL is related to mental distress (e.g., Davidson 2004) and to immune suppression (Sumner et al. 2011), such a finding may partly explain the relationship between stress factors and the URTI reviewed above. Thus, it is possible that HPA hyperactivity and right HL, which may lead to immune suppression, mediate associations between chronic mental stress and URTI risk.

A very important recent study examined the following issue – the mediating role of glucocorticoid receptor resistance (GCRR) in the relationship between chronic stress and URTI. GCRR takes place when immune cells are no longer or are less sensitive to the immunosuppressive effects of cortisol. In fact, we should view the immune system as one that protects us on one hand, but when it is out of control, it results in inflammation and in subsequent complications and even risk of death. The immunosuppressive effects of cortisol and the anti-inflammatory effects of the vagus are regulatory and protective, in the context of inflammation. GCRR has been found to occur in various chronically stressed individuals such as parents of children with cancer (Miller et al. 2002). Indeed, GCRR may result in overproduction of pro-inflammatory cytokines, which plays a role in the manifestation of the signs and symptoms of a URTI (Hendley 1998). GCRR was operationalized as a lack of association between cortisol levels and the number of leukocytes and the lack of association between cortisol and the neutrophil/lymphocyte ratio. Indeed, in their first sub-study, a positive association between cortisol and the neutrophil/lymphocyte ratio and an inverse relation between cortisol and lymphocytes were found in people without major stressors (indicating cortisol regulation of such cells). In contrast, no correlation was found between cortisol and the neutrophil/lymphocyte ratio or between cortisol and lymphocyte percentages in people with major stressors (Cohen et al. 2012). Importantly, and similarly to the previous pattern, cortisol *was* indeed inversely correlated with lymphocyte percentage in people who *did not* develop colds after receiving a viral challenge. In contrast, cortisol was *unrelated* to lymphocyte percentage in those who subsequently *did* develop colds after viral challenge. Thus, this lack of cortisol regulation of immunity characterizes chronically stressed people and those who end up developing the URTI. In their second sub-study, increased GCRR was associated with increases in the pro-inflammatory cytokines IL-6 and TNF-alpha after viral challenge. Together, these results suggest that chronic stress may be associated with colds via inducing CGRR, which then fails to inhibit the inflammatory response that occurs after viral penetration into the body. This would then possibly promote the signs and symptoms of the URTI and possibly its duration (Cohen et al. 2002). These results reveal a fascinating and complex new PNI pathway between stress and colds. However, given the anti-inflammatory role of the vagus nerve (Tracey 2009), and given its reduced activity in chronic stress (Lampert et al. 2016) and its role in brain-immune synchronization (Ohira et al. 2013), future studies need to examine whether GCRR may also result from possible vagal nerve withdrawal in people with chronic stress. This may also result in lack of regulating inflammation and in possibly more enduring URTI and its symptoms.

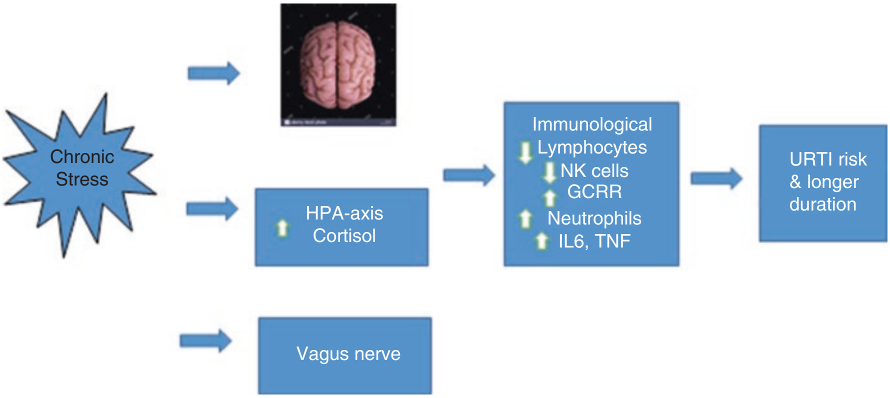

Fig. 6.2 Proposed PNI mechanisms linking stress to the common cold (URTI)

Indeed, concerning the role of the vagus nerve in URTI, one longitudinal study followed 18 athletes during 2 years. They measured weekly their HRV components. They found mostly a reduction in vagal activity as measured in various positions – supine and sitting up (orthostatic position) – just prior to the onset of URTI (Hellard et al. 2011). However, no immune or inflammatory markers were measured, which could have revealed whether the association between HRV and URTI may have been explained by such markers. Figure 6.2 depicts these reviewed PNI mechanisms.

Finally, we considered above the other way around: Effects of the URTI on psychological factors, namely, mood and cognitive performance. But how does this happen? The pioneering study by Reichenberg et al. (2001) sheds light on this question. They administered in a placebo-controlled experimental within-subjects design sub-pyrogenic doses of lipopolysaccharide (LPS, which induces inflammation) or a placebo, to healthy individuals. After LPS, reductions in memory and increases in mental distress were found, which were significantly correlated with levels of circulating pro-inflammatory cytokines. Thus, pathogen-induced inflammation may lead to the common mood and cognitive changes often seen in URTI.

Psychological Interventions for Preventing the URTI

Only a few studies examined the effects of psychological interventions on the incidence or severity of URTI. A groundbreaking study compared the effects of stress management (SM) to guided imagery (GI) and to a control condition, in relation to colds and humoral immunity, in children with frequent colds. The SM included using problem solving for solvable situations and distraction and relaxation for unsolvable situations, in line with the goodness-of-fit hypothesis (Forsythe and Compas 1987). The importance of self-awareness and of positive emotions was also emphasized. The GI included the use of repeated relaxation, as well as self-suggestive imagery concerning IgA as a guard in the mouth, nose, and ears, against invaders. Results showed that duration, but not frequency of colds, and mental distress indices were reduced significantly by both SM and GI, but not in the wait-list control condition. Furthermore, levels of SIgA increased only in the treatment groups. Importantly, the researchers were able to practically replicate their findings in a second sample (Hewson-Bower and Drummond 2001). This research is crucial in several ways. First, it shows causal relations between psychological interventions, immunity, and URTI severity (its duration) because it was an experimental study where the experimental conditions improved on psychological, immunological, and clinical (infectious) outcomes. Second, this could have clinical significance in relation to prevention of colds in children, a population with particularly high infectious disease morbidity. Third, this study replicated its findings, a result which is as important as statistical significance, usually measured simply only by "*p*-values." Replication is by far more important, in my opinion, than reliance on statistical significance alone, for generalization of findings. This is especially important considering the recent study showing sadly that only 36% of studies showing statistical significance in psychology were replicable (Aarts et al. 2015).

A more recent study compared the effects of meditation versus exercise versus a wait-list control on incidence and duration of colds in 149 adults. Amazingly, the illness severity of the meditation group was significantly lower than of controls, and the duration of URTI in the meditation (257 days in total) was significantly lower than in controls (453 days). Finally, meditation participants and exercise participants experienced significantly fewer days off work (16 and 32 days, respectively) than controls (67 days; Barrett et al. 2012). No differences were found in cytokine levels however. This study shows that two forms of behavioral interventions, including meditation, are effective in reducing the severity, duration, and functional consequences of URTI in adults. These findings have both clinical and economic significance and need to be replicated in other cultures, and the underlying mechanisms other than cytokines (e.g., hemispheric lateralization, NK cells, vagal modulation of inflammation) need to be identified.

One study compared the effects of different forms of written disclosure, on multiple psychological outcomes and self-reported URTI symptoms. The participants were randomized to Pennebaker's written emotional disclosure (WED) and guided WED (a counselor provided a day later feedback on writing), an online immediate-guided WED or control writing. Surprisingly, only controls evidenced significant reductions in URTI symptoms. In addition, the guided WED appeared to yield less favorable psychological outcomes than unguided WED (Beyer et al. 2014). However, only self-reported URTI symptoms were used in that study. In addition, it is possible to guide people's writing in advance, without such ongoing guidance and interference, using pre-written-guided WED which asks participants to write chronologically, to label emotions, and to find causality and insight in the event's segments. These instructions aim to shift trauma processing from a limbic, emotional, and somatic level to a more frontal, cognitive, and verbal level, and were found to reduce healthcare visits (Gidron et al. 2002).

Case Study Conclusion

Based on the studies reviewed above, Mr. Jackson is exposed to chronic job stress from overwhelming job demands, conflicts, and little support at home and has great difficulties to disconnect from his work. It is plausible that his low level of lymphocytes may partly result from this chronic stress, increasing his risk for having a cold again and again. Furthermore, the chronicity of his stress may also contribute to reduced vagal activity or to GCRR which may result in higher inflammation, and this could contribute to longer colds. Based on the interventions mentioned above, you could propose to him to undergo job stress management. This will include monitoring his daily stressors and his appraisal of controllability over such events. Then, he could use problem-focused coping for controllable stressors and vagal breathing for uncontrollable stressors, as well as perform that relaxation routinely during the day, during brief 5-minutes breaks. It would also be recommended to repeat assessment of his heart rate variability, immune competence (e.g., lymphocyte count), stress levels, and URTI symptoms over time, to be sure that these have improved.

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Chapter 7
Selected Topics in Pediatric Behavioral Medicine

Case Study

Joyce has asthma since age 3. She is 17 years old. She takes protective medication during asthmatic attacks but, otherwise, is healthy. Her severe attacks occur approximately five times a year, yet, as a GP, you were unable to observe any consistent pattern triggered by environmental changes in pathogens or in other illnesses she may have. However, her father suggests that the attacks follow periods of stress in school and at home, which you dismiss. During the year later, there is no change in her condition. You refer her to an allergologist!

This chapter will investigate and cover several selected topics in behavioral medicine applied to children and adolescents. Children and adolescents constitute a unique group concerning their health care, risk factors, and diseases. This is due to some illnesses having a higher prevalence in childhood such as infectious diseases like chicken pox. At the psychosocial level, children and adolescents are influenced greatly by the family framework and peers, and this needs to be considered when researching and treating their unique psychological factors in relation to prevention, illness onset, and prognosis. In adolescence, peers and peer pressure may become more important, in particularly influencing adolescents' health behaviors. These of course also have crucial implications for prevention and treatment. Because we cannot cover all medical problems in children and adolescents in one chapter, I chose to review a few medical topics through the prism of various psychosocial topics that are crucial to understand in this important age group in behavioral medicine.

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Psychological Factors Relevant to Children and Adolescents

As for adults, psychosocial variables such as life events or daily stressors, coping strategies and social support, as well as personality variables such as hostility or negative attribution style are important to consider in children and adolescents, within behavioral medicine. We shall examine some of these below. However, beyond these variables, children belong to a family network and are heavily influenced by their parents and family, in an interactive style also related to attachment style. Clearly, a major issue in childhood is attachment style. Attachment can be conceptualized as a pattern of emotional and behavioral interactions that develops over time, as the infant and caregiver interact and communicate, especially in response to the infant's needs for comfort and attention (Cassidy and Shaver 1999). Attachment style can then affect the pattern of behaviors with which people relate to or form relationships with others later in life as well. Our previous pattern of relationships formed in childhood affect our pattern of relationships with caregivers, according to Bowlby (1973). Such caregivers can also include medical staff such as doctors or nurses, manifesting the importance of considering attachment style in behavioral medicine. These patterns of interacting with people may be particularly manifested during periods of vulnerability. One such vulnerability period could be during illness or hospitalization of a child, adolescent, or even an adult (Ciechanowski et al. 2004).

Various classification systems exist for describing children and adolescents' attachment styles, some of which are seen in children, others also in adults. These could be grouped broadly into secure and insecure forms of attachment. However, the insecure group can be grouped into three further subgroups, namely, dismissive, fearful, and preoccupied (Bartholomew and Horowitz 1991; Ciechanowski et al. 2004). People with a secure attachment style are thought to have experienced a responsive caregiving and are comfortable in depending on others (Ainsworth et al. 1978) and in independently exploring their environment. People with a dismissing attachment style experienced a more unresponsive caregiving style and become more self-reliant and have difficulties to trust others. Those with a preoccupied attachment style experienced an inconsistent responsive caregiving, and they spend great efforts in relationships, are quite dependent on others' approval, and have a poor self-esteem. Finally, those with a fearful attachment style manifest an approach-avoidance or ambivalent relationship with others, out of fear of rejection, and most probably experienced harsh criticism and an overtly rejecting caregiving. Ainsworth et al. (1978) used the "stranger situation" paradigm to classify infants into these attachment styles. In the paradigm, after a child plays near his or her mother, a stranger enters the room and then the mother temporarily leaves the room. Observers evaluate the infant's response to the stranger and its response to the mother upon her return to the room. The securely attached child prefers the mother over the stranger, may cry when she leaves the room, and calms down upon her return. The child with ambivalent attachment stays close to the mother prior to her leaving, shows approach and avoidance upon her return, and can still cry and even push her away when she returns. Finally, the avoidant child mostly does not cry when left alone, can be comforted by a mother or stranger, and refrains from cuddling when picked up. These patterns are shaped to a large extent by parental responding styles to their children's needs. Berk (1989) found that routine neglect of children by parents can predict angry and ambivalent attachment styles, while physical abuse could lead to avoidant attachment style. Attachment can also have implications for doctor-patient relationships, both from the patient and from the physician's perspectives, and these may also impact patients' health outcomes since doctor-patient communication affects such outcomes (Thompson and Ciechanowski 2003). One more recent study found, for example, that people with a fearful attachment style had significantly lower trust in medical professionals than those with a secure attachment style (Klest and Philippon 2016). This could have relevance to health outcomes since trust in physicians was found to be associated with better patient enablement and quality of life (Ernstmann et al. 2017).

Attachment can play a role in adult couples' responses to stress. One study found that when discussing a conflicting topic, men with highly anxious attachment and women with highly avoidant attachment styles responded with higher cortisol responses to the stressor. Furthermore, men responded with lower cortisol stress responses when their female partner had a secure attachment style (Powers et al. 2006). This study reveals relationships between individuals' attachment style and their biological stress responses but also reveals that one's stress responses can be influenced and even moderated by the attachment style of another significant other sharing the stressful situation. Another study assessed in 212 adult American students their attachment style, attitudes, self-esteem, and risk behaviors. Anxious attachment style was related to negative attitudes and to low self-esteem, which in turn correlated with frequency of drug use (Kassel et al. 2007). This study reveals that attachment style may also affect young people's risky health behaviors.

One factor clearly related to and affecting a child's attachment style is whether the child was adopted, and if so, the timing of adoption during their early developmental stages. Studies on adoption reveal the effects of growing in a secure home which meets a child's needs, on long-term brain development. Tottenham et al. (2010) found that the volume of the amygdala was larger in children adopted later compared to earlier in life. Tottenham et al. (2010) also found a positive correlation between amygdala volume and extent of internalizing and anxious behavior in children. Numerous studies implicate the amygdala's role in processing threat and fear and to be overactive in several mental diseases including post-traumatic stress disorder and depression in adults (e.g., Shin et al. 2004). Together, these studies reveal the long-term important impact attachment styles have on child and adult development and on their responses to challenging situations. Below, we shall review some studies examining the effects of attachment style and other parental responses on children and adolescents' health outcomes in medical contexts.

Developmental Stages and Pediatric Behavioral Medicine

Children obviously experience medical illnesses and procedures differently than adults, also because of developmental differences, which render different abilities to mentally represent, comprehend, learn from such events and regulate one's responses. Childhood is characterized by gradual and important changes in development; however, going into detail on this topic is beyond the scope of this chapter. Indeed, studies have examined health issues in relation to Piaget's model of cognitive development. Briefly, Piaget divided children's development into four stages up to adolescence. The sensorimotor stage (birth–2 years) is the period of sensitivity to movement, sensory development, how to manipulate objects, separating self from the objects/people around, and cause and effect. In the pre-operational stage (ages 2–7 years), children move from babies to "little people." They are first very self-centered and assume everything is centered around them and act accordingly – for example, they use animism – assigning human features to everything as well as to objects (e.g., saying "noty" to a table in which they bumped). This stage also includes symbolism and initial moral reasoning. The third stage called concrete operational stage (ages 7–11 years) includes the development of logical reasoning. Here, however, the child's logic usually requires the presence of objects to reason about them; hence it is concrete. Children also begin to be able to think in a "what if" scenario manner. This of course could affect health behaviors and illness perceptions since they have the potential to imagine and anticipate the possible consequences of unhealthy and healthy behaviors. Finally, in the formal operations stage (ages 11–16 years), the child obtains organized, abstract, and logical thinking. He or she can consider multiple aspects of a problem. This is reflected by operating on information and symbols without the presence of concrete objects.

Esteve and Marquina-Aponte (2012) examined children's perspectives about pain, according to these stages of development. Examining 180 healthy Spanish children, they used a content analysis system to interpret children's perspective about pain. In children between 4 and 6 years old, pain was related to injuries, causality, and the definition of pain. In children aged 7–11, pain was ascribed to specific body parts. Finally, adolescents could also view pain as a learning experience and, generally, being were able to consider the emotional aspects of pain, increased with age. Importantly, the use of cognitive strategies to cope with pain increased with age, in line with a maturing ability to manipulate thoughts and information, crucial in emotional regulation. Knowing these cognitive developmental differences can help health psychologists and doctors understand and treat pain in children, in line with their cognitive developmental stages. This can also help guide researchers and clinicians to examine whether chronic pain disorders in certain children may be related to regression to a younger cognitive developmental stage relative to a patient's chronological age. For example, given the cognitive representation of pain in adolescents, this age group should be able to use cognitive strategies such as reappraisal or acceptance to cope with pain, strategies which are associated with better adaptation to chronic pain (e.g., Viane et al. 2003).

It is possible that illnesses may also impair children's cognitive development. One common health problem in childhood (and adulthood) is stomach aches and its more extreme form of ulcer. This is known to be caused often by the *Helicobacter pylori* infection. One study examined in a cross-sectional design the relationship between levels of the infection and IQ scores of 200 children aged 6–9 years. Inverse and significant associations between levels of the infection and total IQ, verbal IQ, and nonverbal IQ were found, independent of nutritional factors, only among children from a higher socioeconomic background (Muhsen et al. 2011). Though based on a cross-sectional design, the authors speculated that this association may have been found due to the effects of this infection on inducing anemia, known to impair cognitive functions. Despite the methodological limitations, these results are in line with an experimental study showing that administering and acute bacterial infection reduces cognitive abilities in healthy adults (Reichenberg et al. 2001).

Attachment Style and Child and Adolescents' Health Behaviors and Outcomes

Studies examining this issue have generally used cross-sectional designs, and I will provide a few examples in relation to various medical settings and outcomes. One common and often difficult medical procedure for children to undergo is blood tests or venipunctures. Children vary in their level of manifested pain and distress, which can be partly influenced by their attachment styles, among other factors. One study examined the relationship between attachment style and temperament with distress levels during venipuncture, in infants aged 14 months. While levels of distress were not different as a function of attachment style or temperament, each examined alone, children with a combination of both disorganized attachment and temperamental fear had higher distress levels than other children (Wolff et al. 2011). It is possible that in such young age, the venipuncture's strong impact may mask more minor effects of attachment style or temperament alone on children's distress, but that distress could be influenced by a synergism of relatively disorganized attachment together with fearful temperament.

Another study examined attachment style and children's responses to everyday painful incidents (e.g., bumps) and acute pain events like immunizations, in 66 5-year-old children. They found consistent positive correlations between insecure attachment styles (ambivalent and controlling/disorganized attachment) and pain levels and positive correlations between controlling/disorganized attachment style and time to calm down from immunizations (Walsh et al. 2008). This study informs us about the possible role of attachment in pain in children, across several contexts. The results may have implications for understanding the evolution of children's acute and chronic pain conditions and recovery from them. The results can also help in planning interventions aimed at pain reduction, possibly via modification of attachment style early in life.

One recent study examined in 12-month-old children the role of attachment style and temperament in levels of distress while receiving immunizations. The researchers found that temperamental fear moderated the relationship between attachment style and distress duration. Specifically, in children with *low* temperamental fear, secure children regulated their distress levels more slowly than children with avoidant and disorganized attachment styles. In contrast, in children with *high* temperamental fear, avoidant children regulated their distress levels more slowly than those with secure attachment (Horton et al. 2015). Both latter studies show the complexity of this topic and reveal that the role of attachment in medical procedures needs to be considered in the context of children's temperament as well.

An emerging global pandemic is obesity, defined commonly as body mass index (BMI) of >30 kg/m^2. There is a global rise in obesity and particularly in childhood obesity including in countries normally associated with healthy diets (e.g., Lopez and Knudson 2012). One crucial study examined the association between parental attachment style, their manner of emotional regulation when their child manifests negative emotions and their children's food consumption, using mediation analyses. The sample included 497 primary caregivers of children aged 2.5–3.5 years. Results revealed that insecurely attached mothers were more often using punishing and dismissal in repose to their children's negative emotions, while these negative emotion regulation modes of response correlated with more emotion-related feeding and less mealtime routines. These in turn then correlated with more consumption of unhealthy foods by the children. Another analysis revealed that caregiver insecure attachment was associated with food consumption via its relationship with child TV viewing (Bost et al. 2014). These results, though based on a correlation design, revealed that parental attachment styles may impact on children's risk of obesity via affecting their emotion-regulation manners (emotional eating) or via sedentary behaviors (TV viewing) and have vast implications for public health and prevention. These results also propose for long-term studies examining the effects of training parents in attachment skills on their subsequent children's health and well-being.

One increasingly prevalent disease in adolescence is diabetes, with a distinction between Type 1 and Type 2. While in Type 1 diabetes, the body produces insufficient insulin, in Type 2 there is cellular insulin resistance. Insulin is the vehicle via which glucose enters into our cells. Diabetes constitutes a highly important challenge for behavioral medicine, because psychosocial factors could impact one's diabetes-related health behaviors (glucose self-monitoring and diet) as well as the main disease outcome, namely, HbA1C. Importantly, diabetes is on its own a risk factor of heart disease and certain cancers (e.g., Skrivarhaug et al. 2006). Dashiff et al. (2009) examined the relationship between maternal separation anxiety from the child, adolescent cognitive autonomy, and self-care behaviors, in 131 mother-child pairs. They found an association between initial maternal separation anxiety and the adolescent's cognitive autonomy a year later. Cognitive autonomy then predicted self-care a year later as well. This research is important because it demonstrates the delicate and complex associations between adolescents' family variables and their own cognitive schemas and self-care, of crucial pertinence for glucose regulation in diabetes.

Finally, theoretical work has been proposed by Murray (2000) on the role of attachment in the reactions of siblings of children with cancer, to this disease and the changes in family interpersonal dynamics. Future studies should examine empirically the role of attachment in the responses of siblings of ill children who have cancer or other chronic diseases.

Parental Responses to Illness Behavior

In normal circumstances, children usually reside in most cases with their parents. Subsequently, children are exposed to their parents' responses which could reinforce or reduce the frequency of certain behaviors, as in any operant learning process, including illness behaviors. Illness behaviors include facial expressions, body postures and movements reflecting pain, and verbal complaints of pain, as well as help-seeking requests. One factor possibly affecting the manifestation and frequency of such illness behaviors is parents' "rewarding" responses to these illness behaviors. This is termed "illness behavior encouragement." Parents' habitual responses to their children's illness behavior could be pivotal to shaping their child's further illness behaviors and possibly even their health outcomes. This issue is derived from behaviorist models, which view illness behavior (symptom reporting, disability, etc.) as a behavior which can be partly learned and inadvertently reinforced by parental over-responsiveness to a child's illness behaviors, via operant learning. Walker and Zeman (1992) developed a brief questionnaire assessing parental illness behavior encouragement and found that degree of illness behavior encouragement by parents was positively correlated with children's school absence days. However, since that study was cross-sectional, it is impossible to discern whether parents' illness behavior encouragement preceded children's school absence or resulted from it. In contrast, a prospective study examined various psychosocial predictors of adolescents' recovery from oral surgery, one of which included parents' responses to their children's illness behavior, using the same questionnaire developed by Walker and Zeman (1992). Controlling for severity of surgery, higher scores of mothers' illness behavior encouragement significantly predicted a smaller extent of mouth opening in their child a week later. Mouth opening reflected an objectively measured proxy of physical recovery from oral surgery (Gidron et al. 1995).

Finally, Gidron et al. (2003) examined the relationship between illness behavior encouragement and number of attacks in 45 Israeli children with familial Mediterranean fever (FMF). FMF is a genetic auto-inflammatory disease of repeated abdominal painful attacks beginning mostly in childhood. Among other variables tested, scores on the illness behavior encouragement scale were unexpectedly inversely correlated with number of attacks in the past year. However, that study was retrospective.

Taken together, there is some evidence linking parental responses to children's illness behavior and their daily functioning and recovery from surgery. These issues need to be considered in pediatric behavioral medicine and need to be further investigated in future studies in relation to other outcomes in other health conditions such as asthma or diabetes. Furthermore, future intervention studies could examine whether teaching parents to respond to their children's illness behavior appropriately (in moderation and in relation to the illness severity) can influence their children's functional and physical health outcomes.

Children's Stressors and Health Outcomes

Situations perceived to be stressful for children and adolescents can vary as a function of their age and developmental stage, gender, other life and environtmental circumstances, and culture, to name but a few factors. In a study done on two samples of children aged 7–12 years, common children's stressors were found to include being alone, having tests, family fights, boyfriend/girlfriend issues, and too many things to do (Ryan-Wenger et al. 2005). In an earlier study, Compas (1987) found that among adolescents, the three most common daily hassles included family relationships, school achievements, and peer relationships. Another study found that grade transitions in school, pubertal changes, pressure to confirm, and temptations with experimenting with sex and drugs were seen as potential stressors in adolescents (Hauser and Bowlds 1990). A newer study used an open-ended assessment in a small sample of 42 adolescents and found that most adolescents reported school problems, family problems, and problems with peers as stressful (Lohman and Jarvis 2000). Together, these studies show relative consistency especially in adolescents' frequent stressors across time and methods of assessment.

A main limitation in the study of the relationship between stressors and health outcomes is that many have been retrospective or cross-sectional. Such methodology suffers from relying on memory of reporting past stressful events. Furthermore, such studies also may suffer from the possibility of reversed causality: people's health may influence their psychosocial responses (e.g., Reichenberg et al. 2001) and perceptions of present or past recalled situations. Finally, people may need to seek explanations for their health problems and then attribute their health problem to a stressor, while experiencing the event may not have really led to that health problem. The preferred alternative is conducting prospective studies. One important prospective study examined the relationship between existing chronic stress, new severe acute stressors, child and illness factors, and the occurrence and timing of new asthmatic attacks in 90 Scottish children. Severe acute stressors included losing a parent or grandparent due to death or separation, while chronic stress included a long-lasting illness of a family member, marital discord, and bullying. The study found that acute stress significantly predicted new asthma attacks, with a time gap of 4 weeks after the stressors. However, if acute stress occurred together with chronic stress in the background, acute stress significantly predicted risk of new attacks more strongly and with a 2-week gap from the acute event to the asthma attack (Sandberg et al. 2000). This important study not only uses a prospective design and considered relevant confounders (e.g., parental smoking, season of the year), but it shows how combined acute and chronic stressors synergistically interact in relation to the occurrence and more rapid onset of a common illness in children, asthmatic attacks. More studies need to use similar methodologies in other populations and recurrent illnesses such as predicting headaches, diabetic glucose imbalance, or even epileptic attacks.

Another form of stress, which can be particularly emotionally painful for children, is bullying, which is defined as exposure to negative behavior by one's peers

(Storch et al. 2006). Furthermore, children with certain illnesses may be more vulnerable to suffer from bullying as their peers may find their appearance or medical needs to be strange and different. One study examined the relationship between specifically diabetes-related bullying (DRB) and self-management and insulin regulation in 167 children with Type 1 diabetes. DRB was positively correlated with HbA1C (reflecting worse insulin regulation) and inversely related to self-management (glucose testing and diet). Finally, levels of depression partly mediated the relationship between DRB and self-management (Storch et al. 2006). This study is important by showing the possible health consequences of illness-related bullying, a form of chronic stress, in children with a chronic health condition, and because it also shows that the mechanism may be its effects on their mood and poor self-management, crucial for an illness like diabetes.

Social support can have direct positive health effects but may also interact with and thus buffer the effects of stress on health (Cohen and Wills 1985). Thus, in sharp contrast to the previous study, Bearman and La Greca (2002) found that levels of social support related to diabetes from friends (e.g., about administering insulin, glucose self-testing, keeping a meal discipline) were positively correlated with actual blood glucose testing. Both studies on diabetes clearly demonstrate how crucial the psychosocial milieu of children can be for their self-care and even health outcomes in the context of a chronic illness like diabetes.

A more recent and important large study examined the relationship between marital quality and life events during pregnancy, with the risk of newborn children having an infectious disease in the first year of life. The sample included 74,801 parent-children units. Results revealed that prenatal marital dissatisfaction and life events predicted 29 out of 32 separate infections, independent of confounders (e.g., child gender, breastfeeding, parental social support; Henriksen and Thuen 2015). This study follows a long line of research on the effects of prenatal stress on the well-being of the offspring, and such strong and precise prediction has clear and important implications for public health. They also repeatedly reveal the crucial role parental stress may have in the health of infants and children.

Another common health problem in children and adults is pain, in which exposure to stress is often thought to be a possible trigger. However, is there methodologically sound evidence for such claims? To answer this question, I will review two prospective studies. In a two-wave prospective study done in 502 (wave 1) and 289 (wave 2) mostly African-American adolescents, 40% reported weekly headaches and 36% reported weekly episodes of abdominal pain. Psychosocial stressors (e.g., witnessing violence and problem situations), correlated with pain in wave 1, partly mediated via anxiety. However, the longitudinal data supported *a reversed causality* model whereby reporting pain in wave 1 predicted increases in anxiety and victimization in wave 2 (White and Farrell 2006). In another longitudinal study, 1665 children were followed for 1 year in relation to developing headaches and 2040 in relation to developing back pain. In boys, no psychological factor (e.g., dysfunctional coping with stress, catastrophizing, anxiety sensitivity) was predictive of headaches. In contrast, dysfunctional coping with stress and anxiety sensitivity significantly predicted back pains. For girls, dysfunctional coping with stress

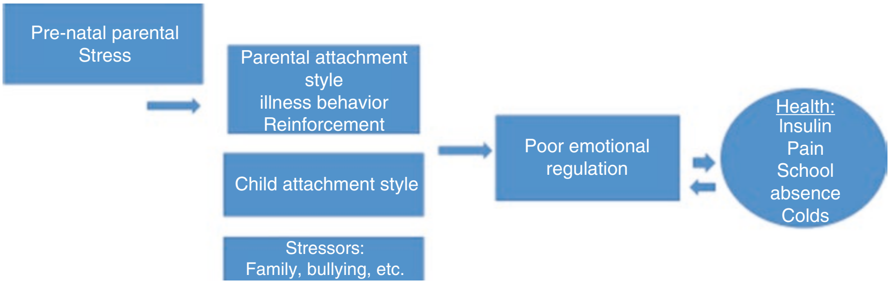

Fig. 7.1 Parental and child psychosocial factors and children's health outcomes

and somatosensory amplification predicted headaches, but these disappeared in a multivariate analysis. For girls' back pain, somatosensory amplification increased the risk of back pain, while, surprisingly, catastrophizing predicted a reduced risk of back pains (Barke et al. 2014). Together, these results propose that psychosocial stress and maladaptive forms of coping may either be risk factors for or consequences of children's pain problems, and both directions of relationships need to be considered, scientifically and clinically. Furthermore, important gender differences exist in this domain. These studies also reveal the importance of testing these associations in prospective studies to identify possible reversed causality. Figure 7.1 depicts part of the associations reported in this chapter and the complexity of results in children and adolescents.

Psychoneuroimmunological Mechanisms Linking Psychosocial Factors with Children's Health Outcomes

As a general framework for this topic, Maunder and Hunter (2008) proposed four mechanisms linking nonadaptive attachment styles with health outcomes in children. These included (1) disturbed recovery from stress; (2) physiological mediators of social relationships, stress, and immunity; (3) health behaviors related to various attachment styles; and (4) risky behaviors that regulate affect such as smoking and alcohol consumption. We focus here on the physiological and immunological mediators. However, concerning the risky behavior route, at least one study found that less securely attached adolescents consumed more drugs or alcohol than did securely attached adolescents (Cooper et al. 1998). Thus, the behavioral route must be considered seriously as well in planning preventative actions and interventions in adolescents.

We shall now examine the physiological route. One study examined cortisol responses of 60 children and adolescents aged 9–18 years, as a function of their attachment style. At awakening, anxious insecurely attached participants had a significantly higher cortisol level than securely attached participants. At 45 minutes post awakening, anxious insecurely attached participants had significantly lower cortisol levels than avoidant insecurely attached participants. Finally, the anxious insecurely attached participants had a significantly *flatter* mean increase in the cortisol awakening response (from baseline to the mean of measures 2–4 post awakening) than insecurely avoidant attached and securely attached participants. The results remained unchanged when effects of age and menarche were considered (Oskis et al. 2011). These complex results demonstrate that the awakening cortisol levels and diurnal changes in cortisol from awakening in anxious insecurely attached children and adolescents are different from avoidant and from securely attached children and adolescents. Such differences may have implications to health conditions which relay on an adequate cellular immune response or on a modulated inflammatory response, both partly under the control of systemic cortisol levels (e.g., Tracey 2009).

Because child well-being is strongly linked to maternal caregiving abilities, mothers' biological correlates of caregiving abilities may be linked to children's attachment styles and responses to illness. One study observed mothers at 4–6 months after giving birth, in their free play with their newborn child, to assess attachment style and mothers' hormonal parameters. At 7–9 months postpartum, plasma oxytocin (OT) levels were measured in the mothers. Interestingly, mothers with low sensitivity to their child had *higher* OT levels than those with high sensitivity to their child. The authors speculated that the high OT levels of low-sensitive mothers may reflect higher stress hormone responses to child-caring stress (Elmadih et al. 2014). However this interpretation is highly speculative because actual stress hormones such as cortisol or ACTH were not measured. Furthermore, infants' physiological stress responses were not measured; thus no direct link between maternal OT and child physiological stress responses could be made. Nevertheless, such studies are needed, and the relationship between maternal OT, attachment styles, and child health outcomes needs to be examined in future longitudinal studies. This is particularly important since OT together with social support have a synergistic effect on calming the stress response in adults (Heinrichs et al. 2003), and this may also occur to children with more supportive mothers who may have higher OT.

A unique and recent study examined prospectively the relationship between parent-child interactions and children's antibody responses to the vaccination against the meningococcal serotype C infection in 164 children aged 9–11 years old. The investigators classified parent-child relationship quality in an interaction task and measured antibody responses 6 months later. A negative/conflictual interaction style predicted a lower antibody response, independent of confounders. Furthermore, while conflictual interactions predicted increased cortisol levels, this increase did not mediate the observed reductions in antibody responses to vaccinations in these children (O'Connor et al. 2015). This study is important because it is methodologically sound using a prospective design, with observed ratings of parent-child interactions, and because it examined an objective immunological factor of crucial importance to child public health, namely, antibody response to a vaccination, at follow-up. The results propose that children exposed to chronic stress or conflictaul attachment with their parents may be at risk for infections due to a less favorable antibody response to vaccinations. However, since changes in the HPA axis (cortisol) did not mediate these results, future studies should examine the role of sympatho-vagal balance as a potential explanatory mechanism in such associations.

The few studies which examined the relationship between attachment style and neuroimmunological indexes mostly included adults rather than children. A unique study examined in adult nurses the relationship between avoidance attachment style and their lymphocyte proliferation and natural killer cell cytotoxicity (NKCC) over 1 year. High avoidance was significantly and inversely related to NKCC (Picardi et al. 2013). While these results may not be generalized to children, they do propose that an avoidance attachment style, which presumably partly originates from childhood, is associated with reduced cellular immunity, which could then put such individuals at risk of certain illnesses such as upper respiratory tract infections and even challenge their body's ability to fight tumor development. Future studies could test these issues.

Attachment may also play a role in adults' inflammatory responses. One study examined levels of the pro-inflammatory cytokine interleukin-6 (IL-6) in 35 couples discussing a supportive or conflictual topic. Participants were classified according to their level of avoidance attachment. Those with avoidant attachment had significantly higher IL-6 during the conflictual than during the supportive discussions (Gouin et al. 2009). Avoidant people also expressed more negative and less positive behavior during the discussions. Together, the latter two studies propose that an avoidant attachment style in adulthood is related to suppressed anti-viral cellular immunity and to enhanced stress-induced inflammation. Both results may have implications for understanding the association between stress, attachment styles, and several health outcomes which either result from suppressed immunity (e.g., colds) or from excessive low-grade inflammation (e.g., diabetes, coronary heart disease).

As seen in other chapters, inflammation is known to contribute tremendously to multiple chronic diseases (summarized in Gidron et al. 2018). Is inflammation associated with child rearing and stressful experiences? A meta-analysis of 20 studies found that childhood maltreatment and experiences were associated with higher inflammatory markers, particularly with C-reactive protein (Coelho et al. 2014). A more recent study found that childhood maltreatment interacted with hemispheric lateralization (HL) in relation to inflammation: right HL was related to adult's higher inflammatory markers, only if they reported moderate to severe childhood maltreatment (Hostinar et al. 2017). These findings have huge implications for public health given the current evidence on the role of inflammation in multiple chronic diseases (Gidron et al. 2018) and given the role of righ-HL in immunity and possibly in poor health outcomes (e.g., Sumner et al. 2011).

One more note toward the future. Scorza et al. (2019) reviewed studies on trans-generational transmission of adverse disadvantages of parents to children, occurring via an epigenetic mechanism – the over- or under-expression of genes – which may play a role in children's vulnerability to adverse health outcomes. Since stress can affect the integrity of the DNA (Gidron et al. 2006), and given the role of stress and attachment in children's health as reviewed above, future studies should examine in humans whether such epigenetic transgenerational transmissions predict increased risks of stress-related diseases and whether various stress management interventions could possibly reverse this novel pathway in children.

The Effects of Psychological Interventions on Children's Health Outcomes

Numerous studies on psychological interventions in behavioral medicine have been conducted on children and adolescents. I shall provide only several examples in relation to the concepts and health conditions mentioned above. We began this chapter with an emphasis on attachment theory and its relationship with health outcomes. Can attachment style be changed? Few health professionals and lay people may know that interventions have been developed and tested empirically, to address precisely this question. Hoffman et al. (2006) tested this issue in 65 high-risk toddler-parent dyads. Their intervention, which included an educational component and learning to perceive and respond adequately to one's child's needs and cues, lasted 20 weeks. Results showed significantly greater shifts in children moving from insecure to secure attachment styles than the opposite shift (Hoffman et al. 2006). This type of intervention is promising because it demonstrates that one of the most basic psychosocial factors which affect children and adults throughout life, namely, attachment style, can be modified and even improved. Given the long-lasting impact attachment has on people's lives, as reviewed above in relation to health, future studies should examine the effects of such attachment modification interventions on preventing adverse mental and physical health outcomes in children and adolescents.

One of the most impressive studies was conducted by Castés et al. (1999). It is outstanding because it is scientifically and clinically important. This study examined the effects of a psychosocial intervention (PSI) on immune and clinical outcomes of 35 children living on an Island in Venezuela, where the prevalence of asthma is extremely high. Asthmatic children received usual treatment (control) or usual treatment and PSI (experimental group). The PSI included learning relaxation methods, guided imagery, and ways to enhance their self-esteem. Results showed that only in the PSI group did the number of asthma attacks and use of bronchodilators decline significantly, while pulmonary functioning improved, compared to baseline measures. These reflected clinically important outcomes. Furthermore, at the immunological level, IgE responses (reflecting an allergic response to a highly relevant allergen) significantly declined in the PSI group, and levels of NK cells and IL-2 receptors on T-cells significantly increased in the PSI group. Taken together, after treatment, the immunological profile of the PSI group resembled that of non-asthmatic children (Castés et al. 1999). The observed immunological changes reflected scientifically important outcomes since they point at the possible mechanism of the PSI. This study also proposes that PSI may have partly reversed the pathological and immunological processes underlying asthma.

Two additional important and different interventions have been conducted in children with different health conditions. One study examined the effects of a social learning intervention aimed at increasing diabetic adolescents' resistance to social pressures, on their insulin regulation. Twenty-one adolescents aged 13–18 years were randomized to discuss medical information about diabetes (edutation control) or to systematically being exposed to social scenarios which challenge children' adherence to their diet (experimental group). In these exposures, adolescents planned how to respond and resist social pressure in such scenarios. After treatment, participants in the social resistance skills group had significantly lower levels of HbA1C (mean = 11.72) than controls (mean = 14.42; Kaplan et al. 1985). This method resembles the psychological inoculation method described in Chap. 3 on adherence, but here the investigators focused on social resistance skills, not on cognitive barriers. Future studies need to systematically test the effects of a psychological inoculation intervention, which focuses on diabetic adolescents' cognitive barriers and social pressures, on their glucose regulation. This may further increase the effects of such a social skills intervention on this health outcome.

Another study mentioned in the chapter on URTI compared the effects of stress management (SM) to guided imagery (GI) and to a control condition, in relation to colds and humoral immunity, in children with frequent colds. The SM included using problem-solving for solvable situations and distraction and relaxation for unsolvable situations. The GI included the use of repeated relaxation, as well as self-suggestive imagery concerning IgA as a guard against invaders. Results showed that duration, but not frequency, of colds and mental distress indices were reduced significantly by both SM and GI, but not in the wait-list control condition. Furthermore, levels of SIgA increased only in the treatment groups. Importantly, the researchers were able to practically replicate their findings in a second independent sample (Hewson-Bower and Drummond 2001). This study has scientific and clinical significance, because it showed the reliability of its clinically significant findings by repeating the study results in the context of a common medical condition in children and revealed its possible immunological mechanism.

Numerous studies have examined the effects of psychological interventions on pain conditions in children. I will provide here one recent and important example. Palermo et al. (2016) conducted a randomized-controlled trial (RCT) on adolescents with various chronic pain conditions (n = 273) and their parents. They were randomized to receive online cognitive-behavioral therapy (CBT) or education control. All assessments were also done online via the Internet, before, immediately after, and 6 months posttreatment. Results revealed significant reductions in pain-related disability and increases in reported sleep quality in the CBT group. Furthermore, parents in the CBT group showed significantly lower scores on inadequate responses to their child's conditions and behaviors. This study is important because it used an RCT design with a large sample of different pain conditions and examined the effects of a psychological intervention on children and their parents. Since it was given online, the findings have implications for performing such interventions on a large-scale basis, with little costs and logistical constraints. Hence, it has potentially vast implications for child public health.

Case Study Conclusion

Given the multitude of scientific findings and guiding frameworks reviewed in this chapter, we can assess and examine the association between Joyce's exposure to and appraisals of stressors in her life, and the onset of her asthma attacks, using a diary method and standardized measures of life events. Furthermore, it would be recommended to administer to her an attachment style scale and examine whether she has insecure attachment. Both factors may predict an increased risk of such attacks, possibly via affecting neuroendocrine modulatory processes of her inflammatory response pertinent to asthma. Then, a stress management training together with learning more secure interpersonal attachment strategies would be advised. Involving her parents to prevent inadvertent illness behavior encouragement may also be warranted.

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Chapter 8
Psychological Aspects of Health and Illness in the Elderly

Case Study

> Mr. Sanderman lives alone in his small house in a little town out of London. He is 87 years old and has very fragile bones. He has fallen a few times and takes just half of the prescribed dose of calcium and medication to counteract his bone density deterioration. He is also very lonely. His family physician proposed him to go out daily, to expose himself to vitamin D, and to exercise his muscles, since these could help with his fragile bone condition. However, due to fear of falling, he resists to go out daily. He goes out only about once in 2 weeks, when his only family member living near him, his granddaughter, visits him. His bones are deteriorating further, and any next fall may increase his disability and depressive mood.

It is with this case which we open this chapter on psychological aspects of health in the elderly. Due to the global aging population, identifying the psychological aspects of health in the elderly and developing evidence-based behavioral medicine interventions for this increasing segment of society become more and more important. It is also after all to them that we owe our lives here.

Basic Biological Processes in Aging

Biologically, aging is a progressive process of tissue loss and accumulating organ dysfunctioning (Flatt 2012). With age, several systems undergo modification such as reduced immune responses (Butenko 1985), increased blood pressure, and reduced cardiac ejection fraction (Lin et al. 2014). In addition, bone mass decreases with age, which is a main contributor to osteoporosis (Elmadfa and Meyer 2008).

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Pertinent to bone loss, there is a decrease in muscle mass with age, termed sarcopenia, resulting in reduced mobility, risk of falling, and subsequent increased risks of morbidity and mortality (Hanna 2015).

One major process that contributes to many of these changes in elderly is cellular senescence, a process of arrest of cellular proliferation. Cellular senescence causes aging and aging-related diseases because of two crucial consequences. First, senescence reduces tissue repair due to inducing cell-cycle arrest of progenitor cells. This reduced cellular replication could harm homeostasis and repair and regeneration of cells and tissues following tissue damage (Childs et al. 2015). Second, senescence produces pro-inflammatory signals (Childs et al. 2015), which contribute to multiple diseases prevalent in the elderly such as cancer and heart disease (Ross 1999; Voronov et al. 2003). Furthermore, various medical treatments (e.g., radiation) and normal tissue homeostatic processes can all lead to cell senescence. However, a major sign of aging-related cell senescence is the reduction in cellular immune competence, which otherwise could eliminate damaged (e.g., cancer) cells and viruses. In addition, accumulating senescent cells contribute to age-related illnesses. Senescence is not only a sign or result of aging, but it is causally related to aging because deletion of its underlying promoters slowed down aging in mice models (Baker et al. 2011). One group of molecular signatures of cell senescence is the senescence-associated secretory phenotype (SASP) which includes pro-inflammatory cytokines and matrix-degrading molecules (Childs et al. 2015). SASP molecules play a causal role in diseases such as atherosclerosis, diabetes, and cancer (e.g., Thangavel et al. 2011; Sikora et al. 2014).

At the neuronal level, neuronal loss occurs less than expected and appears to be most present at the hippocampus and cortex. This pattern of neuronal loss explains the obvious age-related decline in memory and executive functioning, respectively. The subtle changes in the interneuronal synapses, neurotransmitters, receptors, cell structure, and three intracellular main processes (the mitochondria, reactive oxygen species, and intracellular calcium homeostasis) and changes in electrical transmission, all contribute to cognitive decline (Shankar 2010).

Alzheimer's disease (AD) is a degenerating condition, where accumulating neuronal damage in regions that mediate memory and cognition leads to memory loss, reduced cognitive functioning, and communication abilities (Rubio-Perez and Morillas-Ruiz 2012). Two crucial processes occur in AD, reflecting the neural hallmark of the disease. First, extracellular Aβ plaques develop in response to production of the amyloid precursor protein (APP) cleavage, which marks cellular injury. Second, neurofibrillary tangles develop intracellularly, where the tau (τ) protein normally responsible for transportation of cell products to axonal ends binds and forms tangled threads (Rubio-Perez and Morillas-Ruiz 2012). Following exposure to Aβ plaques, neuronal astrocytes and glia cells (monocytes residing in the brain) respond by producing inflammatory markers, growth factors, and adhesion molecules (e.g., Wyss-Coray 2006). The combination of the Aβ plaques, neurofibrillary tangles, and reactive inflammation contributes together to neuronal dystrophy. Importantly, inflammatory responses can then lead to reproduction of the APP, which then leads to further Aβ plaque development, and to hence a vicious chronic inflammatory and neurodegeneration process (Rubio-Perez and Morillas-Ruiz

2012). Finally, excessive inflammatory responses via glia cells can also damage nearby healthy cells (Luo et al. 2010), possibly accelerating neurodegeneration.

Thus, we see how aging-related changes in the body and in the brain, and particularly cell senescence and inflammation, contribute to age-related diseases. It is on this background that we shall begin to consider the psychological aspects of aging, and we shall return below to some of these biological processes when we shall link them to the psychosocial ones via PNI.

Psychological Models in Gerontology

These models have evolved over the past 50 years. I will briefly describe a few of them, as these provide a conceptual framework for understanding and trying to choose variables for predicting health conditions and treating them among the elderly. The interested reader is invited to read the important review by Schroots (1996). Cumming and Henry (1961) proposed the disengagement theory, claiming that old age is characterized by withdrawal from previous social roles due to a mental withdrawal and focus on oneself. However, this quite pessimistic view does not account for those elderly people who resist such changes and maintain highly active lifestyles. This model can be contrasted with the activity model by Havighurst (1948) which claims that older people substitute past roles with new ones, appropriate to their age, abilities, and social functions, to maintain a sense of self. This more positive model focuses on elderly people's abilities and adaptiveness to the changes in their life circumstances and capacities. Thus, this model looks at changes in resources and focuses on activities rather than on losses and withdrawal.

A newer model is the life span and development model (Baltes 1987; Baltes et al. 1991). This model sees development and aging as synonyms of behavior change across the life span. The model then posits seven claims: (1) individuals differ on their degree of having normal, pathological, or optimal aging; (2) aging is very heterogeneous; (3) elderly people have hidden reserves; (4) the range of reserves and adaptivity declines with age; (5) individual and social knowledge can compensate for age-related reductions in fluid intelligence; (6) the balance between gains and losses declines with age; and (7) the self remains a resilient system of coping and integrity. Stemming from this model was the term "successful aging." This model is quite structured, it proposes individual differences, and it focuses on seeing aging as part of development, with its uniqueness as well.

One of the more recent models in gerontology is Tornstam's (1989) model of gerotranscendence (GT). In the GT model, elderly are seen as shifting from a materialistic and rational perspective to a cosmic and transcendent one. According to the GT model, throughout life we can change our perspective and attitudes toward people and time, as a function of events and maturity. With age, and particularly among the elderly, people may perceive proximity to people or to moments in time, which are distant from them. Thus, elderly people may feel close to loved ones who may have passed away or close to people who are geographically distant from them like their children or grandchildren living in another country. Similarly, they may feel close and relate to events which have long past. A recent study found that university students found such approaches, depicted within scenarios, as more unusual and concerning than elderly people found (Buchanan et al. 2015). That study revealed that the concepts of the GT model are familiar to elderly. One interesting question and assumption about GT has been that GT may increase due to adverse negative life events. Indeed, in a unique prospective Dutch study on 1569 participants aged 58–89, occurrence of negative life events predicted increases in GT (its cosmic element). Similarly, absence of negative life events predicted reductions in this GT element (Read et al. 2014).

The models described here can guide us when evaluating an elderly person's functioning and when trying to identify factors which affect such functioning and other health outcomes. For example, the life span and development model proposes to examine a person's resources, and these could also predict their recovery from surgery or a disease (see Chap. 9). The GT model reflects cognitive resources which may also predict people's coping with and adaptation to diseases and to changes in aging.

Stress and the Importance of Control and Perceived Control in the Elderly

As outlined in Chap. 1, a basic factor affecting the outcomes of stressors is one's appraisals or perceptions of control over situations which one perceives to be stressful. Let's first examine whether the types of stressors elderly people report are different from those of young adults? Aldwin et al. (1996) examined precisely this issue within the Normative Aging Study data. They interviewed 74 early mid-life adults (aged 45–54 years), late mid-life adults (aged 55–64 years), young old adults (aged 65–74 years), and old-old adults (74 +). All participants were men. Both elderly groups reported significantly more often health problems and general hassles as daily stressors. In contrast, they reported less often work stressors, for obvious reasons. The old-old age group reported significantly less use of instrumental coping strategies, less use of interpersonal hostility to cope with stress, and significantly less use of cognitive reframing than the mid-age adult group. This study informs us about the stressors elderly people face and their ways of coping with them. We shall now focus on the issue of control, because older age is characterized by a reduced sense of control, and this is linked to the health problems reported by Aldwin et al. (1996) which often impede objectively the level of control a person has over daily activities.

The importance of control over events and life circumstances is a central issue in stress and adaptation for any life stage and in particular for the elderly. This issue is well reviewed by Mallers et al. (2013) concerning the elderly. Perceptions of control over a situation constitute an important element of the secondary appraisal stage in models of stress, coping, and adaptation (Taylor 1995). One should distinguish between objective control, the degree to which we can actually regulate our environment via selective responses (Rodin 1990), versus subjective control, our cognitive appraisals of what conditions are needed to obtain control over a situation or over our environment (Mossbarger 2005). Numerous studies over several decades, domains, and cultures have shown the importance of perceived control. For example, a large-scale study done on Asian adults in several countries found that those with higher perceived control had lower blood pressure levels (Barrett et al. 2015). In a study conducted with 336 Turkish survivors of the Marmara earthquake, perceived control was significantly and inversely correlated with general distress, intrusions, and avoidance (Sumer et al. 2005), relevant to post-traumatic stress disorder. In one experimental study, spouse caregivers of demented partners were exposed to an acute stress of giving a speech, and their sympathetic response was measured by levels of norepinephrine. Those with higher levels of perceived mastery, reflecting greater perceived control and self-efficacy, had lower norepinephrine levels post-stress (Roepke et al. 2008). However, perceived control may not always be adaptive. In a rare and important study done on 171 women surviving a sexual assault, perceived control was assessed in relation to four time points together with distress. Perception of past personal control, which reflected self-blame, was related to higher distress levels. In contrast, perceived control over the recovery process was indeed adaptive (Frazier 2003). This pattern of results echoes the "goodness-of-fit hypothesis" (Forsythe and Compas 1987) but extends it to perceived control instead of coping: it is possible that having a sense of control predicts better adaptation in objectively controllable but not uncontrollable situations.

It is crucial to investigate perceived control in the elderly because of the obvious reduction in one's objective amount of control over one's physical and mental capacities with age and over one's environment. The latter can be expected from the consequences of retirement and decline of meeting colleagues, peers, and family members with age, though the extent of such consequences can be individual. Furthermore, elderly people who accept stereotypes about the elderly as a period of decline only have indeed lower levels of perceived control (Rodin and Langer 1980). However, from conceptual and philosophical points of views, increasing elderly people's perceived control reflects a shift from viewing them as being socially detached and withdrawn to viewing them as active people with new life challenges, purposes, and reasons for personal growth.

The pioneering study of Langer and Rodin (1976) and Rodin and Langer (1977) aimed to examine the effects of increasing objective and perceived control on elderly people' s well-being and even on their survival. Residents in an elderly home were randomized to have choice over arranging their furniture, move around, see friends, and take care of a plant given to them (control-enhancement experimental group) or were told that the staff is there to take care of them, to help them, and to care for the plant given to them ("external control," control group). During an 18-month follow-up, the health of those in the control-enhancement group was better, and fewer of them died, compared to the usual care and more passive control group. This pioneering study was crucial as it stimulated change in the vision of elderly people and their needs. Furthermore, it showed how simple but theory-driven psychological interventions without any costs can yield important health outcomes in the elderly. Such studies need to be replicated, and their underlying mechanisms should be revealed.

In a more recent study, Ruthig et al. (2007) investigated the mediating role of perceived control and optimism, in the effects of falling on health outcomes, in 231 elderly people. Only in the old-old subgroup (≥85 years old) did perceived control (and optimism) mediate the relationship between falling and well-being. These results also propose the important need to increase perceptions of control in elderly people following falling, an unfortunately common and debilitating problem in that age group. Furthermore, these results highlight the importance of perceived control even in the old-old elderly group of people.

Another large-scale study conducted in 926 elderly people tested the factors associated with fear of falling (FF) and with fear-induced activity restriction, reflecting disability. Nearly 50% reported at least some FF, of whom nearly 65% reported fear-induced activity restriction. Levels of perceived mastery, a concept highly related to control and self-efficacy, were inversely related to FF and to fear-induced activity restriction (Deshpande et al. 2008). We shall return in greater detail to the topic of falling in the elderly, below. Together, these studies reveal the centrality of control and perceived control in the elderly, in their health and well-being, and call for more research and applied intervention research on this important topic.

Psychological Aspects of Dementia

The prevalence of dementia is high and increasing due to the aging population. Alzheimer's disease (AD), the most common cause of dementia, affects 27 million people worldwide (Wimo et al. 2006), and its prevalence is expected to triple by 2050 (Hebert et al. 2003). One important psychological aspect of dementia is the occurrence of behavioral and psychological symptoms in dementia (BPSD). It is estimated that up to 90% of demented patients will experience BPSD sometime during their illness. BPSD include agitation and irritability, aberrant motor responses, anxiety, elation, depression, apathy, disinhibition, delusions and hallucinations, sleep, and appetite changes (Cerejeira et al. 2012). Furthermore, BPSD have an independent prognostic value in relation to hospitalization and medication use. The neuropsychiatric inventory is among the most important semi-structured interview used for assessing the BPSD (Cummings 1997). Importantly, these symptoms are among the crucial reasons caregivers decide to institutionalize their demented family member (Cerejeira et al. 2012). Interestingly, one study found an inverse relationship between severity of dementia and changes in BPSD (Aalten et al. 2005). It is possible that more severe forms of dementia are characterized by patients shifting to be more silence and to have withdrawal behavior which replace the symptoms typical of BPSD. Fronto-temporal lobe degeneration, a specific type of neural degeneration seen in dementia, is typically associated with various aspects of the BPSD (Cerejeira et al. 2012). This pattern of neurological degeneration may explain the disinhibition and emotional regulation difficulties in BPSD, which rely on intact prefrontal functioning (Gross 1998).

Do psychological factors predict prognosis in dementia? In a study on 116 demented people, life events were assessed during 6 months, and patients' survival was subsequently examined over 5 years. Life events, and particularly experiencing depression, significantly predicted a reduced chance of survival during follow-up (Butler et al. 2004). The uniqueness of this study is that life events and psychological factors were assessed serially over 6 months, and the study was of course prospective. This study also showed the dynamics of stress and adaptation, where depression, often seen as an outcome of stress, can then be a stressor and risk factor on its own right. Another unique longitudinal study measured both psychological stress and cognitive performance repeatedly over time, in cognitively healthy individuals (N = 61) and in people with mild cognitive impairments (N = 41), aged 65–97 years. Cortisol levels were also measured. While higher event-related stress levels over time were associated with faster cognitive decline, higher cortisol levels were related to slower cognitive decline, both found only in the group with mild cognitive impairment (Peavy et al. 2009). These results propose an adverse prospective association between event-related stress levels and cognitive deterioration over time, while they propose a possible protective role of cortisol in cognitive performance of elderly people with mild cognitive impairment. These results are in line with possible boosting effects cortisol may have in memory formation (Guenzel et al. 2014). Nevertheless, more research of this type is needed to examine whether psychological factors predict the progression of dementia and whether other mediators such as inflammation play a role in such associations.

Psychological Aspects of Falling in the Elderly

The prevalence of falling in elderly is high. In one study conducted in Brazil, 38.6% reported to have fallen (Fhon et al. 2013). In a study conducted in 2273 elderly Italians in 30 communes, the prevalence of falls in the past 12 months was 28.6%, among whom 43.1% had fallen two or more times. The risk factors of falling included stroke, visual difficulties, diabetes, urinary incontinence, physical inactivity, and consumption of antianxiety medication (Mancini et al. 2005).

This issue is of huge importance in the elderly because it produces a vicious circle of fears and disability. After falling and suffering from possible fractures, elderly people may develop more fears of walking and of doing daily activities independently, consequently being more disabled which could then weaken their muscles and further increase their risk of falling and so on. Among the risk factors of falling are hazards at home and sarcopenia, weak muscle functioning (e.g., Pfortmueller et al. 2014). Another aspect involved in this vicious circle is vitamin D deficiency. Vitamin D is important for bone strength (e.g., Ahmadieh and Arabi 2011). Furthermore, lack of vitamin D is related to mental health: in a large sample of 1618 elderly people, those with low levels of vitamin D had higher levels of depression (Lapid et al. 2013). People with depression may remain more homebound, and one important source of vitamin D is sun exposure. Indeed, low levels of vitamin D have been associated in some studies with less sun exposure (Roomi et al. 2015). Another factor which affects levels of vitamin D is diet, with particularly fish oil fat being a significant source of vitamin D (Lehmann et al. 2015). All these factors together may contribute to little vitamin D and subsequently to weaker bones and further risk of falling. Another biological factor which may contribute to falling is soluble vascular cellular adhesion molecule 1 (sVCAM-1). Indeed, high levels of sVCAM-1 are related to cerebral blood flow dysregulation and even significantly predict falling in the elderly (Tchalla et al. 2015a). Furthermore, levels of sVCAM-1 were positively associated with depressive symptoms in the same elderly sample (Tchalla et al. 2015b). These can then increase the risk of falling due to poorer cerebral control of movement and of attention to hazards. Falling, in turn, can then further increase fear of recurrent falling as both were highly prevalent in the elderly (e.g., Fhon et al. 2013), resulting in remaining more at home and then again in less sun exposure and subsequently low vitamin D levels, lack of physical activity, all which again increase the risk of more falling in a vicious circle.

One factor which could maintain this vicious circle is negative reinforcement: avoiding to go out prevents falling out of the home, and this prevention reinforces elderly people's remaining at home. However, not all studies confirm such a vicious circle chain. One prospective Australian study found that baseline fear of falling did not predict future risk of falling and nor did past falling predict future fear of falling (Clemson et al. 2015). However, depression did predict risk of future falling, while cognitive impairment and reduced social activity (among other factors) predicted future fear of falling (Clemson et al. 2015). Others have also proposed similar vicious circle models (e.g., Rolland 2011) but have not taken all psychobiological factors mentioned here into account. Figure 8.1 thus proposes a more comprehensive model, based on the corroborative evidence proposed here.

Fig. 8.1 A proposed psychobiological model of fear of falling

Psychoneuroimmunological Mechanisms

This chapter covers multiple topics in psychogeriatrics. For this reason, this section on psychoneuroimmunology will provide some general findings as well as specific examples which may help understand the relationship between psychosocial factors and health and illnesses in the elderly. One important pathway by which elderly people may be more ill from conditions which depend on or progress with inflammation is reduced vagal activity. First, as repeatedly mentioned in this book, the vagus nerve physiologically reduces inflammation, either via activating the HPA-axis or via a splenic pathway activating specific splenic T-cells producing acetylcholine, which bind to the alpha-7 nicotinic acetylcholine receptor on monocytes, resulting in less inflammation (Rosas-Ballina et al. 2011; Tracey 2009). Many chronic diseases progress via low-level inflammation which could then be moderated by vagal nerve activity (De Couck et al. 2012; Gidron et al. 2018). In contrast, many studies show that vagal nerve activity, indexed by HRV, reduces with age (Abhishekh et al. 2013), and reduced HRV is also associated with depression (e.g., Patron et al. 2012). Thus, lower HRV could mediate associations between, for example, depressive symptoms and progression of heart disease or of cancer (e.g., Chida et al. 2008), diseases which prevail among elderly people. Indeed, Carney et al. (2005) found that the association between depression and post-MI mortality was partly mediated by reduced HRV. Future studies should examine such mediation in the elderly and then examine whether increasing HRV improves both depression and health outcomes in that age group.

Caregiving, as a family member, for a demented person, constitutes a chronic stressor. This is stressful because of the amount of tasks and hours caregiving demands per day, because it reduces actual control over one’s own life, and because the care recipient often provides no observable positive reinforcement (if severely demented). One study compared endothelial dysfunction, a precursor of coronary artery disease, in 55 caregivers of demented spouses and in 23 married non-caregiver control elderly people. Endothelial dysfunction was measured using the flow-mediated dilation (FMD) method. Levels of FMD were significantly lower in caregivers of a moderate-severely demented spouse than in caregivers of mildly demented spouses and compared to non-caregiver controls. Importantly, within the caregiving group, number of years of caregiving was significantly and inversely related to FMD ($r = -0.465$, $p < 0.001$; Mausbach et al. 2010). Such a finding can explain the association between psychosocial stress and cardiac events in the elderly (e.g., Liu and Waite 2014).

A very unique and scientifically significant study examined the relationship between executive functioning (EF), stress exposure, stress responses, stress recovery and stress restoration, and finally with status of frailty (pre-frailty versus non-frail). Stress exposure was assessed by a questionnaire, stress responses were measured by heart rate changes due to a lab stressor, and stress recovery was the rate of recovery in HR following the stressor. Finally, stress restoration was operationalized by levels of interleukin-6 (IL-6) and sleep quality. As expected, EF levels were lower in pre-frail than in non-frail individuals. Importantly, this relationship was statistically mediated by higher perceived stress and IL-6 and poorer sleep quality (Roiland et al. 2015). This important study not only reveals the possible role of stress and inflammation in neurodegeneration of frail people, but it considers these relationships within a "holy triangle" framework including the brain (EF), psycho-biological changes (stress, IL-6, and sleep quality), and a health outcome relevant to the elderly, namely, frailty.

Telomere length (TL) is a biomarker of cellular aging or senescence and thus is highly relevant for understanding processes in the health and illness of the elderly. TL has been associated with chronic stress, in several studies beginning with the pioneering study of Epel et al. (2004). One study on caregivers of elderly people, which was not restricted to but also included elderly caregivers, examined caregiving parameters and TL. Among caregivers, more hours/week and greater perceived strain were significantly related to shorter TL (Litzelman et al. 2014). This could be one mechanism by which psychological factors contribute to onset of disease and to accelerated aging in the elderly. However, this conclusion must be viewed with caution because a recent study found that change in TL did not significantly predict change in cognitive or physical abilities in the Lothian birth cohort study (Harris et al. 2016). Nevertheless, reduced TL is a predictor of death and diseases in the elderly (e.g., Fitzpatrick et al. 2011).

Taken together, these studies propose that psychosocial factors, which predict or affect the health of the elderly, may do so via affecting their levels of inflammation, vagal nerve activity, EF, and TL. The vagus nerve, indexed by HRV, is related to most of these factors (e.g., Gidron et al., 2018; Tracey 2009) and thus could be a major mediator in such relationships and an important therapeutic target in the elderly.

Psychological Interventions in Psychogeriatrics

We now end this chapter with reviewing some intervention studies done in the elderly. Some were mentioned above in the topic of perceived control. Here I shall review others, which I selected to reflect different types of interventions with different foci and outcomes, all relevant to elderly people. A relatively recent study examined the effects of a support intervention guided by the gerotranscendence (GT) framework, in an RCT design. They used measures of GT, depression, and life satisfaction as outcomes, before and 1 week after the intervention or control groups were administered. The GT-based intervention included sharing experiences of universe-, self- and social transcendence, as well as ways to increase life satisfaction via the GT framework. Results revealed significant increases in GT and life satisfaction in the experimental group as well as lower levels of depression posttreatment in the experimental group, compared to controls (Wang et al. 2011). The novelty of this study is its use of the GT model as a framework for a supportive treatment and the use of outcomes of huge relevance to the well-being of elderly people.

An older but scientifically elegant and clinically important study used the framework of learned helplessness, but in the positive sense, and tested whether control over one's social support can improve the well-being of elderly (Schulz 1976). Elderly people were randomized to one of four conditions: no extra social support, to receive social support at times informed to them in advance (predictability), to receive social support at times and durations of their own choice (full control), or to received support at random times and durations. The second and third groups were matched such that duration and frequency of visits were the same. Being able to predict visits and having self-control over such visits positively affected the elderly people's well-being (Schulz 1976). This study is scientifically and clinically important because it demonstrates the robustness of the learned helplessness model and since its clinical implications for reducing elderly people's loneliness are crucial.

Concerning fall prevention, many studies have been conducted on this important issue. An outstanding study was conducted by van het Reve and de Bruin (2014) in which they compared a strength-balance (SB) intervention with a SB-cognitive (SBC) intervention which additionally included computerized cognitive training exercises. In both groups, there were significant reductions in fear of falling and in *actual* falls; however, performance speed, gait initiation, and attentional tests were improved significantly more in the SBC group. While providing important and impressive results, this study lacked a real control group. Nevertheless, its results inform us on the importance of teaching elderly people both strength-balance exercises and cognitive training, to improve movement and attentional skills needed to prevent falls.

One intervention study compared people who commonly practice love-kindness meditation (LKM) to controls. LKM encourages people to meditate and perform good gestures to other people. Telomere length (TL) was found to be significantly longer in woman practicing LKM than in controls (Hoge et al. 2013). However, this study did not use an RCT design and it was not performed only in elderly people. Despite this limitation, this study may have implications for the ability of interventions used in behavioral medicine to affect numerous health outcomes in general and in elderly specifically, because of the known relationship between TL and morbidity and mortality (e.g., Fitzpatrick et al. 2011).

One repeatedly tested type of intervention, suitable for elderly people, is the life-review intervention. Various methods and protocols exist. For example, Chippendale and Bear-Lehman (2012) randomly assigned elderly people to an 8-week share your life story intervention or to a wait-list control condition. In the life-review intervention, participants wrote about their lives and also shared this with other participants and received positive feedback. Depressive symptoms were significantly less prevalent in the life-story group than in controls after treatment. Another unique study, pertinent to elderly people near death, compared terminally ill cancer patients undergoing a brief life-review intervention to similar patients undergoing a supportive control condition. In the life-review intervention, patients answered questions including their most vivid and impressive life memories, important moments, influential people, and moments of pride in their lives. Results revealed significantly greater improvements on multiple outcomes such as hope, a sense of life completion, and suffering in the life-review condition than in controls (Ando et al. 2010). Such an intervention is so often needed and is even more rarely tested scientifically. The study by Ando et al. (2010) thus provides unique evidence, and its results could be easily applied for the benefit of patients and possibly their close ones, to come to a better sense of closure in such delicate human moments.

Finally, in an older study, a relatively simple method of relaxation, namely, progressive muscular relaxation (PMR), combined with guided imagery, was tested among 45 elderly people (mean age = 74 years). They were randomly assigned to either PMR + guided imagery, to social contact only, or to no treatment. The interventions were provided individually, three times/week for 1 month. PMR includes the contraction and relaxation of body segments from legs to eyebrows, individually. Immune and psychological outcomes were assessed at baseline, soon after the interventions and 1 month after the interventions. Natural killer cell activity (NKCA) increased significantly only in the PMR + guided imagery condition, and this was paralleled by improvements in distress in that group as well (Kiecolt-Glaser et al. 1985). This study provides compelling evidence that a simple but skill-oriented systematic relaxation intervention provides actual immunological and psychological benefits, over and above merely providing support to elderly people. Too many so-called “support groups” need to upgrade themselves to incorporate such skills.

Case Study Conclusion

Mr. Sanderman has a history of past falls and bone fragility and is quite depressed. He has a clear fear of falling. Due to these, he entered in a vicious circle of fearing more falls, and remains at home, which is negatively reinforced by avoiding the falls when going out. However, the long-term price of this behavior is reduced exposure to sun-related vitamin D and reduced exercise, thus making his bones and muscles more fragile and more prone to future falls, etc. His depression also increases the risk of nonadherence to medication and to other adverse health outcomes. To change this, we need to disconnect the association he has developed between going out and falling, using systematic desensitization. This could be done by a set of meetings in which we gradually make more vivid images and actions related to walking outside followed by deep relaxation, rather than followed by anxiety responses. In addition, using “psychological inoculation,” one could address any exaggerated and overwhelming negative thought distortions he may hold about the risks of walking outside. For example, he could be told the following challenging sentence: “ALL your friends who walk outside FALL EVERYTIME they go out!” If he would refute this sentence by saying, for example, “No, no, my best friend Charles takes daily walks and he is fine!”, this would help to eventually break his cognitive distortions underlying his fear of falling. Finally, the walking could be done with his granddaughter. These steps could hopefully break the vicious circle and increase his sun exposure and subsequent vitamin D levels and muscular use, thus increasing his bone and muscular strength and self-efficacy to perform such activities. Hopefully, he would then fall less due to his progress.

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Chapter 9
Effects of Psychological Predictors and Interventions on Recovery from Surgery

Case Study

Ms. Cohen from Tel-Aviv is 44. She has undergone a finger operation after she broke it while playing football with her youngest son. She is a leading pianist and teacher. She has had hundreds of students and has given many concerts worldwide. However, she has not played the piano for about 5 weeks and has become quite sad from this. You invite her for a routine post-operative checkup as the treating orthopedist, 4 weeks after surgery. However, you are surprised to discover that the wound healing is extremely delayed and the bone reunion has not even begun. She promised you that during the last 4 weeks, she has not attempted to play the piano at all. You believe her but cannot explain this retarded recovery, despite having operated many patients with similar broken fingers!

The Biology of Wound Healing and Surgical Recovery

Surgery involves deliberate cutting often of skin, muscles, and at times bones as well, in the attempt to reach and repair an ill organ or tissue. Following such an assault to the body, healing gradually takes place. Wound healing includes two main steps – regeneration and repair. Regeneration involves non-injured healthy cells replacing the specific damaged ones. Regeneration includes several known processes and stages, which can be generally grouped into four: bleeding, inflammation, proliferation, and remodeling. In reality, because each phase stimulates the subsequent one, these are in fact interlinked. The bleeding phase is time-limited and lasts on average about 4–6 hours, depending on the type of tissue damage and how vascular it is. For example, muscles may take longer to stop bleeding than other tissues (Watson 2012). The inflammatory phase begins a few hours post injury and

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peaks at between 1 and 3 days later. It is an essential response, initially. Pro-inflammatory cytokines such as interleukin-1 (IL-1), IL-8, and tumor necrosis factor (TNF) alpha are pivotal in protecting the injured tissue from infection, preparing the injured tissue for repair and in activating phagocytic responses. Such cytokines, released from the recruited immune cells, help epithelial cells and fibroblasts to remodel the injured cite (Barbul 1990). The proliferative phase begins 24–48 hours post injury, peaks at 2–3 weeks later, but lasts 4–6 months post injury. During this phase, collagen increases to produce the scar response at the injury site. The last phase is remodeling, which begins a week post injury and results in a scar which attempts to act like the damaged tissue it replaced. This is more like a long-term mend of the initially damaged tissue. This is a naturally occurring response, however, if delayed or absent due to repeated trauma, therapeutic aid is needed (Watson 2012).

In recent decades, surgeons have more and more used minimally invasive procedures by inserting thin tubes called trocars in a procedure called laparoscopy. One of the trocars will be the path by which a miniature camera will be inserted for the staff to view the surgical site on a TV screen, while the others will serve for inserting the surgical knifes. Laparoscopy results in less pain, fewer incisions, and smaller length of stay in hospital (Baltayiannis et al. 2015). However, in many types of specific laparoscopy surgeries, these claims are not supported by empirical evidence because either there are no significant differences in preventing disease relapse or complications or because of paucity of randomized controlled trials (e.g., Kuhry et al. 2008).

During surgery, a physiological stress response emerges, which includes sympathetic nerve system activation, endocrine stress responses including pituitary hormone release, and insulin resistance. Immunological and hematological changes include the acute-phase reaction and cytokine production, increased neutrophils, and lymphocyte proliferation (Desborough 2000). For example, one study found levels of norepinephrine to rise by 60% during third molar teeth extraction (Goldstein et al. 1982). These responses were thought to initially catabolize body fuels needed for survival after injury. However, current surgical practice does not require such a response. At the endocrine level, corticotropin and cortisol are released; arginine vasopressin secretion affects the kidneys; insulin is diminished, all which increase catabolism to provide energy sources. Furthermore, these changes increase abilities to maintain water and salt (Desborough 2000). Activation of the HPA-axis stimulates the sympathetic nervous system, resulting in increased epinephrine and norepinephrine. These lead to typical cardiovascular changes including tachycardia and hypertension. However, the functioning of other visceral organs such as the liver and pancreas is also modified by direct sympathetic innervation or from these neurohormones spilling into the circulation (Desborough 2000). In addition to increases in cortisol and adrenocorticotropic hormone (ACTH) within minutes after surgery, levels of growth hormone also increase due to surgical stimulation. Interestingly, the usually regulatory functional negative feedback of cortisol in the HPA-axis seems not to function postoperatively, resulting in elevations of both cortisol and ACTH (Desborough 2000). Increased cortisol has metabolic and immunological effects. It can lead to lipid breakdown, and it can reduce inflammation (macrophage and neutrophils in stressed sites) and leads to a temporary state of insulin resistance (Desborough 2000).

Surgical stress elicits three primary pro-inflammatory cytokine responses: interleukin-1 (IL-1), tumor necrosis factor alpha (TNF-alpha), and IL-6. While initially IL-1 and TNF-alpha are secreted by monocytes and macrophages in the damaged regions, elevated IL-6 results from these cytokine increases and has systemic effects (Sheeran and Hall 1997). After major operations such as colorectal or vascular surgery, levels of cytokines can remain high up to 48–72 hours postoperatively. Importantly, both IL-1 and IL-6 can stimulate production of ACTH and subsequently of cortisol. However, cortisol can then reduce inflammation systemically (Desborough 2000). As mentioned above, the inflammatory response is part of the natural post-operative regeneration response. While it is essential, it must remain short-lasting and not too strong over time, because it predicts post-operative complications. For example, a recent study found that levels of the general inflammatory marker C-reactive protein (CRP) at day 4 after gastrectomy for gastric cancer was a significant and independent predictor of post-operative complications in Korean patients (Kim et al. 2017). The vagus nerve, as mentioned throughout this book, could of course protect against such outcomes, if its anti-inflammatory reflex functions adequately (Tracey 2009).

Another issue, belonging more to the neurobiology of surgery but relevant to behavioral medicine and neuroscience, is the phenomenon of awareness during surgery and full anesthesia. This is clearly potentially a traumatic experience for patients since only their muscle paralyzing agent works, while their analgesic and drug for blocking consciousness do not work. The patient may be in major pain without the ability to communicate this at all. However, the prevalence of this is thankfully quite low – between 0.13% and 1.0% for definite awareness and between 0.00084% and 9.82% for definite + probable awareness (Mashour and Avidan 2015). Part of the variability in these rates is due to the type of surgery, type and amount of anesthesia, and the manner of assessing awareness during anesthesia. In an older study conducted at the Soroka University Hospital of Ben-Gurion University, Israel, we examined the issue of unconscious awareness during anesthesia and its determinants. We exposed patients undergoing laparoscopic cholecystectomy to neutral or emotionally negative word pairs during surgery (e.g., table-chair; ugly-disgusting) and then asked patients postoperatively to provide word associates to cue words either presented during surgery or not (control words). Postoperatively, the reaction time (RT) to provide word associates was faster if the word cue was presented during surgery *and* was emotionally negative. Furthermore, level of brain activity during surgery, as measured by electroencephalography, correlated significantly only with the post-operative RT to providing word associates to emotionally negative word cues presented during surgery (Gidron et al. 2002). Given those results, future studies need to examine the incidence of emotionally negative words said by the surgical staff during surgery and its effects on patients' implicit intrasurgical awareness and post-operative recovery and well-being.

Do Psychological Factors Predict Recovery from Surgery?

Before answering this question, let's examine what research teaches us about patients' concerns before surgery. Yoon et al. (2012) inquired about this issue in 303 Korean patients about to undergo arthroscopic rotator cuff surgery (in the shoulder). Using structured questionnaires, they found that patients were concerned about having post-operative scars and pain, the duration of recovery, whether they will have a caregiver in hospital, and for women – having difficulties to comb one's hair. Among elderly patients, concerns included reduction in functional abilities, payments for the surgery, and whether the surgeon was too young (Yoon et al. 2012). One study focused on heart-focus anxiety – attention to and fears about one's heart, among patients undergoing cardiac surgery, compared to matched controls, undergoing orthopedic surgery. Among the cardiac patients, total cardiac anxiety as well as fear, avoidance, and attention were significantly higher than in controls, before surgery (Hoyer et al. 2008). These results clearly show the concerns and fears which preoccupy patients before surgery, which need to be addressed by health psychologists and medical staff. As we shall see below, many of these issues can be dealt with by providing patients sufficient information.

Do people with different psychological states and traits recover differently from the same surgery? If so, how can this happen, and does this have implications for helping people recovery better? In this section, we will review some studies on the relationship between pre-surgical psychological factors and recovery from surgery. I will review only prospective studies which statistically controlled for background factors (e.g., age, gender) and medical factors (e.g., severity of surgery, disease severity). Controlling statistically for these factors is essential for examining the unique contribution of psychological factors to recovery from surgery.

One important systematic review (Mavros et al. 2011) examined the relationship between pre-surgical psychological variables and objectively measured indices of recovery. They examined the following psychosocial factors: distress, depression, loneliness, state and trait anxiety and anger, anger expression and anger control, perceived stress, social support, optimism, intra-marital relationships, religiousness, expectations about recovery, health locus of control, worry about the surgery, examination-induced stress, and, finally, stress from a demented relative (reflecting background chronic stress). Recovery included wound healing and inflammation, cytokines in wound fluid, clinician-assessed recovery, and post-operative complications. They found 13 studies, with all except one being prospective, which included in total 1347 patients. While some studies found that anxiety or distress did significantly predict poorer recovery, others did not. Mode of anger expression (in versus out) did not predict recovery, but better anger control predicted faster wound healing. State anger predicted poorer outcomes but not in all studies. Intra-marital hostility predicted slower wound healing. Decreased religiousness predicted more post-operative complications. Little pain expectations predicted less swelling, while external health locus of control predicted faster wound healing. Two studies found that vigilant (versus avoidant) coping predicted more complications. Depression was related to slower healing but was unrelated to post-operative complications. Optimism was found in this review to have a minimal association with recovery, and social support was not predictive of recovery. This pattern of results suggests that in several types of surgeries, certain psychosocial factors do predict recovery from surgery. Importantly, the pattern also proposes that an avoidant coping and external health locus of control may have beneficial roles in surgical recovery. How can this be explained? In line with the goodness of fit hypothesis (Forsythe and Compas 1987), it is possible that because the surgical context is not under patients' control, more emotion-focused coping and "letting go" responses may be more adaptive in such uncontrollable situations.

A major systematic review of 29 studies which used a prospective design and which statistically controlled for known clinical prognostic factors was conducted by Rosenberger et al. (2006). They found that attitudinal and mood factors strongly, significantly, and independently predicted recovery from surgery. In contrast, personality factors had a far weaker prognostic role in recovery from surgery. These results point at the prognostic role of situational factors relevant to the surgical context such as patients' attitudes and specific pre-surgical mood compared to the role of more general and robust issues such as their personality. This is important since it is more possible to influence patients' specific mood and attitudes about surgery than to alter their more stable and global personality patterns, to try and improve their recovery from surgery.

Another review of studies focused specifically on psychological predictors of recovery from anterior cruciate ligament (ACL) reconstitution surgery (ligament below the knee), often hurt among dancers or people doing intensive sports. They found eight prospective studies meeting their inclusion criteria. The review found evidence for higher self-efficacy and optimism to predict better functional outcomes post-operatively. Some of these relationships may have been due to these factors (including social support) predicting also adherence to post-operative recommended rehabilitation exercises, suggesting a biobehavioral mediation. Surprisingly, the review found no evidence for the predictive validity of pain catastrophizing or of fear of re-injury with recovery (Everhart et al. 2015). This review has important implications for treatment because people can be taught how to increase their self-efficacy via mastery and to increase their optimism via receiving realistic information about the post-operative phases. I shall now review several studies as specific examples for the predictive role of psychosocial factors in recovery from surgery.

Two older studies examined the role of expectancies and optimism in recovery from surgery. Scheier et al. (1989) assessed dispositional optimism using the Life Orientation Test (LOT; Scheier and Carver 1985) in 51 patients before undergoing coronary artery bypass surgery. I review this study because of its very high level of scientific methodology and crucial findings, despite it being relatively old. Patients' myocardial responses during surgery, functional capacity in the days after surgery in hospital, and their quality of life 6 months later were assessed. Importantly, disease severity, number of grafts performed (partly reflecting surgical severity), and an index of several heart disease risk factors (e.g., smoking) were statistically controlled for. First, optimists had significantly fewer new Q-waves than pessimists, indicating less chance of a myocardial infarction perioperatively. Furthermore, optimists took fewer post-operative days to begin to walk in the hospital than pessimists. Finally, optimists were significantly more likely than pessimist to resume more life domains at 6 months after surgery (Scheier et al. 1989). This methodologically sound study revealed the wide role optimism may play in physical and functional short- and long-term recovery from cardiac x^2 surgery.

Another study examined in Canadian adolescents whether psychosocial factors predict recovery from oral surgery. Similar to Scheier et xx^{22} al. (1989), Gidron et al. (1995) statistically controlled for extent of surgery, using a validated scale completed by the oral surgeons. They examined whether expectancies concerning recovery from surgery, coping, negative affect, and mothers' illness behavior encouragement predicted adolescents' mouth opening, pain, and disability a week after surgery. Illness behavior encouragement reflected mothers' inadvertent reinforcement of children's illness behavior (e.g., pain complaints) via extra attention or gifts when seeing such behaviors. Results showed that optimistic expectancies were positively and significantly related to post-operative mouth opening and inversely related to disability, independent of surgical severity. Illness behavior encouragement significantly and inversely predicted mouth opening, independent of surgical severity. Coping strategies did not predict recovery at all. That study revealed the importance of optimistic expectations and parental responses to children's symptoms in relation to adolescents' recovery from surgery.

In contrast to that last study, a more recent one examined whether pre-surgical pain catastrophizing coping and anxiety predicted post-operative hand grip, hand motion range, and disability in $n = 93$ patients undergoing surgeries after hand fractures (Roh et al. 2015). While pre-surgical catastrophizing and anxiety predicted all three outcomes at 3 months after surgery, the effects of anxiety alone seemed to remain constant at 6 months. These findings are important because health psychologists and clinicians can target pre-surgical pain catastrophizing via cognitive restructuring and try to reduce anxiety via cognitive restructuring and relaxation methods. The section below will inform us about the effects of psychological interventions on recovery from surgery.

One major variable often tested in behavioral medicine is self-efficacy, which reflects one's perceived ability to perform certain behaviors, usually in face of challenges or barriers. A systematic review of 8 studies and 967 patients examined whether self-efficacy predicts functional recovery and well-being after hip or knee replacement surgery. Results found no conclusive evidence for the predictive validity of pre-surgical self-efficacy. In contrast, post-operative self-efficacy did predict post-operative functional recovery (Magklara et al. 2014). It is possible that only after experiencing the surgery and being exposed to one's actual initial post-operative ability, does self-efficacy predict longer-term recovery.

Of relevance to the last issue, a few studies examined whether post-operative factors predict long-term recovery after surgery. One study focused on sense of coherence (SOC; Antonovsky 1993). To remind the reader, SOC reflects a person's comprehensibility (understanding and linking one's events in life), manageability (meeting one's planned tasks or goals), and meaningfulness (viewing one's efforts as worthwhile). In a study on 74 Finnish patients with lumbar spinal stenosis who underwent decompressive surgery, levels of SOC 3 months after surgery significantly predicted levels of pain, disability, life satisfaction, and depression over a 5-year follow-up (Pakarinen et al. 2017). This study as well as the review on self-efficacy mentioned above inform us to also consider the predictive power of post-operative psychological resilience factors in the long-term recovery of patients, especially those undergoing surgery with possible long-term disability.

Several studies have examined the predictive role of social support in recovery from surgery as well. I chose to cite a unique one, because of the outcome variable it focused on – post-operative delirium (Do et al. 2012). Post-operative delirium is characterized by lack of awareness of the environment and attentional changes, and its symptoms include disorientation, hallucinations, and memory dysfunction, among others. In their study, $n = 106$ patients undergoing orthopedic surgeries were assessed preoperatively for social support and other factors. Results showed that patients' satisfaction with their social support, not the size of their supporting network, significantly predicted post-operative delirium, independent of age, type of surgery, preoperative medications, and depression. Metaphorically speaking, those unhappy with their support network disconnected more from their environment after surgery.

To summarize this section, please look at Fig. 9.1. Using Taylor's (1995) framework of stress, coping, and adaptation, I mapped some of the main findings reviewed above on this framework. The figure shows that numerous external and internal resources and impediments predict various aspects of recovery from surgery. Furthermore, appraisal variables (e.g., optimistic expectancies) and coping also predict recovery from surgery. All these can serve as targets for preparing patients for surgery, our next topic.

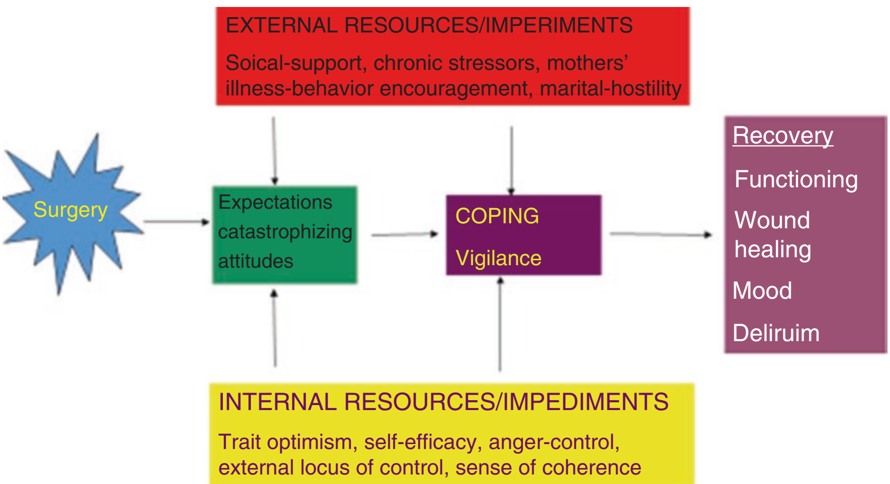

Fig. 9.1 Mapping reviewed studies on predictors of recovery from surgery on Taylor's framework of stress, appraisal, coping, and adaptation (Taylor 1995)

Psychoneuroimmunology of Surgery

Do pre-surgical stress responses predict perioperative physiological responses? One study found that levels of pre-surgical anxiety were associated with lower intraoperative cortisol levels, independent of preoperative cortisol levels (Pearson et al. 2005). I hypothesize that highly anxious people may have a dysfunctional HPA axis (Faravelli et al. 2012), possibly resulting in lower intraoperative cortisol. This may then lead to insufficient control of excessive immune responses, which may result in more post-operative inflammation and complications. However, such a hypothesized chain of reactions needs empirical support.

Another study examined whether pre-surgical cortisol levels predicted new cases of post-operative depression, following abdominal aortic aneurysm surgery. In addition to a history of past depression, baseline levels of pre-surgical cortisol significantly predicted baseline and post-operative depression levels at the 9-month follow-up (King et al. 2015). While not focusing on a physical index of recovery, this study shows that pre-surgical cortisol, a stress hormone, is a risk factor of long-term post-operative depression. This also demonstrates the bi-directional relationships between psychological and physiological factors, a point to consider in behavioral medicine.

One important study examined the relationship between surgery-induced immune cell redistribution and post-operative recovery in patients undergoing knee surgery (Rosenberger et al. 2009). Surgery induces a physiological stress response and temporary increases in immune trafficking (Viswanathan and Dhabhar 2005), which is thought to be initially adaptive if it is brief and not excessive. In a study on $n = 57$ patients, recovery was assessed with the Lysholm rating scale which considers knee functioning, mobility, pain, and ability to perform daily activities. Results revealed that levels of lymphocyte and neutrophil redistribution significantly and positively predicted recovery following knee surgery at week 1. Levels of lymphocyte redistribution alone also significantly and positively predicted recovery at week 24. Though these results are relevant to the relationship between perioperative surgical physiological stress and recovery, the researchers did not assess any psychological variable. This could have enabled them to conclude that their observed surgical stress-induced immune redistribution may have mediated effects of psychological factors on recovery from surgery. Future studies should examine this issue.

In contrast to the latter study, one study performed a more comprehensive examination. The researchers followed patients undergoing two forms of spinal cord surgery and assessed their levels of pain, stress, and anxiety, as well as their natural killer cell activity (NKCA) and interleukin-6 (IL-6) levels, all measured 1 week before, the day before surgery, a day after, and 6 weeks after surgery. Results revealed that levels of mental distress were associated with reduced NKCA before surgery. After surgery, levels of NKCA continued to decline, while IL-6 rose. Finally, in the recovery stage, levels of NKCA increased and returned to or above baseline levels, paralleled by reductions in mental distress and pain (Starkweather et al. 2006). These results reveal important associations between psychological and immunological processes taking place perioperatively and propose to test whether some of the associations between adverse psychological factors and post-operative recovery may occur via reduced antiviral immunity (NKCA) or via initially increased inflammation (IL-6).

Post-operative recovery also includes healing of skin wounds. In an early and pioneering study, Kiecolt-Glaser et al. (1995) compared the speed of experimentally-induced wound healing of women who took care of a chronically ill demented spouse (caregivers with chronic stress) to the healing of matched controls. In addition, levels of mRNA-interleukin-1 (mRNA-IL-1), reflecting a potential local inflammatory response, were measured at the wound site, because inflammation is initially crucial for wound healing. Results revealed that the wounds of women with chronic stress took significantly more days to heal than those of controls. Furthermore, the chronically stressed group produced lower levels of mRNA-IL-1, reflecting less synthesis of IL-1, which is crucial in recruiting other immune effectors and growth factors essential for wound healing.

We began this chapter with a case depicting depressive symptoms and impaired bone healing following a fracture. But are they scientifically related? Yirmiya et al. (2006) conducted a pioneering study in which they induced depression in rodents, via chronic mild stress. Astonishingly, depression led to reduced bone mass, reduced osteoblasts, and less connectivity density. These changes were mediated by sympathetic activity, since they could be inhibited by the beta-blocker propranolol. These results provide causal relationships between depressive-like behavior, sympathetic activity, and impaired bone structure and thus could partly explain how depressive states may affect recovery from surgery (Mavros et al. 2011), particularly orthopedic surgeries or recovery from fractures.

Finally, one study examined whether various parameters reflecting heart rate variability (HRV) predict cardiac outcomes in patients undergoing pulmonary surgery. In a study on $n = 117$ patients, 16% developed atrial fibrillation (AF) after surgery. Results revealed that patients with higher HRV (RMSSD) and lower change in HRV (between the afternoon, night, and day) were at higher risk for post-operative AF (Ciszewski et al. 2013). It is possible that surgical stress demands some vagal withdrawal, and its absence may be unhealthy. In contrast, constant HRV without change in HRV may reflect lack of adaptation to changing circumstances, which also predicts poor post-operative outcomes. On the other hand, a study on $n = 34$ patients after abdominal surgery found positive correlations between the LF/HF HRV ratio (partly reflecting sympatho-vagal activity) and post-operative pain intensity (Chang et al. 2012). These results propose that after surgery, a relatively higher sympathetic to vagal response (though LF-HRV only partly reflects sympathetic activity) may be associated with more post-operative pain. It is possible that reduced vagal activity and slightly enhanced sympathetic activity together contribute to insufficient inhibition of post-operative inflammation, possibly resulting in more pain (De Couck et al. 2014).

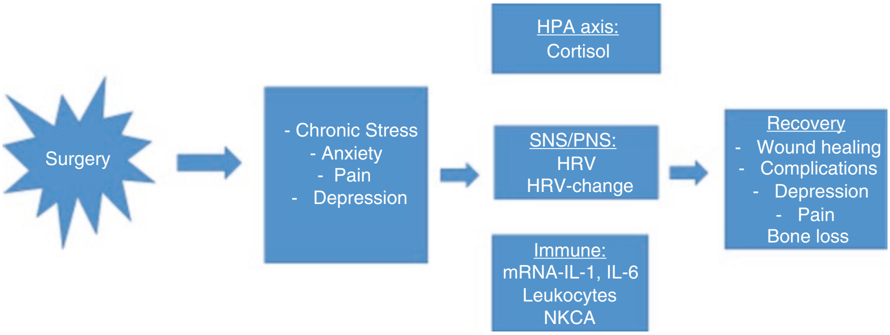

Fig. 9.2 Psychoneuroimmunological processes in wound healing and recovery from surgery

Figure 9.2 depicts a model integrating these studies into PNI processes which possibly mediate the effects of psychological factors on recovery from surgery, wound healing, post-operative pain, and depression.

Effects of Pre-surgical Psychological Interventions on Recovery from Surgery

Several systematic reviews and meta-analyses were conducted on this important issue, which consolidate the many studies and provide the evidence-based backbone of this topic. An older systematic review was conducted by Johnston and Vögele (1993) on 38 identified studies examining the effects of psychological preparation on recovery from surgery. The outcomes included pain, pain medication, negative affect, satisfaction, length of stay (LOS), behavioral and clinical indices of recovery, physiological indices, and other outcomes. Two types of psychological preparation, namely, procedural information and behavioral instructions concerning what patients can do perioperatively, positively affected all outcomes. Furthermore, relaxation methods positively affected all outcomes except for behavioral outcomes. Other forms of preparation such as hypnosis and sensory information had no effects on recovery. This review points at the need to disseminate such types of preparation, especially procedural information, behavioral instructions, and relaxation in surgical departments and in medical education.

A more recent review was conducted specifically on "mind-body" preparatory interventions which included relaxation, hypnosis, and guided imagery. The researchers found 20 studies conducted on a total of 1297 patients before various surgeries. All the intervention types affected psychological outcomes, while only guided imagery also affected post-operative analgesic intake. No intervention affected LOS. However, the review concluded that the methodological quality was not high (e.g., not reporting allocation concealment and blindness procedures), which must improve (Nelson et al. 2013).

The most recent and comprehensive review on this topic has been done in a Cochrane report by Powell et al. (2016). Their review included 105 studies with 10,302 patients. Beyond all types of psychological interventions, significant effects were found for reducing post-operative pain, LOS, and negative affect. However, again, the authors concluded that in most cases, the level of evidence was low. Despite the methodological limitations, it is important to note the large amount of scientific research done on this topic. Keeping these important limitations in mind, I will now provide a few examples, each reflecting unique forms of intervention, some less known to researchers and clinicians.

One study allocated $n = 82$ patients to psychological preparation or to usual care, before hip replacement surgery. The psychological preparation included information about the surgical procedure, expected sensations patients may experience, coping skills such as using distraction and relaxation, and seeking support. The psychological preparation led to significantly less use of post-operative intramuscular analgesics, fewer post-operative days when patients began to walk in the hospital, and a shorter LOS, compared to controls (Gammon and Mulholland 1996). Importantly, the reductions in LOS were on average 3 days, which could also have immense economic impact for health-care institutions and for patients.

A unique study took a neuropsychological approach and investigated the effects of post-operative repetitive transcranial magnetic stimulation (rTMS) of the left prefrontal cortex, compared to sham stimulation, on several outcomes. Left-rTMS was given at a frequency of 10 Hz, 10 seconds on, and 20 seconds off, for 20 minutes. Results revealed that left-rTMS led to significantly lower levels of morphine prescribed up to discharge than sham stimulation (Borckardt et al. 2008). While such an intervention cannot be applied in all settings, this does call for more research and proposes that activating the left prefrontal cortex may reduce post-operative analgesic requirements. This study also has profound implications for the topic of neuromodulation of health and disease, a topic I think will be a next domain in the future decades.

Often, surgical staff listen to music, to help regulate their stress levels. However, surgical patients are never offered the same, one reason presumably being that they are thought not to be aware of their environment and to have no sensory experiences during full anesthesia. However, the studies reviewed above do suggest full awareness, though rare, and possibly more frequent implicit awareness during full anesthesia (Gidron et al., 2002). One study tested the effects of patients "listening" to music during and after surgery, versus controls without music, on patients' recovery outcomes. Patients in the music group reported significantly lower post-operative anxiety and required less intubation time than controls (Twiss et al. 2006). However, one limitation is that the control group should have also had earphones on their heads during surgery, to partly control for such effects. Nevertheless, such an intervention, if replicated, can have important implications.

Given the role of social support in behavioral medicine in general and in predicting recovery from surgery specifically (e.g., Do et al. 2012), and since a basic way of increasing self-efficacy is by vicarious learning, I review here a study which examined the effects of modelling on patients' recovery from surgery. In this study, $n = 56$ patients about to undergo coronary artery bypass graft surgery were randomized to receive social modelling or to a control group. In the social modelling condition, patients met a previous patient who underwent the same surgery and who provided them information and emotional support, modelled how a similar patient can do well, and tried to increase patients' self-efficacy to perform lifestyle changes by modelling the changes they performed in their own lives. Significantly greater reductions in anxiety and increases in self-efficacy and in activity levels were reported by patients receiving the modelling condition than by controls, at various follow-ups (Parent and Fortin 2000). This study may be easily applied in many places where "veteran patients" may actually be unemployed, thus empowering both them and new patients.

A relatively recent study (Chartrand et al. 2017) was done in children and their parents. Here the models were children's parents. Child-parent dyads ($n = 123$ dyads) were randomized to receive usual care or usual care + an educational DVD. The DVD also included information on expected child responses and coping strategies to help children relax. Children underwent dental or ear, nose, and throat surgeries. Parents of children in the experimental group showed greater knowledge of and actual use of distraction and relaxation in the recovery room than control parents. Finally, these differences were paralleled by less pain in children of the experimental group than controls. These results show that in children, preparing parents can have important effects on their adequate care for their child and on the child's actual short-term recovery from surgery.

Finally, two forms of "social-environmental" interventions have been tested as well. First, residing in a room before surgery, with patients who already underwent surgery, was related to lower anxiety levels, shorter LOS, and faster post-operative mobility, than when residing in a room with patients who were also before their own surgery (Kulik and Mahler 1989; Kulik et al. 1993). Second, an old study found that patients who underwent cholecystectomy and had a view of nature from their rooms had shorter LOS, required less analgesics, and were rated by nursing staff less "negatively" than those without such nature views (Ulrich 1984).

While it is not always logistically possible to have such social-environmental factors for all patients occur, these studies propose two important points: first that the social and ecological environment play a role in patients' recovery. Second, such simple "interventions" could be implemented especially for patients about to undergo a complex surgery or to those with risk factors found in the studies reviewed above (e.g., low self-efficacy).

Case Study Conclusion

Upon assessment, using the brief 10-item center for epidemiological studies depression scale, you realize that Ms. Cohen shows symptoms of depression, possibly due to her thoughts and fears from not being able to play the piano as before. You read the study by Yirmiya et al. (2006) on depression and bone healing and decide to prescribe her five sessions of cognitive-behavioral therapy for her unrealistic thoughts and low mood. Her sleeping improves; her depressive symptoms decline in parallel with a now faster bone reunion and healing 3 months later. In the last follow-up, at 6 months, she returns to your office smiling, accompanied by a recording of her recent show with her piano in London.

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Chapter 10
Emergency Mental Health After Traumatic Events

Case Study

> Mr. El-Hakim arrived to your GP clinic a week after having a minor traffic accident. He complains of nightmares about the accident and was found to often have rapid heart rate, though he was only mildly injured. You ask him if he spoke to anyone about the accident, and you show your surprise at his childish behavior – after all, he was only mildly injured! He replies that he spoke to a psychologist in the hospital a day after the accident. She "debriefed" him. You then wonder why he is not getting better after psychological debriefing; after all, he expressed his feelings about the event to her and had someone listen to his story. You offer him sleeping pills, which help him sleep, but he still complains of many flashbacks about the accident during the day. You have no more ideas how to help him!

Introduction

Throughout life, we may be exposed to various types of events, some of which can be life threatening and "traumatic." These can include arriving at emergency room following a traffic accident, witnessing someone else suddenly injured, being severely physically assaulted, or even learning about the unfortunate and unexpected death of a loved one. *What makes an event traumatic and what percentage of the population has been exposed to traumatic events? If we follow the pioneering work of Speisman et al.* (1964)*, showing the centrality of the cognitive appraisal in determining the stress response, and if we follow the definition of the DSM-IV* (1994)*, an event is traumatic only when it is **perceived** as threatening, if it elicits fear, and if people experience helplessness or lack of control soon after it commences. Unfortunately, most of the research does not inquire about these*

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perceptions and feelings when trying to estimate the lifetime experience of traumatic events (TE). Thus, we must keep this limitation in mind when reading the following findings. One study conducted in 1779 Italian adults found that 56.1% reported at least one TE in their life (Carmassi et al. 2014). Similarly, a six-country European study found that 63.6% of Europeans reported at least one potential TE in their life (Darves-Bornoz et al. 2008). *Given the lack of assessment of appraisals in many studies on TE, it is impossible to estimate whether these results truly reflect TE. Thus, I shall refer to them as potentially TE, connoting that their real degree of adversity depends possibly on the appraisals of fear, threat, and helplessness, and thus, these are potential TE. Nevertheless, these figures, even if overestimated, suggest that there may be a need to help many people deal with potential TE. However, what are the consequences of potential TE such as a traffic accident, exposure to someone else's injury or even death, experiencing violence or even rape, and other life-threatening events? Studies show that except following a rape, most people exposed to potentially TE* ***do not*** develop long-term adverse mental health problems in the form of post-traumatic stress disorder (PTSD; Norris 1992). This suggests that many people are possibly very resilient, a fact that is often overlooked. We shall return to this important concept below.

Three typical mental health consequences may occur in three phases following a potentially TE. First, people may develop signs of the acute stress reaction (ASR) up to 48 hours from the TE. In the ASR, people may report emotional symptoms of fear, show behavioral signs of withdrawal, avoidance or panic, cognitive signs of confusion and dissociation, and physiological signs of rapid heart rate (HR) or even sweating. It is very hard to estimate from the literature the true prevalence of the ASR because many investigators confuse it with the acute stress disorder (ASD) which formally begins 48 hours after the triggering TE, up to a month later. Thus, many researchers use measures of the ASD to assess the ASR soon after the TE or assess the ASR beyond 48 hours from the event. The ASD includes a cluster of intrusive symptoms (e.g., reliving the event, having emotions similar to those during the event, experiencing flashbacks about the TE), a cluster of avoidance symptoms (e.g., avoiding to think about the TE or to be exposed to reminders of the TE) and having emotional numbing (emotional flatness), and physiological arousal (e.g., difficulties concentrating and sleeping). These same symptoms constitute the post-traumatic stress disorder (PTSD), which formally begins a month after the potentially TE.

What is the prevalence of PTSD following different TE? *One large-scale study done in the USA was conducted on 5877 people aged 15–54 years. That study found a lifetime prevalence of PTSD of 7.8%. Astonishingly, in one third, the symptoms never resolved (Kessler et al. 1995). The latter finding proposes that many people either get no therapy or receive ineffective treatment. Sadly, as we shall see below, many, too many interventions, are not scientifically based and, hence, are not* ***evidence-based, as is the case in many paramedical domains****. Studies show that following traffic accidents, approximately 20% will develop PTSD; after direct exposure to terrorist attacks, 28% develop PTSD; after a myocardial infarction, between 5 and 10% develop PTSD; and sadly, after rape, approximately 45% may* *develop PTSD (*e.g.*, Norris* 1992*; Gidron* 2002*). Because of the high prevalence of traffic accidents and the relatively high prevalence of subsequent PTSD, traffic accidents are the leading cause of PTSD in developed countries. However, it is important to note that the above prevalence rates differ sharply between studies done within the same type of TE because of differences in sampling (*e.g.*, treatment-seeking or all survivors of a TE) and differences in assessment methods (interviews yield lower rates than questionnaires) and time since the TE (Richardson* et al. 2010*).*

Given the amount of migrants and refugees today, this vulnerable population is becoming increasingly important to investigate and help because many have been exposed to repeated and severe TE. One study examined the prevalence of PTSD in forcefully displaced children. At baseline, 20–48.7% of children had post-traumatic stress symptoms, and 12 years later, this rate was still high – 35% (Tam et al. 2017). In a study we conducted among displaced Burundian students living in Rwanda during a humanitarian mission, 51% had probable PTSD, based on a brief questionnaire (Gidron and Arke unpublished data). These alarmingly high figures call for international en masse mental health aid, using evidence-based methods.

Emergency mental health is also relevant for people arriving at the emergency room (ER) for nonmental health reasons, rather for physical conditions. This is because often the unexpected reason for which people arrive at the ER (e.g., traffic accident, fracture) together with its potential threat on life and possible elicited helplessness can elicit the ASR, which in some people could develop into the ASD and later into PTSD, as we shall see.

Theoretical Models Explaining Responses to Traumatic Events

Multiple theoretical models have been developed, to explain why people exposed to potentially TE may develop PTSD and other adverse mental health outcomes (e.g., depression). Each approach then proposes relevant interventions. The behaviorist approach uses a classical conditioning approach and posits that associations are formed between the initial TE (unconditioned stimulus; e.g., a red Audi hitting a man) and resembling and associated stimuli (conditioned stimuli – any red car), which can subsequently elicit the conditioned response (e.g., rapid HR, flashbacks) themselves. Stemming from this approach would be to expose people to conditioned stimuli in a safe environment and to cut their association with the unconditioned stimulus, until their conditioned reaction disappears. We will review the treatment method of prolonged exposure below, which represents such theorizing.

The cognitive model posits that distorted cognitions concerning the triggering TE, and especially thoughts concerning resembling but nontraumatic stimuli, contribute to and maintain PTSD symptoms. Thus, for example, if a person thinks that *any* red car must be as dangerous as the original red Audi that hit him or her, then every exposure to a red car could elicit PTSD symptoms. These would also result in a pathologic avoidance of any red car or even other red stimuli. The derived therapy would then of course be cognitive behavioral therapy, in which such exaggerated and overwhelming cognitions would be challenged and replaced by more realistic ones, coupled by relaxation techniques.

A psychosocial resources model was developed by Hobfoll (1989) which is called the conservation of resources (COR) model. This model proposes that under severe TE, people try to minimize their loss of psychosocial resources and increase gains. Such resources include material ones (e.g., home), interpersonal resources (e.g., family, friends), and intrapersonal resources (e.g., self-efficacy, optimism). Indeed, one study was done in 1196 Palestinians exposed to occupation-related violence in the Israeli-Palestinian conflict. They were interviewed over three time periods. The outcome was engagement – looking forward to one's day and being absorbed in one's job. Trauma exposure was predictive of less engagement via loss of resources at time 1. Furthermore, resources at time 2 were an important predictor of engagement at time 3 (Hobfoll et al. 2012). In a study done among Turkish survivors of an earthquake, levels of human and material loss were positively associated with general distress and with intrusive symptoms (Sumer et al. 2005).

Finally, a neurocognitive model includes cognitive control and the vagal nerve and proposes that the vagus nerve, whose activity is indexed by heart rate variability (HRV), may exert cognitive control via inhibitory cortical control. PTSD is characterized by uncontrollable intrusive thoughts about the TE. In contrast, HRV is positively correlated with frontal activity (Thayer et al. 2012), and the latter can exert inhibitory control over intrusive thoughts (Gillie and Thayer 2014). Indeed, HRV is also associated with better inhibition of unwanted thoughts in laboratory tasks (Gillie and Thayer 2014). From this model we can derive interventions which aim to increase frontal and vagal control over trauma-relevant intrusions. The vagal nerve, being the main branch of the parasympathetic nervous system, would also reduce sympathetic stress responses that are often high in PTSD. The main intervention described below aims precisely to achieve both.

Risk Factors for Developing PTSD After Traumatic Events

Numerous risk factors for PTSD have been investigated, and I shall focus only on those identified in prospective studies, as these are methodologically stronger. Many of these studies were reviewed by DiGangi et al. (2013). Focusing on physiological predictors, some studies found that higher heart rate (HR) in the ER significantly predicts increased risk of PTSD (e.g., Shalev et al. 1998), though one study found the opposite (Blanchard et al. 2002). A meta-analysis of cross-sectional studies on this topic confirmed an increased HR in PTSD (Nagpal et al. 2013). Other studies found that pre-trauma startle responses, the physiological "jumping" response to triggers like noise, predicted PTSD (e.g., Guthrie and Bryant 2005). Finally, one study examined whether pre-deployment levels of heart rate variability (HRV), a vagal nerve index (Kuo et al. 2005), in two relatively large samples of American soldiers ($N = 1415$, $N = 745$), predicted future risk of PTSD after deployment. The rates of PTSD in the samples were 3.3–4.2%. Indeed, a high ratio of low-frequency/high-frequency HRV (LF/HF) predicted an increased risk for PTSD. In addition, the prevalence of PTSD in soldiers with high pre-deployment LF/HF was 15.8%, while the prevalence of PTSD in soldiers with a low pre-deployment LF/HF ratio was only 3.7% (Minassian et al. 2015). The low-frequency HRV reflects sympathetic and parasympathetic activity (not only sympathetic, as is commonly thought), while the high-frequency HRV component reflects parasympathetic activity only. Thus, a cautious interpretation of this finding is that an increase in the relative sympatho-vagal tone may predict risk of PTSD, when exposed to potentially TE such as military combat experiences. The meta-analysis of Nagpal et al. (2013) on cross-sectional studies supports a reduced vagal tone (HF-HRV) in PTSD. These results support the cognitive vagal model of PTSD (Gillie and Thayer 2014).

What about neurocognitive factors? Several studies have shown that peri-traumatic dissociation, a state involving misperception of time during the TE, out-of-body experience (e.g., viewing one's body from above), and amnesia concerning important aspects of the TE, predicts quite consistently the development of PTSD. Indeed, a meta-analysis of 35 studies supported this conclusion (Breh and Seidler 2007). Another related neurocognitive factor is limited cognitive capacity (e.g., IQ, autobiographical memory, processing speed), which was found to predict PTSD (Betts et al. 2012; Koenen et al. 2007).

Personality factors such as hostility (e.g., van Zuiden et al. 2011), negative affect and neuroticism (Parslow et al. 2006; Weems et al. 2007), and trait dissociation (Hodgins et al. 2001) were found to predict PTSD as well. In their review of studies on forcefully displaced children which mostly included prospective studies, exposure to TE, older age, and prior psychopathology were significant risk factors of post-traumatic stress symptoms in such children (Tam et al. 2017).

An important meta-analysis of prospective studies found that pre-trauma cognitive factors (e.g., negative self-appraisal, autobiographical memories), certain coping styles (e.g., rumination, little repression), personality (e.g., neuroticism, hostility), psychopathology (e.g., depression), psychophysiological factors (e.g., arousal), and social-ecological factors (e.g., family of origin, little social support) predicted PTSD (DiGangi et al. 2013).

Past Interventions for Preventing PTSD

This is the sad aspect of this chapter, not only because of the failure to prevent PTSD but, more crucially, because these results reveal what happens if we do *not* base interventions on empirical evidence using randomized controlled trials (RCT). This must be the scientific, ethical, and clinical pillar for guiding clinicians in which treatment to use (any type of treatment) for any clinical problem, following the Oxford pyramid of evidence. In an important study done on Dutch patients following traffic accidents (TA), Brom et al. (1993) randomly assigned people to either a “coping and working through” intervention or to no treatment (control), a month after the TA. Both groups showed reductions in PTSD symptoms. Two limitations of this study are the lack of clarity and explication of what coping and working through actually were and that the interventions may have been provided too late following the TA. It is possible that we need to intervene in the ASR phase to prevent PTSD.

Perhaps the most commonly used form of intervention, sadly still used in many countries, is psychological debriefing. This type of intervention originated from military practice, where the commander of a platoon or unit “debriefed” soldiers returning from the battlefield, to inquire about the combat procedures in order to learn lessons and improve performance in future missions. In therapeutic debriefing, people are asked to detail the event they experienced, their reactions are normalized (e.g., “everyone would experience such sadness as you do”), and the therapist would provide legitimacy to patients’ reactions. Importantly, the debriefing emphasizes provision of empathy and support from the therapist, and especially there is an emphasis on expression of emotions. Does debriefing work, can it prevent PTSD when provided soon after a potentially TE? Sadly, despite its continuing common worldwide use, many studies have showed that it fails to prevent PTSD. Even worse, several studies found debriefing to increase the risk of PTSD. For example, Mayou et al. (2000) randomized people soon after traffic accidents to receive debriefing or no psychological treatment and followed patients for 3 years. Remarkably, those assigned to debriefing had significantly worse PTSD and pain symptoms and lower levels of quality of life than controls, at the 3-year follow-up. Importantly, this grim result especially occurred in those with initially high levels of intrusions, a risk factor and part of the PTSD cluster of symptoms (Mayou et al. 2000). These results demonstrate that debriefing is ineffective, is possibly even harmful, especially for people with initially high intrusions.

Five systematic reviews and meta-analyses have been conducted, with all concluding that debriefing has *no effect* on preventing PTSD, and some evidence even exists that this method can increase the risk of PTSD (e.g., Mayou et al. 2000; Rose et al. 2002; Van Emmerik et al. 2002). Two critiques often posed by the proponents of debriefing are that this method should be given by professionals trained in debriefing and that it possibly suits more professionals with preparation for TE such as soldiers or rescue providers than nonprofessionals. However, one study examined the effects of debriefing over 2 years in a sample of 62 debriefed and 133 controls, all rescue help providers following an earthquake in Australia. Again, there was no evidence for any benefits from debriefing in such a population as well (Kenardy et al. 1996). Why has debriefing failed, and, more importantly, why are many clinicians still providing it? The first question is a scientific one, while the second one relates to multiple professional issues which are not in the scope of this book. However, I would like to question the ethical and clinical legitimacy in continuing to provide debriefing when so much stringent evidence guides clinicians to *cease* from using it. This is medical negligence.

A Voyage into the Neuropsychology of Trauma Leading to an Alternative Intervention

It is possible that understanding the neuropsychology of how traumatic memories are encoded in people who later develop versus do not develop PTSD is a key for developing better preventive interventions. Van der Kolk et al. (1985) and Van der Kolk and Fisler (1995) found that compared to traumatic events which people recall in a fragmented, somatic, and emotional manner, stressful yet non-traumatic events are recalled with a narrative (beginning, middle and end), using more reasoning. This informs us about qualitative differences in which traumatic events are encoded and recalled in memory, being more chaotic, physical, and emotional. Furthermore, studies have compared people with PTSD to those without PTSDs on the emotional stroop test. In this test, people are asked to name the color of words which are emotionally negative and related to the traumatic event (e.g., “car” following a traffic accident), general emotionally negative words (e.g., “fear”), and emotionally neutral words (e.g., “tap”), matched on word length and frequency of use in the language. This is a test of the inhibition element of executive functioning. In several studies across cultures and ages, it was shown that people with PTSD take significantly longer time to name the color of words which are relevant to the traumatic event than either generally emotionally negative or neutral words, while people experiencing such events, who did not develop PTSD, do not show such differences (e.g., Foa et al. 1991; Moradi et al. 1999). These results propose that people with PTSD have difficulties to inhibit their attention to trauma-relevant external reminders (e.g., the word “car” on a screen) in favor of attending to neutral information like colors of words (e.g., orange). As mentioned above, this can be understood as reflecting poorer executive inhibitory control, as measured by the emotional stroop test. Poorer executive control has been proposed to underlie PTSD together with reduced vagal nerve activity (Gillie and Thayer 2014). An analogy from the (Jewish) kitchen for cooking “Hamin” before the Sabbath can help to understand this issue. Religious people do not use electricity or fire during the Sabbath, since these acts reflect “work.” Thus, they cook Hamin beforehand and maintain it warm on a hot plate which is never off, throughout the Sabbath. Similarly, the traumatic memory is on low fire and does not require much external stimuli to turn it on. Any external reminder of the event requires little neuronal effort to activate the traumatic schema, which seems not to be “turned off” in people with PTSD, as seen on the emotional stroop studies.

Finally, the most direct and compelling evidence comes from studies using functional neuroimaging techniques, which reveal what may occur in the brains of people with and without PTSD, while processing trauma-relevant information. In these studies, people following traumas with versus without PTSD are compared to no-trauma healthy controls. While undergoing brain scanning, they listen and/or see stimuli related to their traumatic event or are exposed to neutral stimuli. Brain activity is then measured during exposure to both sets of stimuli. Numerous studies have found that people with PTSD show greater activation in multiple brain regions but especially in the amygdala, a threat-processing region of the limbic system. In contrast, "traumatized" non-PTSD controls show greater activation in the prefrontal cortex (e.g., Bremner et al. 1999; Liberzon et al. 1999; Shin et al. 2004). Furthermore, levels of PTSD symptoms are positively correlated with amygdala activity levels and are inversely correlated with prefrontal activity levels (Shin et al. 2004). Importantly, one study examined the size of the hippocampus and its connectivity with frontal regions, in $n = 33$ Israeli soldiers, before and after exposure to military threats. In some of them, the hippocampus decreased, while in others it increased in size, after military service. Reductions in its size were associated with higher PTSD levels after military threat exposure. Of greatest importance was the finding that among those with decreased hippocampi, extent of hippocampal connectivity to the ventromedial prefrontal cortex (vmPFC) was lower than in those with increased hippocampi, after military threat exposure (Admon et al. 2013a). This could be crucial since the vmPFC can reduce levels of helplessness (Amat et al. 2008), often experienced in traumatized people. In a review of these studies, Admon et al. (2013b) found that abnormal amygdala and dorsal anterior cingulate activity predict PTSD. In contrast, after developing PTSD, one finds reduced connectivity between the hippocampus and the vmPFC. These findings teach us that people who develop PTSD over-activate the threat-relevant brain region (amygdala) before the TE and after it, they encode the traumatic memory in a more emotional manner reflecting threat as well (amygdala). Furthermore, they fail to activate sufficient connectivity between the memory-related region (hippocampus) and a frontal region which modulates helplessness, namely, the vmPFC (Amat et al. 2008). In addition, this weak connectivity can also reflect a disconnection between traumatic memories and a higher level of processing in the frontal cortex. In contrast, "traumatized" healthy controls encode the event in a more rational manner reflected by greater frontal activity. This is crucial because the frontal cortex can exert inhibition of amygdala activity (Quirk and Gehlert 2003), a capacity which people with PTSD seem to particularly lack (Shin et al. 2004).

What does all this inform us for the prevention of PTSD? It is possible that in order to prevent PTSD, we need to *shift the processing* of the traumatic event from a limbic, emotional, chaotic, and somatic manner to a frontal, cognitive, coherent, organized, and verbal manner (Gidron et al. 2001). Furthermore, by making the memory more logical, we may increase the hippocampal-vmPFC connectivity which is lacking in PTSD (Admon et al. 2013b). Simplistic emotional catharsis or simple retelling the TE may not necessarily achieve this processing shift and could instead kindle an already overactive amygdala, which may explain the failure of debriefing (e.g., Rose et al. 2002).

What hints exist in the clinical scientific literature that may guide us in achieving this processing shift? In the context of providing "flooding therapy," a form of systematic behavior desensitization, to women who developed PTSD after the tragic experience of rape, Foa et al. (1995) found that the more women's story became chronologically organized from session to session, the lower their PTSD symptoms were. Chronological organization, which may require some meta-cognition, probably reflects a higher level of brain processing than processing in the amygdala. However, since many traumatized people may recall their event in a fragmented manner (Van der Kolk and Fisler 1995), they may need guidance in chronologically organizing its segments soon after experiencing it. Furthermore, Hariri et al. (2000) found that asking people to match emotionally negative faces to similar faces activated the amygdala, while matching emotionally negative *words* to corresponding faces activated the prefrontal cortex. This suggests that rather than asking people to repeat their traumatic story and try to "reconnect to" or to "re-experience" their emotions and somatic experiences, which may tragically reactivate an already overactive amygdala, they need to learn to verbally label such emotions and somatic experiences. This may shift their processing to the prefrontal cortex. Finally, in the context of content analyses of written disclosure studies of past traumas, Pennebaker and Francis (1996) found that using words which reflected *insight and causality* (e.g., "now I understand that"; "I was in my car because I picked up my child") predicted the health benefits of written disclosure. These results inform us that people may need help to *understand* and report the *causal links* between their event's segments and the causes of their emotional and somatic experiences (e.g., "I was very sad because I could not see my daughter at that moment"). These understandings, based on converging empirical findings, guided me in the development of the memory structuring intervention (MSI).

The Memory Structuring Intervention: A Neuroscience-Based Method

The aims of the memory structuring intervention (MSI) have been laid out in previous publications (Gidron et al. 2001, 2007). Briefly, the MSI aims to shift trauma processing from a limbic, emotional, and somatic level of processing to a frontal, cognitive, verbal, and coherent level, as stated above. It aims to provide patients with greater control over their traumatic memory. In the MSI, patients first recall their event. Each time they say a somatic or emotional word (e.g., pain, fear), they are asked to elaborate these verbally and to provide a reason (e.g., "what do you mean by fear, and why were you afraid?"). In parallel, the counselor numbers to him/herself on paper the events' segments in their real order of occurrence. In stage 2, the counselor repeats the story, in its *chronological order*, using precise *labels* for the emotions and sensations and linking the segments and the emotional and somatic experiences to their respective *causes* (e.g., "You were afraid because you did not see your daughter in the car"). The patient is then told that we cannot change the event but can try to help him/her change *the way they remember* the event. They are then asked to report it in its chronological order, using precise labels for their recalled emotions and sensations and linking the segments and the emotional and somatic experiences to their respective causes. In stage 3, the patient retells the event following these three principles and the way the counselor told it. Typically, the patient appears now as a journalist recalling the event.

Sadly, many people who work in this domain stop here, without providing any scientific evidence for their new intervention technique, in the name of being "experts." I am no expert, and, more importantly, the Oxford pyramid of evidence, which should guide the adoption of any clinical work, puts expert opinion at the bottom of the pyramid. In contrast, randomized controlled trials (RCT) and meta-analyses of RCT are at the top of the pyramid. The MSI is not there yet, but we are working toward that level of evidence.

In the first small RCT, we randomly assigned 17 Israelis following a traffic accident, to either the MSI or to supportive listening (control), both conducted over the phone within 24–48 hours after the accident. Each group received two phone calls. Three months later, their symptoms were assessed by a person blind to their group status (your servant here). The inclusion criteria were having a mild injury following an accident, with HR > 94 BPM, a risk factor of PTSD in some studies (Shalev et al. 1996) and without loss of consciousness or head injury. At 3 months, the MSI group reported significantly lower levels of total PTSD and lower levels of intrusion and arousal than controls (Gidron et al. 2001). However, the sample was very small. A subsequent study followed a highly similar protocol but included *n* = 34 patients. Sadly, we found no group differences, and instead we observed a statistically significant group x gender interaction in relation to PTSD. In men, the MSI group reported significantly higher PTSD symptoms than controls, while in women, the MSI group reported significantly lower PTSD symptoms than controls (Gidron et al. 2007). First, these execpected and unfortunate results demonstrate the firm ethical, clinical, and scientific need to rely on scientific evidence when developing new interventions, instead of on "dogma" or "models." Second, it is partly possible that additional intervention elements are needed for men. In boys, an initially elevated sympathetic response predicts future risk of PTSD, while this was not seen in girls (Delahanty et al. 2005). Men may require an additional intervention element to first reduce their sympathetic response prior to restructuring the recall of their traumatic event by the MSI. Activating the vagus nerve, the main branch of the parasympathetic nervous system, can of course reduce excessive sympathetic activity (Saku et al. 2014). Adding a vagal enhancing element is warranted because PTSD is associated with reduced vagal activity (e.g., Green et al. 2016) and because vagal activation may lead to increased frontal over limbic activity (Dietrich et al. 2008), which may reverse the pattern often seen in PTSD (e.g., Shin et al. 2004). A meta-analysis on brain correlates of heart rate variability (HRV), the vagal nerve index, found that HRV is quite strongly positively correlated with prefrontal cortical activity (Thayer et al. 2012). Vagal activation can be simply achieved by slow-paced breathing (Wheat and Larkin 2010), and systematic long-term vagal breathing alone was found to reduce PTSD symptoms (Tan et al. 2011). In a larger RCT done on *n* = 124 patients in the North of Israel, we observed in the emergency room that those assigned to the MSI + VB reported significantly lower levels of pain and anxiety and had lower HR, than controls. This pattern of results was similar in men and women (Gidron et al. 2018).

Subsequently, in a more recent study, we compared an expanded MSI + vagal breathing (VB) to a supportive listening control condition in *n* = 25 people arriving to an ER in Brussels. All patients were at risk for PTSD because they reported in two consecutive times ≥5 or increased from below to above that cut-off in their levels of stress on a 1–10 scale (Dekel and Kutz 2004). In this last RCT, patients assigned to the MSI + VB reported significantly lower levels of perceived traumatic experiences (fear, threat, and helplessness) an hour later, compared to controls. We also measured a proxy of frontal activity by measuring verbal fluency (Wagner et al. 2014). After treatment, the MSI + VB showed significantly higher levels of verbal fluency than controls. Importantly, across groups, increases in verbal fluency significantly and positively correlated with reduced perceived traumatic experience and HR (Gidron et al. 2018). Future studies need to examine the long-term effects of the MSI + VB on PTSD prevention and whether the mechanism includes activation of the prefrontal cortex over limbic brain activity, using neuroimaging. Effects of the MSI + VB on exisiting PTSD also need to be tested.

Additional Methods in Emergency Mental Health

The MSI + VB was developed to be provided to people within hours or up to 48 hours after the TE, to reduce the acute stress response and prevent PTSD. However, there are two important additional intervention windows, one at zone 1, where the scene occurs, and the second one days-weeks after the TE, where long-term recovery should take place. In zone 1, people may show a multitude of reactions from proactive responses (helping oneself and others) to severe anxious reactions, dissociation, and even catatonia. My colleague Moshe Farchi has developed and preliminarily tested an intervention called the Six C's model (Hantman and Farchi 2015). This model is based on concepts such as Antonovsky's (1993) sense of coherence, Bandura's self-efficacy (Bandura 1977), and Taylor's work on the neural correlates of psychosocial resources (Taylor et al. 2008). In her work, Taylor administered to people a multitude of scales (e.g., meaning of life, autonomy, mastery) and found them to load onto one factor they termed psychosocial resources (PR). PR was inversely correlated with stress-induced cortisol and with amygdala activity but positively correlated with right ventrolateral prefrontal cortical activity (Taylor et al. 2008). Together, these studies propose that we need to help people in a potential TE to be active and coherent and to have a sense of direction and purpose, which are forms of using one's PR. These contract the current emphasis on pittying and empathy which could possibly only weaken people's PR and adaptation to a TE. Farchi developed a simple protocol for nonprofessionals, reflected by the Six C's model. In this model, the helper is asked to Communicate, Cognitively (rather than emotionally) (e.g., what is your name? How many people arrived with you?), provide simple Challenging tasks to people (e.g., Please tell me how many people are hurt in the accident?), provide people Control (e.g., do you want to call your wife or children?), remain Committed to them (e.g., You stay with them until further help will arrive), and help them put the story in a chronological or Continuous order (e.g., you drove to work, you then had an accident, now we are here, and you will soon go to hospital to undergo checkups and receive treatment). In pilot work during one of the wars between Israel and Gaza, people seeking help for mental distress showed statistically significant reductions in mental distress following the Six C's intervention method. However, no proper control group was used (Farchi et al. 2018). Though this intervention method must be tested in an RCT, the ethical and clinical constraints in doing so in zone 1 prohibit running such an RCT. We are now running a lab experiment to test the Six C's method. Nevertheless, this intervention shows promise for emergency mental health providers at the first minutes and hours following a potentially TE. In addition, this method can be utilized by nonprofessionals, crucial for mass disasters.

Finally, it is important to help people return to normal functioning in the days-weeks after the TE. This is especially crucial in mass disasters where a perturbed nation needs to stand on its feet. However, many barriers, whether objective or subjective, may prevent recovery. For example, following an earthquake, loss of property, water, and electricity constitute a major objective barrier for recovery. Furthermore, people may have quite realistic perceptions of lack of control and lack of ability to pursue their daily activities. When such perceptions are global and exaggerated, these can paralyze a person's actions and contribute to disability. Education and awareness alone may be insufficient. How can we help people overcome these perceptions, to accelerate recovery and rehabilitation? The method of "psychological inoculation" (PI), described in the chapter on adherence, can be used. To remind the reader, in PI, we expose people to exggerated challenging sentences (the "vaccine") which reflect their unrealistic perceptions (e.g., "there are no jobs at all") or which reflect social pressures they face, both which impede recovery. They then practice to systematically refute them (the "antibody response"). In a study we conducted over the phone during a war context between Israel and Gaza, we exposed people to supportive listening (control) or to PI. In the PI, we challenged their perceptions of lack of control and coping abilities (e.g., "You have no idea what to do once there is a siren"), which they had to systematically refute (e.g., they could say: "Of course I know what to do, I go into the shelter or hide behind the living room wall"). The war was going on for months and in fact for many years in sporardic waves. Results revealed that PI reduced perceptions of helplessness only in men, while these were reduced by supportive listening in women (Farchi and Gidron 2010). Though not a scientific study, I wll report on the adoption of PI in one unique context. In the context of a humanitarian mission in Nepal, we were sent by the Israeli humanitarian organization Natan. We learned that soon after the earthquake, someone drove in Kathmandu the capital, telling people that there will be another earthquake in the magnitude of 9.5 on the Richter scale! Many people ran into the streets, and the perpetrator went into peoples' houses to steel property! Beyond this immoral and disgraceful act, his words made "waves" across Nepal, and many people believed there will be such a strong earthquake. Inspired by two football coaches in our mission, and equipped with reliable information obtained from Sushil, a highly educated Nepali social worker who was my right hand, I decided to use the PI method. A reliable American geological Internet site informed us of a very low chance for such an earthquake. We then trained our trainees (teachers, psychologists) in using the PI method via throwing a ball (making the refutation of sentences more physical), where the challenger would say “There is a big chance for a big earthquake” and then the trainees threw the ball towards to trainee, and the trainee refuter would reply: “There is a small chance for a big earthquake.” This was also broadcasted by us on Nepali radio, to reach out to many citizens. We did not evaluate the effectiveness of such an intervention, but applied it and its evidence-based basis in a humanitarian context.

Case Study Conclusion

As we have seen in the scientific literature reviewed here, Mr. El-Hakim’s injury severity does not predict his risk of having PTSD. First, as a clinician, whether GP or psychologist, you need to use standardized tools for assessing PTSD and any other clinical entity whether psychosocial or medical, over the course of treatment, always. Kimerling et al. (2006) developed a brief and valid 7-item scale for assessing PTSD. It also provides a cut-off for diagnosing “probable PTSD.” The scale should be used before and after the intervention, preferably twice before and twice after, reflecting an AABB design. Second, numerous studies and five reviews show that debriefing does not prevent and may even cause PTSD (e.g., Mayou et al. 2000). Third, in any clinical action, we must be guided by RCT and their systematic review, in accordance with the Oxford pyramid of evidence. One possibility would be to give Mr. El-Hakim the vagal breathing + MSI intervention, to enable him to structure his memory and gain more control over it. Both should also help him sleep better and reduce his sympathetic responses. Finally, we should never laugh at or judge patients’ symptoms! They have the right to respond with any symptomatology, and it is the caregiver’s responsibility to provide the best care, based on the highest level of scientific evidence and humane approach.

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